



LTE and NR Core Network

Version: 2026-06-12

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1 Introduction

LTEMME is a LTE EPC (Evolved Packet Core) implementation. It has a built-in MME (Mobility Management Entity), SGW (Serving Gateway), PGW (Packet Data Network Gateway), PCRF (Policy and Charging Rule Function), HSS (Home Subscriber Server), EIR (Equipment Identity Register) and ePDG (evolved Packet Data Gateway). It can easily be used with the Amarisoft LTE eNodeB to build a highly configurable LTE test network.

Depending on your software license, it also includes a NR 5GC (5G Core Network). It has build-in AMF (Access and Mobility Management Function), AUSF (Authentication Server Function), SMF (Session Management Function), UPF (User Plane Function), UDM (Unified Data Management) and 5G-EIR (5G Equipment Identity Register).

2 Features

2.1 EPC

- LTE release 18 compliant.
- Implements one EPC with built-in MME, SGW, PGW, PCRF, HSS and EIR.
- Supports several eNodeBs with standard S1 interface (S1AP and GTP-U protocols).
- NAS integrity check and encryption using the AES, Snow3G and ZUC algorithms.
- Support of USIM cards using the XOR, Milenage or TUAK authentication algorithm.
- Handling of UE procedures: attach, authentication, security configuration, detach, tracking area update, service access, radio bearer establishment, paging.
- Multi-PDN support and built-in dynamic ERAB setup for easy VoLTE/IMS testing.
- Transparent access to the IP network (no external Serving Gateway or PDN Gateway is necessary).
- Configurable access point name, IP range, DNS and E-RAB QoS.
- Support sending of Public Warning System messages (ETWS/CMAS).
- IPv6 support.
- Configurable logging system for all channels with built-in text decoders.
- Remote API using WebSocket.
- Command line monitor.
- PSM and eDRX support.
- Group WUS support.
- Supports several IMS servers with Rx interface.
- Support of NB-IoT RAT.
- Support of control plane CIoT EPS optimization.
- Non-IP data delivery CIoT feature.
- Attach without PDN connectivity CIoT feature.
- User management via internal database without any external HSS.
- Support of optional S6a interface with external HSS.
- Support of optional S13 interface with external EIR.
- Support of optional SGsAP interface with external VLR/MS.
- Support of optional SBcAP interface with external CBC.
- Support of broadcast and multicast PDN options.
- Support of DCNR UEs.
- Support of LCS-AP.
- Support of Ethernet PDN connectivity.

2.2 5GC

- NR release 18 compliant.
- Implements one 5GC with built-in AMF, AUSF, SMF, UPF, UDM and 5G-EIR.
- Supports several gNodeBs, ng-eNBs or N3IWFs with standard NG interface (NGAP and GTP-U protocols).
- NAS integrity check and encryption using the AES, Snow3G and ZUC algorithms.

- Support of USIM cards using the XOR, Milenage or TUAK 5G-AKA authentication algorithm.
- Handling of UE procedures: registration, authentication, security configuration, deregistration, service access, radio bearer establishment, paging.
- Multi PDU sessions support and built-in dynamic QoS flow setup for easy VoNR/IMS testing.
- Transparent access to the IP network (no external UPF is necessary).
- Configurable access point name, IP range, DNS and QoS flows.
- IPv4, IPv4v6, IPv6 and unstructured PDUs support.
- Configurable logging system for all channels with built-in text decoders.
- Remote API using WebSocket.
- Command line monitor.
- MICO, active time and eDRX support.
- Supports several IMS servers with Rx interface.
- Support of NB-IoT, LTE and non-3GPP RAT.
- User management via internal database without any external HSS.
- Support of broadcast and multicast PDU session options.
- Support sending of Public Warning System messages (ETWS/CMAS).
- Support of N12 interface with external AUSF.
- Support of N8 and N13 interface with external UDM.
- Support of N17 interface with external 5G-EIR.
- Support of N20 interface with external SMSF.
- Support of N50 interface with external CBC.
- Support of N62 interface with external AF.
- Support of network slicing.
- Support of control plane CIoT 5GS optimization.
- Non-IP data delivery CIoT feature.
- Support of NL1 interface.
- Support of N5 interface with the following restriction: IMS_SBI feature is declared by the PCF but credit management is not supported.
- Support of Ethernet PDU sessions.
- Support of MBS broadcast sessions.

3 Requirements

3.1 Hardware requirements

- LTEMME can run on the same PC as the Amarisoft eNodeB/gNodeB if a simple and compact solution is needed. Otherwise, any reasonably recent PC with at least one Gigabit Ethernet port is acceptable.
- A test USIM card should be plugged into the UE. Test USIM cards from Anritsu are supported by the default configuration. Other test USIM cards should work as well provided they implement the dummy XOR authentication algorithm and that their IMSI and secret key are known. USIM cards using the Milenage or TUAK algorithm are also supported.

3.2 Software requirements

- A 64 bit Linux distribution. Fedora 42 is the officially supported distribution. The following distributions are known as compatible:
 - Fedora 22 to 42
 - Cent OS 7
 - Ubuntu 14 to 24

Your system requires at least GLIBC 2.17.

4 Installation

[Quick installation instructions are also given in the Amarisoft eNodeB/gNodeB documentation in case LTEMME is installed on the same PC as the eNodeB/gNodeB].

The network access thru the Gigabit Ethernet port must be correctly configured.

LTEMME can be run directly from the directory when it was unpacked. No need for explicit installation.

4.1 Local network configuration

LTEMME will set up a new virtual network interface `tun0` where each UE has a specific IP address. If you want them to connect to your local network (and Internet), you need to set up IP forwarding and masquerading.

An example is found in the `lte_init.sh`:

Syntax:

```
sudo ./lte_init.sh [-6] <ifname>

sudo ./lte_init.sh default
sudo ./lte_init.sh -6 eth1
```

4.2 Linux setup

4.2.1 Packages

LTEMME uses the SCTP protocol for which the necessary packages are not usually installed. In order to install them, do as root user:

- Fedora

```
dnf install lksctp-tools kernel-modules-extra
```

- Ubuntu

```
sudo apt-get install lksctp-tools linux-image-extra-3.13.0-24-generic
```

Note that `linux-image-extra` package name may differ depending on your kernel version.

To verify that SCTP kernel module is running, do as root user:

```
checksctp
```

If it reports that the protocol is not supported,

- check if you have a `/etc/modprobe.d/sctp-blacklist.conf` file
- edit it to comment the 'blacklist sctp' line

Then reboot the PC in case the Linux kernel was upgraded too.

4.2.2 OpenSSL

LTEMME has been compiled against openssl version 3.5.4.

If your system does not have compatible version installed you may have this error message at startup:

```
error while loading shared libraries: libssl.so.3: cannot open shared object file: No such file or directory
```

To overcome this problem, you may:

- Copy `libssl.so.3` and `libcrypto.so.3` from `libs` subdirectory of your release tarball. If you have installed software with automatic install script, this should have been done automatically.
- Compile and install proper openssl version yourself

In case of persisting issue, raise a ticket from our support site at <https://support.amarisoft.com/> with the information provided by below commands executed in LTEMME directory:

```
uname -a
ls -l
ldd ./ltemme
openssl version
```

4.2.3 NGHTTP2

LTEMME has been compiled against nghttp2 version 1.41.0.

If your system does not have compatible version installed you may have this error message at startup:

```
error while loading shared libraries: libnghttp2.so.14: cannot open shared object file
```

To overcome this problem, you may:

- Copy nghttp2.so.14 from `libs` subdirectory of your release tarball.
If you have installed software with automatic install script, this should have been done automatically.
- Install libnghttp2 with your package manager
- Compile and install proper nghttp2 version yourself

In case of persisting issue, raise a ticket from our support site at <https://support.amarisoft.com/> with the information provided by below commands executed in LTEMME directory:

```
uname -a
ls -l
ldd ./ltemme
```

4.3 License key installation

LTEMME needs a license key file to run. *It is associated to your PC, so if you replace it or change its hardware configuration you must contact Amarisoft to get a new license key.*

The following steps are needed to get this license file:

- Run LTEMME:

```
./ltemme config/mme.cfg
```


It says that the license key is not present and prints a 16 digit hexadecimal code.
- Send by mail to delivery@amarisoft.com this hexadecimal code to your contact at Amarisoft. You will get back the `ltemme.key` license key file.
- Copy the `ltemme.key` file to the `${HOME}/.amarisoft/` directory (`${HOME}` is the home directory of the `root` user). You can use the shell variable `AMARISOFT_PATH` to change this path.

Once the license key is installed, ltemme should start normally.

4.4 Initial testing

- Edit the file `config/mme.cfg` to set the bind address of the GTP-U interface. Normally it is the address of the default Ethernet of the PC (you can see it with `ifconfig`). You can also set the address of the DNS (`dns_addr` property). You don't need to change the other parameters for an initial test.

- LTEMME creates one virtual network interface where the UE traffic is redirected. A modification of the default routing rules and iptables is usually needed if you want to redirect the UE traffic to the local network and Internet. The script `lte_init.sh` in the Amarisoft LTEMME package gives an example of setup to configure a NAT to access the Internet.
- Start the program as root with:

```
./ltemme config/mme.cfg
```

[The root access is only needed to set up the Linux virtual interface.]
- The command line interface is used to monitor the operation of LTEMME and to change the logging options. Use `help` to get the list of commands and `quit` to stop the program.
- Use `enb` to list the connected eNodeBs and `gnb` to list the connected gNodeBs.
- In addition to using the log file, you can monitor the traffic between LTEMME and the eNodeBs or gNodeBs with Wireshark. The LTE specific traffic is filtered by putting `s1ap || gtp` in the `filter` input area. The NR specific traffic is filtered by putting `ngap || gtp` in the `filter` input area.
- For optimal performance, it is better to avoid fragmenting the GTP-U packets. So the Ethernet interfaces used between the eNodeBs or gNodeBs and LTEMME should be configured to have a MTU of at least 1564 (assuming the UEs use the standard MTU of 1500). You can verify with Wireshark whether the GTP-U packets are fragmented.

5 Configuration reference

5.1 Configuration file syntax

The main configuration file uses a syntax very similar to the Javascript Object Notation (JSON) with few extensions.

1. Supported types:
 - Numbers (64 bit floating point). Notation: 13.4
 - Complex numbers. Notation: 1.2+3*I
 - Strings. Notation: "string"
 - Booleans. Notation: true or false.
 - Objects. Notation: { field1: value1, field2: value2, }
 - Arrays. Notation: [value1, value2,]
2. The basic operations +, -, * and / are supported with numbers and complex numbers. + also concatenates strings. The operators !, ||, &&, ==, !=, <, <=, >=, > are supported too.
3. The numbers 0 and 1 are accepted as synonyms for the boolean values false and true.
4. {} at top level are optional.
5. " for property names are optional, unless the name starts with a number.
6. Properties can be duplicated.

If properties are duplicated, they will be merged following [JSON merge rules], page 9, with overriding occuring in reading direction (last overrides previous).

Ex:

```
{
  value: "foo",
  value: "bar",
  sub: {
    value: "foo"
  },
  sub: {
    value: "bar"
  }
}
```

Will be equivalent to:

```
{
  value: "bar",
  sub: {
    value: "bar"
  }
}
```

7. Files can be included using *include* keyword (must not be quoted) followed by a string (without :) representing the file to include (path is relative to current file) and terminating by a comma.

Arrays can't be included.

Merge will be done as for duplicate properties.

If *file1.cfg* is:

```
value: "foo",
include "file2.cfg",
foo: "foo"
```

And *file2.cfg* is:

```
value: "bar",
foo: "bar"
```

Final config will be:

```
{
  value: "bar",
  foo: "foo"
}
```

8. A C like preprocessor is supported. The following preprocessor commands are available:

#define var *expr*

Define a new variable with value *expr*. *expr* must be a valid JSON expression. Note that unlike the standard C preprocessor, *expr* is evaluated by the preprocessor.

#undef var

Undefine the variable *var*.

#include *expr*

Include the file whose filename is the evaluation of the string expression *expr*.

#if *expr* Consider the following text if *expr* is true.

#else Alternative of **#if** block.

#elif Composition of **#else** and **#if**.

#endif End of **#if** block.

#ifdef var

Shortcut for **#if defined(var)**

#ifndef var

Shortcut for **#if !defined(var)**

In the JSON source, every occurrence of a defined preprocessor variable is replaced by its value.

9. Backquote strings: JSON expression can be inserted in backquote delimited strings with the ``${expr}` syntax. Example: `'abc${1+2}d'` is evaluated as the string `"abc3d"`. Preprocessor variables can be used inside the expression. Backquote strings may span several lines.

5.1.1 JSON merge rules

Merge overriding direction depends on context, i.e source may override destination or the opposite.

JSON merge is recursive for Objects and Arrays.

Example, merging

```
{
  foo: { value: "bar" },
  same: "one",
  one: 1
}
```

with

```
{
  foo: { value: "none", second: true },
```

```

    same: "two",
    two: 1
}

```

Will become:

```

{
  foo: { value: "bar", second: true },
  same: "one",
  one: 1
  two: 1
}

```

assuming first object overrides second one.

In case of Array merging, the final array length will be the maximum length of all merged arrays.

For each element of the final array, merge will be done considering defined elements only.

Ex:

```

{
  array: [0, 1, 2, { foo: "bar" } ],
  array: [3, 4],
  array: [5, 6, 7, { bar: "foo" }, 8 ]
}

```

Will be merged to:

```

{
  array: [5, 6, 7, { foo: "bar", bar: "foo" }, 8 ],
}

```

5.2 Properties

log_filename

String. Set the log filename. If no leading /, it is relative to the configuration file path. See [Log file format], page 131.

log_options

String. Set the logging options as a comma separated list of assignments.

- *layer.level=verbosity*. For each layer, the log verbosity can be set to **none**, **error**, **info** or **debug**. In debug level, the content of the transmitted data is logged.
- *layer.max_size=n*. When dumping data content, at most **n** bytes are shown in hexa. For ASN.1, NAS or Diameter content, show the full content of the message if **n > 0**.
- *layer.payload=[0|1]*. Dump ASN.1, NAS, SGsAP or Diameter payload in hexadecimal.
- *layer.key=[0|1]*. Dump security keys (NAS and RRC layers).
- *layer.crypto=[0|1]*. Dump plain and ciphered data (NAS and PCDP layers).
- *layer.verbose=[0|1]*. If **layer** is **ipsec**, dump all packets filtering informations.
- *time=[sec|short|full]*. Display the time as seconds, time only or full date and time (default = time only).
- *time.us=[0|1]*. Dump time with microseconds precision.
- *file=cut*. Close current file log and open a new one.

- `file.rotate=now`. Move and rename to the same directory or to the directory pointed by `file.path` and open a new log file (Headers are kept).
- `file.rotate=size`. Every time log file size reaches *size* bytes, move and rename to the same directory or to the directory pointed by `file.path`, and open a new log file (Headers are kept).
Size is an integer and can be followed by K, M or G.
- `file.rotate=#count`. Everytime number of logs in log file reaches *count*, move and rename to the same directory or to the directory pointed by `file.path`, and open a new log file (Headers are kept).
Size is an integer and can be followed by K, M or G.
- `file.path=path`. When log rotation is enabled (`file.rotate` set), rename and move current log to this path instead of initial log path.
- `append=[0|1]`. (default=0). If 0, truncate the log file when opening it. Otherwise, append to it.

Available layers are: `nas`, `ip`, `s1ap`, `ngap`, `gtpu`, `rx`, `s6`, `cx`, `s13`, `sgsap`, `sbcap`, `lcsap`, `lppa`, `n12`, `n13`, `n8`, `n17`, `n50`, `n5`, `n11`, `nrppa`, `epdg`, `ikev2`, `ipsec`, `n20`, `n62`

log_sync Optional boolean (default = false). If true, logs will be synchronously dumped to file.

Warning, this may lead to performances decrease.

gtp_addr

String or array of strings. Set the IP address (and an optional port) on which the GTP-U packets are received. The default port is 2152. It is normally the IP address of the network interface connected to the core network.

Syntax:

- "1.2.3.4" (use default port)
- "1.2.3.4:5678" (use explicit port)
- "2001:db8:0:85a3::ac1f:8001" (IPv6 address and default port)
- "[2001:db8:0:85a3::ac1f:8001]:5678" (IPv6 address and explicit port)

gtp_ext_addr

Optional string or array of strings. Set the IP address on which the eNodeB should transmit the GTP-U packets. It is the same as `gtp_addr` by default. It can be different if LTEMME is behind a NAT. It should have the same number of entries as the `gtp_addr` parameter.

gtp_payload_mtu

Optional integer (range 68 to 16384, default = 1500). MTU in bytes for the GTP-U payload. Do not forget to update the network interface MTU accordingly for optimal performance. See [Initial testing], page 6. If you want to use another value than the default one, you likely want to change the `mtu_ipv4` and/or `ipv6_mtu` parameters also.

gtp_use_packet_bundling

Optional boolean (default = false). Concatenate multiple GTP-U PDUs within a single UDP datagram. Be careful, this is a non-standard option that must not be activated if the peer is not an Amarisoft eNodeB/gNodeB/N3IWF with this option activated.

s1ap_bind_addr

Optional string or array of objects.

If the object is a string, it contains the IP address and optional port on which the

S1AP SCTP connection is bound.

If the object is an array, it contains the following parameters:

bind_addr

String. IP address and optional port on which the S1AP SCTP connection is bound.

gtp_addr String. IP address and optional port of the GTP-U interface associated with this S1AP interface. It must correspond to an entry of the **gtp_addr** object.

slap_time_to_wait

Optional enumeration: 0, 1, 2, 5, 10, 20, 60 (default = 0). Defines the value in seconds of the S1AP time to wait information element sent in the S1AP S1 setup failure or eNB configuration update failure message. The value 0 means that the IE is not sent.

ngap_bind_addr

Optional string or array of objects.

If the object is a string, it contains the IP address and optional port on which the NGAP SCTP connection is bound.

If the object is an array, it contains the following parameters:

bind_addr

String. IP address and optional port on which the NGAP SCTP connection is bound.

gtp_addr String. IP address and optional port of the GTP-U interface associated with this NGAP interface. It must correspond to an entry of the **gtp_addr** object.

ngap_time_to_wait

Optional enumeration: 0, 1, 2, 5, 10, 20, 60 (default = 0). Defines the value in seconds of the NGAP time to wait information element sent in the NGAP NG setup failure or RAN configuration update failure message. The value 0 means that the IE is not sent.

plmn

String. PLMN identity of the MME (5 or 6 digits). It should match one of the PLMN identities broadcasted by the eNodeB or gNodeB.

cag_support

Optional boolean (default = false). Applicable to 5GC only. Indicates if the AMF supports the CAG feature. CAG feature cannot be declared as supported in a SNPN (ie when nid item is present).

nid

Applicable to 5GC only in case of Stand-Alone Non-Public Network (SNPN). The NID (as defined in 3GPP TS 23.003 12.7 Stand-Alone Non-Public Network Identifier) associated with the AMF PLMN identifies the SNPN. It contains the following objects defining the NID:

nid_value

10 digits NID value.

assignment_mode

Optional enumeration: self, coordinated_1, coordinated_2, according to 3GPP TS 23.003 clause 12.7 (default = self).

mme_group_id

Optional integer, range: 0 to 65535. Set the MME group ID.

- mme_code** Optional integer, range: 0 to 255. Set the MME code.
- mme_name** Optional string. MME name used for S1AP signalling. If absent no MME name is used.
- amf_region_id**
Optional integer, range: 0 to 255. Set the AMF region ID. If not present, the value is derived from the **mme_group_id** value. If present, it must match the value derived from the **mme_group_id** value if it is present, using the rules defined in 3GPP TS 23.003 chapter 2.10.2.2.2.
- amf_set_id**
Optional integer, range: 0 to 1023. Set the AMF Set ID. If not present, the value is derived from the **mme_group_id** and **mme_code** values. If present, it must match the value derived from the and **mme_code** values if they are present, using the rules defined in 3GPP TS 23.003 chapter 2.10.2.2.2.
- amf_pointer**
Optional integer, range: 0 to 63. Set the AMF Pointer. If not present, the value is derived from the **mme_code** value. If present, it must match the value derived from the **mme_code** value if it is present, using the rules defined in 3GPP TS 23.003 chapter 2.10.2.2.2.
- truncated_amf_set_id**
Optional integer, range: 0 to 7. Set the truncated AMF Set ID length for Control Plane CIoT 5GS optimization reestablishment procedure.
- truncated_amf_pointer**
Optional integer, range: 0 to 5. Set the truncated AMF Pointer length for Control Plane CIoT 5GS optimization reestablishment procedure.
- amf_name** Optional string. AMF name used for NGAP signalling. Default is set to amarisoft.amf.5gc.mnc<MNC>.mcc<MCC>.3gppnetwork.org.
- amf_nf_instance_id**
Optional 16 bytes hexadecimal string.
Defines the random number to use to create the AMF NF instance id.
Allows to keep the same value of AMF NF instance id even after a restart of the AMF.
If not present, the AMF instance id will be generated as described in IETF RFC 4122 paragraph 4.4.
- eps_5gs_interworking**
Optional enumeration: none, without_n26, with_n26 (default = none). Defines whether inter RAT mobility between EPS and 5GS is supported or not, and whether N26 interface is supported or not. Note that interworking with N26 is required to perform handover between EPS and 5GS.
- eps_assign_5gs_resources_without_n1_mode_capability**
Optional boolean (default = false). If set to true and if the UE does not indicate supporting N1 mode in its UE network capability IE, the EPC will still allocate 5GS QoS rules and Qos flow descriptions if the UE indicates a PDU session ID in its ESM PDN connectivity request message. Only applicable when **eps_5gs_interworking** is set to "with_n26".
- eplmn_list**
Optional array of strings (1 to 15). List of equivalent PLMNs used for NAS, S1AP and NGAP messages.

relative_capacity

Optional integer. Range: 0 to 255. Default : 50. Set the MME or AMF relative capacity value used for MME or AMF load balancing in S1AP S1 Setup Response, S1AP MME Configuration Update, NGAP NG Setup Response and NGAP AMF Configuration Update messages.

nas_cipher_algo_pref

Array of integers. Set the preferred algorithms for NAS encryption in decreasing order of preference. If none match the UE capabilities, then EEA0/5G-EA0 (no encryption) is selected.

List of supported algorithms:

Value	Algorithm
1	EEA1/5G-EA1 (Snow 3G)
2	EEA2/5G-EA2 (128 bit AES)
3	EEA3/5G-EA3 (ZUC)

If encryption is necessary, for best performance use AES (EEA2/5G-EA2) as first choice if your CPU supports the AES NI Intel instruction set (available starting from Sandy bridge CPUs). Otherwise use Snow3G (EEA1/5G-EA1) or ZUC (EEA3/5G-EA3).

nas_cipher_algo_null_allowed

Optional boolean (default = true). If set to false, the use of NULL ciphering algorithm (EEA0/5G-EA0) is forbidden unless the UE performs an emergency registration.

nas_integ_algo_pref

Array of integers. Set the preferred algorithms for NAS integrity check in decreasing order of preference. If none match the UE capabilities, then EIA0/5G-IA0 (no integrity check) is selected.

List of supported algorithms:

Value	Algorithm
1	EIA1/5G-IA1 (Snow 3G)
2	EIA2/5G-IA2 (128 bit AES)
3	EIA3/5G-IA3 (ZUC)

For best performance, use AES (EIA2/5G-IA2) as first choice if your CPU supports the AES NI Intel instruction set (available starting from Sandy bridge CPUs). Otherwise use Snow3G (EIA1/5G-IA1) or ZUC (EIA3/5G-IA3).

nas_integ_algo_null_allowed

Optional boolean (default = true). If set to false, the use of NULL integrity algorithm (EIA0/5G-IA0) is forbidden unless the UE performs an emergency registration.

tun_setup_script

String. Set the path of the shell script to set up the virtual network interface. Script is called for each PDN connectivity or PDU session with following parameters:

1. Interface name
2. PDN or PDU session index
3. Access Point Name
4. Type: 'ipv4', 'ipv6' or 'ethernet'

If type is 'ipv4', the next parameters are:

1. IP address: interface address
2. First IP address
3. Last IP address
4. Subnet mask

If type is 'ipv6', the next parameters are:

1. Link local address
2. Interface IP address
3. First IPv6 prefix
4. Last IPv6 prefix
5. Subnet mask

If type is 'ethernet', there are no extra parameters

t3402 Optional integer (default = -1). Value in seconds of the T3402 or T3502 timer. -1 means that the timer value is not transmitted in attach accept or TAU or registration accept so that the UE uses the default value (12 minutes).

t3412 Optional integer (default = 1800). Value in seconds of the T3412 (TAU update) timer. -1 means that the timer is deactivated. This is the value sent to the UE in NAS signalling, unless the UE is requesting the use of a longer timeout with T3412 extended value information element.

t3412_low_priority

Optional integer (default = t3412 value). Value in seconds of the T3412 (TAU update) timer if the UE indicates NAS signalling low priority. -1 means that the timer is deactivated. This is the value sent to the UE in NAS signalling, unless the UE is requesting the use of a longer timeout with T3412 extended value information element.

t3512 Optional integer (default = 1800). Value in seconds of the T3512 (periodic registration) timer. -1 means that the timer is deactivated. This is the value sent to the UE in NAS signalling, unless the UE is requesting the use of a longer timeout with requested T3512 value information element.

t3501 Optional integer in range 1-30 (default = 5). Value in seconds of the MANAGE UE POLICY COMMAND timer in the PCF.

n3gpp_dereg_timer

Optional integer (default = 3240). Value in seconds of the non-3GPP de-registration timer. This is the value sent to the UE in NAS signalling.

purge_timer

Optional integer (default = -1). Value in seconds of the purge timer started when the UE gets deregistered. When it expires, the UE context is deleted. -1 means that the timer is deactivated.

psm Optional boolean (default = true). If set to false, MME will ignore the PSM request sent by the UE.

mico_support

Optional boolean (default = true). If set to false, AMF will ignore the MICO request sent by the UE.

registration_area_alloc_ind

Optional integer (default = 0). Sets the Registration Area Allocation Indication bit in the 5GMM MICO indication IE. 0 means 'all PLMN registration area not allocated' and 1 means 'all PLMN registration area allocated'.

sprrt_support

Optional boolean (default = false). If set to true and if **mico_support** is set to true, the AMF will accept the use of the strictly periodic registration timer.

t3412_extended_forced

Optional integer (default = -1). Value in seconds of the T3412 extended timer if UE uses PSM. If different from -1, the MME will ignore the value requested by the UE and will send this one instead.

force_t3412_extended_ie

Optional boolean (default = false). If set to false, the MME selects the greatest T3412 value between the one configured in the MME and the one requested by the UE for PSM (unless **t3412_extended_forced** is set), and it does not send the T3412 extended IE if the value can be encoded as a GPRS timer IE. If set to true, the MME accepts a T3412 value requested by the UE smaller than the configured one, and the T3412 extended IE is always sent.

requested_t3512_forced

Optional integer (default = -1). Value in seconds of the T3512 timer if UE uses MICO. If greater than -1, the AMF will ignore the value requested by the UE and will send this one instead. If set to -2, the AMF will accept a T3512 value requested by the UE smaller than the configured one.

t3324_forced

Optional integer (default = -1). Value in seconds of the T3324 timer if UE uses PSM or MICO. If different from -1, the MME or AMF will ignore the value requested by the UE and will send this one instead. -2 means that the timer is deactivated.

t3346 Optional integer (default = -1). Value in seconds of the T3346 timer. The timer is transmitted in the reject messages if the EMM or 5GMM cause is #22 (congestion) and the value is not -1.

t3442 Optional integer (default = 0). Value in seconds of the T3442 timer.

t3448 Optional integer (default = -1). Value in seconds of the T3448 timer. The timer is transmitted if the value is different from -1 and the UE indicates its support in the UE network capability information element.

t3460 Optional integer (default = 6). Value in seconds of the T3460 or T3560 timer.

t3460_wb_ce

Optional integer (default = 24). Value in seconds of the T3460 or T3560 timer for UE operating in WB-S1/CE or WB-N1/CE mode.

lower_bound_timer

Optional integer (default = -1). The timer is transmitted in the reject messages if the EMM or 5GMM cause is #78 (PLMN not allowed to operate at the present UE location) and the value is not -1.

t3560_ng_ran_sat

Optional integer (default = 11). Value in seconds of the T3560 timer for UE operating in NR(MEO) or NR(GEO) satellite RAT.

- 5gmm_backoff_timer**
Optional integer (default = -1). Value in seconds of the 5GMM DL NAS transport back-off timer. The timer is transmitted if the value is not -1. -2 means that the timer is deactivated.
- edrx**
Optional boolean (default = true). If set to false, the core network will ignore the eDRX request sent by the UE.
- edrx_ptw_wb_s1**
Optional integer (0 to 15, default = 3). 4 bits Paging Time Window length for WB-S1 and WB-N1 UEs as defined in 3GPP TS 24.008 chapter 10.5.5.32.
- edrx_ptw_nb_s1**
Optional integer (0 to 15, default = 3). 4 bits Paging Time Window length for NB-S1 and NB-N1 UEs as defined in 3GPP TS 24.008 chapter 10.5.5.32.
- edrx_ptw_nr**
Optional integer (0 to 31, default = 3). 8 bits Paging Time Window length for NR connected to 5GCN UEs as defined in 3GPP TS 24.008 chapter 10.5.5.32.
- edrx_cycle_forced**
Optional integer (-1 to 15, default = -1). 4 bits E-UTRAN or NR eDRX cycle length duration as defined in 3GPP TS 24.008 chapter 10.5.5.32. If different from -1, the core network will ignore the value requested by the UE and will send this one instead.
- gwus_support**
Optional boolean (default = true). Group WUS support.
- gwus_prob_forced**
Optional enumeration: -1, 0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100 (default = -1: paging probability not forced). Force group WUS paging probability for all the UEs. The MME will ignore the value requested by the UE and will send this one instead.
- ims_list**
Optional array. Each entry is an object defining connection to Amarisoft IMS server. This is useful for SMS over SG or 3GPP mode of IMS server when Rx interface is not used.
Each entry has following members:
ims_addr IP address of Amarisoft IMS server.
bind_addr IP address of network interface to use for IMS connection.
- ims_vops_eps**
Optional boolean (default = false). Set the IMS voice over PS session in S1 mode supported bit of the EPS network feature support field in the NAS attach accept message (VoLTE).
- ims_vops_5gs_3gpp**
Optional boolean (default = false). Set the IMS voice over PS session over 3GPP access indicator of the 5GS network feature support IE of the NAS registration access message. See 3GPP TS 24.501 table 9.11.3.5.1.
- ims_vops_5gs_n3gpp**
Optional boolean (default = false). Set the IMS voice over PS session over non-3GPP access indicator of the 5GS network feature support IE of the NAS registration access message. See 3GPP TS 24.501 table 9.11.3.5.1.

emc_bs	Optional boolean (default = false). Set the emergency bearer services in S1 mode supported bit of the EPS network feature support field in the NAS attach accept message (VoLTE, Release 9).								
emc	Optional integer (default = 0). Set the emergency service support indicator for 3GPP access bits of the 5GS network feature support IE in the NAS registration accept message. See 3GPP TS 24.501 table 9.11.3.5.1.								
emc_n3gpp	Optional boolean (default = false). Set the emergency service support indicator for non-3GPP access bits of the 5GS network feature support IE in the NAS registration accept message. See 3GPP TS 24.501 table 9.11.3.5.1.								
emf	Optional integer (default = 0). Set the emergency service fallback indicator for 3GPP access bits of the 5GS network feature support IE in the NAS registration accept message. See 3GPP TS 24.501 table 9.11.3.5.1.								
epc_lcs	Optional boolean (default = false). Set the Location services indicator via EPC supported bit of the EPS network feature support field in the NAS attach accept message.								
5gs_sms_over_nas	Optional boolean (default = true). Defines if 5GC should indicate the support of SMS over NAS in the 5GMM registration accept message, if the UE indicated its support in the 5GMM registration request message.								
emergency_number_list	Optional array of objects. Defines a list of emergency numbers to be sent to the UE in the NAS Attach Accept, Tracking Area Update Accept or Registration Accept messages. Each object must contain the following parameters: <table> <tr> <td>category</td><td>Integer. Bitmask of the category bits as defined in 3GPP TS 24.008 table 10.5.135d (bit 1: police, bit 2: ambulance, bit 3: fire brigade, bit 4: marine guard, bit 5: mountain rescue, bit 6: manually initiated eCall, bit 7: automatically initiated eCall).</td></tr> <tr> <td>digits</td><td>String. Emergency number.</td></tr> </table>	category	Integer. Bitmask of the category bits as defined in 3GPP TS 24.008 table 10.5.135d (bit 1: police, bit 2: ambulance, bit 3: fire brigade, bit 4: marine guard, bit 5: mountain rescue, bit 6: manually initiated eCall, bit 7: automatically initiated eCall).	digits	String. Emergency number.				
category	Integer. Bitmask of the category bits as defined in 3GPP TS 24.008 table 10.5.135d (bit 1: police, bit 2: ambulance, bit 3: fire brigade, bit 4: marine guard, bit 5: mountain rescue, bit 6: manually initiated eCall, bit 7: automatically initiated eCall).								
digits	String. Emergency number.								
extended_emergency_number_list	Optional object. Defines a list of extended emergency numbers to be sent to the UE in the NAS Attach Accept, Tracking Area Update Accept or Registration Accept messages. The object must contain the following parameters: <table> <tr> <td>validity</td><td>Enumeration (country or plmn). Validity of the extended emergency number list.</td></tr> <tr> <td>emergency_numbers</td><td>Array of objects. Each object must contain the following parameters: <table> <tr> <td>digits</td><td>String. Emergency number.</td></tr> <tr> <td>sub_services</td><td>Optional string. Emergency number sub-services.</td></tr> </table> </td></tr> </table>	validity	Enumeration (country or plmn). Validity of the extended emergency number list.	emergency_numbers	Array of objects. Each object must contain the following parameters: <table> <tr> <td>digits</td><td>String. Emergency number.</td></tr> <tr> <td>sub_services</td><td>Optional string. Emergency number sub-services.</td></tr> </table>	digits	String. Emergency number.	sub_services	Optional string. Emergency number sub-services.
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digits	String. Emergency number.								
sub_services	Optional string. Emergency number sub-services.								
cp_ciot_opt	Optional boolean (default = false). If true, enable control plane CIoT optimization (if supported by the UE).								

- attach_without_pdn**
Optional boolean (default = false). If true, enable attach without PDN functionality (if supported by the UE).
- fifteen_bearers**
Optional boolean (default = false). If true, enable the use of 15 EPS radio bearers (if supported by the UE).
- apn_oi**
Optional string (default = mncABC.mccXYZ.gprs where ABC is the PLMN MNC and XYZ the PLMN MCC). Defines the APN/DNN Operator Identifier. An empty string removes the APN OI from the APN.
- network_name**
Optional string (default = empty). Set the network name in the EMM information or configuration update command message.
- network_short_name**
Optional string (default = empty). Set the network short name in the EMM information or configuration update command message.
- emm_information_time_enable**
Optional boolean (default = true). Include the time and time zone in the EMM information or 5GMM configuration update command message.
- emm_information_enable**
Optional boolean. Default = true if **network_name** or **network_short_name** are not empty. If true, send the EMM information message after the NAS attach complete message or the 5GMM configuration update command message after the 5GS registration accept message.
- ext_emm_cause**
Optional integer (range -1 to 15). If EMM cause is 15 in attach reject or tracking area update reject message and the value is different from -1, extended EMM cause IE is sent. The values 0 to 15 correspond to the bits specified in 3GPP TS 24.301 chapter 9.9.3.26A.
- attach_reject_error**
Optional integer (range 0 to 255, default depending on scenario).
Force value of EMM reject cause in NAS attach reject message.
- tracking_area_update_reject_error**
Optional integer (range 0 to 255, default depending on scenario).
Force value of EMM reject cause in NAS tracking area update reject message.
- service_reject_error**
Optional integer (range 0 to 255, default depending on scenario).
Force value of EMM reject cause in NAS service reject message.
- pdn_connect_reject_error**
Optional integer (range 0 to 255, default depending on scenario).
Force value of ESM reject cause in NAS PDN connectivity reject message.
- pdn_disconnect_reject_error**
Optional integer (range 0 to 255, default depending on scenario).
Force value of ESM reject cause in NAS PDN disconnect reject message.

bearer_resource_allocation_reject_error

Optional integer (range 0 to 255, default depending on scenario).

Force value of ESM reject cause in NAS bearer resource allocation reject message.

bearer_resource_modification_reject_error

Optional integer (range 0 to 255, default depending on scenario).

Force value of ESM reject cause in NAS bearer resource modification reject message.

registration_initial_reject_error

Optional integer (range 0 to 255, default depending on scenario).

Force value of 5GMM reject cause in NAS registration reject message (for 5GS registration type 1 or 4).

registration_mobility_periodic_reject_error

Optional integer (range 0 to 255, default depending on scenario).

Force value of 5GMM reject cause in NAS registration reject message (for 5GS registration type 2 or 3).

5gs_service_reject_error

Optional integer (range 0 to 255, default depending on scenario).

Force value of 5GMM reject cause in NAS service reject message.

pdu_session_establishment_reject_error

Optional integer (range 0 to 255, default depending on scenario).

Force value of 5GSM reject cause in NAS PDU session establishment reject message.

pdu_session_release_reject_error

Optional integer (range 0 to 255, default depending on scenario).

Force value of 5GSM reject cause in NAS PDU session release reject message.

pdu_session_modification_reject_error

Optional integer (range 0 to 255, default depending on scenario).

Force value of 5GSM reject cause in NAS PDU session modification reject message.

5gmm_dl_nas_transport_error

Optional integer (range 0 to 255, default depending on scenario).

Force value of 5GMM reject cause in NAS DL NAS transport message.

eps_user_unknown_reject_cause

Optional integer (range 0 to 255, default = 8).

EMM cause sent in the NAS attach reject message when the IMSI is unknown in the HSS.

5gs_user_unknown_reject_cause

Optional integer (range 0 to 255, default = 3 or 75 if SNPN).

5GMM cause sent in the NAS registration reject message when the SUPI is unknown in the UDM.

attach_result_mode

Optional string (default = auto). Set attach result of attach accept message.

Can be:

auto This is standard LTE behavior.

eps_only If set and UE is sending combined EPS/IMSI attach, the MME will answer with EPS only in attach accept message (EMM cause will be CS domain not available).

combined If set and UE is sending EPS only attach, the MME will answer with combined in attach accept message.

combined_sms_only

If set the UE will only be attached for EPS and SMS.

additional_update_result

Optional integer (range -1 to 3, default = 2). Set the value of additional update result in NAS attach accept and tracking area update accept messages.

If set to -1, the additional update result won't be set.

sms_services_status

Optional integer (range -1 to 7, default = 0). Set the value of the SMS service status information element in the NAS attach accept and tracking area update accept messages. It is sent if the value is different from -1 and the UE performed an EPS service with SMS only procedure and does not have SMS service allowed.

network_policy

Optional integer (range -1 to 15, default = -1). Set the value of the network policy information element described in 3GPP TS 24.301 chapter 9.9.3.52. The value -1 means that the IE is not transmitted.

imeisv_request_in_smc

Optional boolean (default = true). Ask for the UE IMEISV in the NAS security mode command message. Must be enabled if **multi_sim** is set to **true**. IMEISV will always be requested if a S13 or N17 connection is defined, or if **me_db** object is defined.

coarse_location_information_support

Optional boolean (default = false). If set to true, and if a NB-IoT UE accessing the core network via a NTN cell supports the feature, the MME will request the coarse location information in the NAS security mode command message.

authentication_mode

Optional string (default = auto). Set NAS authentication procedure behavior.

Can be:

auto The MME or AMF performs authentication procedure unless the UE is already successfully authenticated.

force The MME or AMF forces a new NAS authentication procedure even if the Attach Request or Registration Request was already successfully authenticated

skip The MME or AMF skips the NAS authentication procedure and uses EIA0/EEA0 or 5G-IA0/5G-EA0 algorithms. This needs to be supported on UE side also.

dummy_authentication_autn_mac

Optional boolean (default = false). If set to true, the network will send an invalid AUTN MAC value in the NAS authentication request message.

authenticate_known_emergency_supi

Optional boolean (default = false). If set to true, the network will authenticate known IMSI/SUPI during an emergency registration procedure and reject the UE if it fails.

secure_known_emergency_supi

Optional boolean (default = false). If set to true and if **authenticate_known_emergency_supi** is set to true, known IMSI/SUPI during an emergency registration procedure will use the integrity and ciphering algorithm matching the network and UE configuration. Otherwise it will use a null integrity and ciphering algorithm.

restrict_ec_wb

Optional boolean (default = false). Sets restriction on enhanced coverage for WB-S1 and WB-N1 UEs.

restrict_ec_nb

Optional boolean (default = false). Sets restriction on enhanced coverage for NB-S1 and NB-N1 UEs.

skip_smc_proc

Optional boolean (default = false). If set to true, the MME or AMF will not perform a NAS security mode control procedure and will send all messages as plain. This needs to be supported on UE side also.

force_identity_request

Optional boolean (default = false). If set to true, the network will perform a NAS identity request procedure even if the GUTI in the attach request or the 5G-GUTI in the initial registration request is already known.

allocate_new_guti_in_tau

Optional boolean (default = true). If set to true, a new GUTI is allocated during each Tracking Area Update procedure. If set to false, the UE keeps the GUTI allocated during the attach procedure.

force_guti_in_tau

Optional boolean (default = false). If set to true, GUTI IE will be systematically present in Tracking Area Update Accept message even if it does not change.

attach_reject_filter

Optional nullable object. Represent UE to reject when trying to attach to EPS.

Each property name represent IMSI. Use of wildcard "*" with an IMSI prefix is allowed to match IMSI range (Ex: 0010112456*). If multiple filters are matching, the one with the longest prefix will be used.

Each property value is an integer defining the redirection type as described in *rrc_redirect* eNB configuration.

The configuration may be removed using:

`attach_reject_filter: null`

Example:

```
attach_reject_filter: {
  "*": 0,
  "0010112345678": 1
  "00101123456*": 2
}
```

Will reject UE with IMSI *0010112345678* using redirection configuration 1 and all other UEs using redirection configuration 0.

emm_procedure_filter

Optional object. Allows to define the MME behavior for a list of EMM procedures. Each property name represents an EMM procedure. The ones currently supported are **attach**, **attach_with_security_protection**, **tracking_area Updating**, **detach**, **service_request**, **identity**, **authentication**, **security_mode_control** and **nas_transport**.

Each property value is an object containing the following fields:

action Enumeration (**treat** (UE message is processed), **ignore** (UE message is ignored) or **reject** (UE message is rejected)).

t1 Optional integer. If set, the **reject** of **ignore** filter is applied **t1** times. If not set, the filter is applied until it is modified.

send_status_on_reject

Optional boolean. It set and if **action** is set to reject an EMM status message is sent.

By default all procedures are treated.

Example:

```
emm_procedure_filter: {
  attach: {
    action: "treat"
  },
  service_request: {
    action: "reject",
    t1: 1
  }
}
```

5gmm_procedure_filter

Optional object. Allows to define the AMF behavior for a list of 5GMM procedures.

Each property name represents a 5GMM procedure. The ones currently supported are **registration_initial**, **registration_initial_with_security_protection**, **registration_mobility_periodic**, **service_request**, **identity**, **authentication**, **security_mode_control**, **generic_ue_update_command**, **nas_transport_n1_sm**, **nas_transport_sms**, **deregistration**, **control_plane_service_request**, **network_slice_specific_authentication** and **nas_transport_lpp**.

Each property value is an object containing the following fields:

action Enumeration (**treat** (UE message is processed), **ignore** (UE message is ignored) or **reject** (UE message is rejected)).

t1 Optional integer. If set, the **reject** of **ignore** filter is applied **t1** times. If not set, the filter is applied until it is modified.

send_status_on_reject

Optional boolean. It set and if **action** is set to reject a 5GMM status message is sent.

By default all procedures are treated.

Note that **nas_transport_n1_sm** filter must be used together with the **apply_nas_transport_n1_sm_filter** DNN parameter.

Example:

```
"5gmm_procedure_filter": {
```

```

        registration_initial: {
            action: "treat"
        },
        service_request: {
            action: "reject",
            ttl: 1
        }
    }
}

```

qci_dscp_mapping

Optional array of objects. Allows to define a specific IP differentiated services code point for a given QCI/5QI. QCI/5QI not explicitly configured use the default DSCP value 0.

Each object must contain the following properties:

qci	Integer (range 1 to 254). QCI or 5QI value.
dscp	Integer (range 0 to 63). DSCP value.

rate_bucket_duration

Optional. Range 500 to 5000 (default = 2000). Duration in ms for the average bit rate estimation. It is used to enforce the APN and Session Aggregate Maximum Bit Rate.

dcnr_support

Optional boolean (default = false). Set it to true to enable Dual Connectivity with NR support.

dcnr_implicit_support

Optional boolean (default = false). If set to true, the MME will not send the 2nd byte of the EPS network feature support IE because of DCNR. Can be useful to test the UE behavior.

cpu_core_list

Optional array of integers. Defines the list of CPU cores indexes on which LTEMME will run.

If not set, LTEMME may use all cores.

Note that the number of cores depends on Linux scheduler and LTEMME configuration.

cn_assistance_info_support

Optional boolean (default = false), applicable to 5GC only. If set to true, the AMF will send a Core Network Assistance Information in the Initial Context Setup message.

This is mandatory to have a functional RRC Inactive support in the RAN.

ecc_params

Optional object. Set the ECC network configuration for the SUPI protection and de-concealment of the SUCI. Applicable to 5GC only. It contains the following objects:

A	Optional array of objects. Set the home network private key for profile A protection scheme.
----------	--

home_nw_private_key

String. Set the home network private key in hexadecimal format (32 bytes).
--

home_nw_key_id

Optional integer in range 0 to 255 (default = 1). Set the home network key identifier.

B Optional array of objects. Set the home network private key for profile B protection scheme.

home_nw_private_key

String. Set the home network private key in hexadecimal format (32 bytes).

home_nw_key_id

Optional integer in range 0 to 255 (default = 2). Set the home network key identifier.

Here is the procedure to generate a private/public key-pair:

Profile A:

```
openssl genpkey -algorithm x25519 -out key.pem
openssl pkey -in key.pem -text
```

Profile B:

```
openssl ecparam -genkey -name secp256r1 -out key.pem
openssl ec -in key.pem -noout -text -conv_form compressed
```

nf_ssl_certificate

Optional string. Applicable to 5GC only. If set, forces SSL for NF interfaces. Defines CA certificate filename.

nf_ssl_key

Optional string. Applicable to 5GC only. Mandatory if nf_ssl_certificate is set. Defines CA private key filename.

Here is the procedure to generate the private key file key.pem and the certificate file cert.pem:

```
openssl req -new > cert.csr
openssl rsa -in privkey.pem -out key.pem
openssl x509 -in cert.csr -out cert.pem -req -signkey key.pem --days 365
```

nssai Applicable to 5GC only. Optional array. List of S-NSSAIs served by the AMF. Default content is sst: 1 (eMBB).

Each entry will set a S-NSSAI value as defined below:

sst Integer (range 0-255). Slice Service Type.

sd Optional integer (range 0-0xFFFFFE). Slice Differentiator.

default_nssai

Applicable to 5GC only. Optional array. List of default S-NSSAIs served by the AMF.

Can only take S-NSSAIs contained in the non-default list above. If not present, takes the same content as the non-default list. See [nssai], page 25.

nssai_inclusion_mode

Applicable to 5GC only. Optional enumeration (none, A, B, C, D, default = none). NSSAI inclusion mode value to send in message Registration accept.

nssai_subject_to_nsac

Applicable to 5GC only.

Optional array. List of the S-NSSAIs subject to NSAC. Can only take S-NSSAIs

contained in the list of S-NSSAIs served by the AMF. Each entry of the array contains the following objects:

snssai	S-NSSAI value.						
sst	Integer (range 0-255). Slice Service Type.						
sd	Optional integer (range 0-0xFFFFFE). Slice Differentiator.						
access_type	Optional array of enumeration: 3gpp, non-3gpp (default value 3gpp).						
eps_counting	Optional boolean (default = false). If set to false, indicates that the network slice is subject to NSAC only in 5GS. If set to true, indicates that the network slice is subject to NSAC in EPS and 5GS.						
ue_max_count	Optional integer (range: 1 to MAX_INT). If present, the control of the number of UEs registered to the S-NSSAI is activated.						
pdu_session_max_count	Optional integer (range: 1 to MAX_INT). If present, the control of the number of PDU sessions registered to the S-SNNAI is activated.						
backoff_timer	Optional integer (default = -1). Value in seconds of the S-NSSAI backoff timer. If present and not set to -1, this value is transmitted in the IE Extended rejected NSSAI.						
nsag	Applicable to 5GC only. Optional array of object needed to configure the NSAGs in the AMF PLMN and equivalent PLMNs. Each element of the array contains: <table> <tr> <td>id</td><td>Integer in range 0..255. NSAG identifier.</td></tr> <tr> <td>priority</td><td>Optional integer in range 0..255. NSAG priority.</td></tr> <tr> <td>tai_list</td><td>Optional array of objects (up to 65535). See [tai_list_5gs], page 49. If tai_list is not present, the NSAG is valid in the AMF PLMN and equivalent PLMNs.</td></tr> </table>	id	Integer in range 0..255. NSAG identifier.	priority	Optional integer in range 0..255. NSAG priority.	tai_list	Optional array of objects (up to 65535). See [tai_list_5gs], page 49. If tai_list is not present, the NSAG is valid in the AMF PLMN and equivalent PLMNs.
id	Integer in range 0..255. NSAG identifier.						
priority	Optional integer in range 0..255. NSAG priority.						
tai_list	Optional array of objects (up to 65535). See [tai_list_5gs], page 49. If tai_list is not present, the NSAG is valid in the AMF PLMN and equivalent PLMNs.						
eap_tls	Optional object applicable to 5GC only. Shall be present if EAP-TLS method is used in the UE database. It contains the following objects: <table> <tr> <td>certificate</td><td>Define the server certificate filename.</td></tr> <tr> <td>private_key</td><td>Define the server private key filename.</td></tr> <tr> <td>ca_certificate</td><td>Define the CA certificate filename. It contains a list of root certificates to authenticate the user.</td></tr> </table>	certificate	Define the server certificate filename.	private_key	Define the server private key filename.	ca_certificate	Define the CA certificate filename. It contains a list of root certificates to authenticate the user.
certificate	Define the server certificate filename.						
private_key	Define the server private key filename.						
ca_certificate	Define the CA certificate filename. It contains a list of root certificates to authenticate the user.						
tai_lists	Optional array of objects (up to 65535). Applicable to EPC only. Defines an array of TAI lists. When the UE initiates registration in a tracking area, the TAI list containing this tracking area,						

if available, is provided to the UE in the NAS Attach Accept and Tracking Area Update Accept messages.

Each object must contain the following parameters:

plmn	String (5 or 6 digits). Must match the PLMN identity of the MME See [plmn], page 12, or an equivalent PLMN See [eplmn-list], page 13.
tacs	Array of up to 16 integers in range 1 to 0xFFFF, except 0xFFFE. Each element defines a 2 bytes long tracking area code.

tai_lists_5gs

Optional array of objects (up to 65535). Applicable to 5GC only. Defines an array of TAI lists.

When the UE initiates registration in a tracking area, the TAI list containing this tracking area, if available, is provided to the UE in the Registration Accept message. Each object must contain the following parameters:

plmn	String (5 or 6 digits). Must match the PLMN identity of the MME See [plmn], page 12, or an equivalent PLMN See [eplmn-list], page 13.
tacs	Array of up to 16 integers in range 1 to 0xFFFFF, except 0xFFFFFE. Each element defines a 3 bytes long tracking area code.

forbidden_eps_tacs

Array of up to 16 integers in range 1 to 0xFFFF, except 0xFFFE. Applicable to EPC only. Forbidden tracking areas codes in the MME PLMN (See [plmn], page 12) or an equivalent PLMN (See [eplmn-list], page 13).

areas_list_5gs

Optional array of objects. (up to 65535). Applicable to 5GC only.

Gives the list of tracking area codes associated to each area code.

This array is used in the interface AMF-UDM (See [allowed_5gs_tais], page 46, See [forbidden_5gs_tais], page 47). Each object must contain the following parameters:

code	Integer in range 0 to 0xFFFF defining the area code (operator specific).
tacs	Array of integers in range 1 to 0FFFFFF, except 0FFFFFFE. List of the tracking areas codes in the area.

operator_defined_access_categories

Optional array of objects. Operator-defined access categories as defined in 3GPP TS 24.501 paragraph 9.11.3.38. Applicable to 5GC only. Each object contains the following parameters:

precedence	Integer in range 0 to 255. Precedence value.
access_category	Integer in range 32 to 63. Operator defined access category value.
nssai	Optional array of S-NSSAIs. See [nssai], page 25.
dnn_list	Optional array of strings.
standard_access_category	Optional integer (possible values 0, 1, 2, 3, 4, 5, 6, 7, 10). Access category number of the related standardized access category.

akma_kaf_lifetime

Optional integer in range: 1 to 3650 (default = 365). Applicable to 5GC only. AKMA Application Keys lifetime in days as defined in 3GPP 33.535 paragraph 5.2 AKMA key lifetimes.

<code>com_addr</code>	Optional string. Address of the WebSocket server remote API. See [Remote API], page 67. If set, the WebSocket server for remote API will be enabled and bound to this address. Default port is 9000. Setting IP address to <code>::</code> will make remote API reachable through all network interfaces.
<code>com_name</code>	Optional string. Sets server name. MME by default
<code>com_ssl_certificate</code>	Optional string. If set, forces SSL for WebSockets. Defines CA certificate filename.
<code>com_ssl_key</code>	Optional string. Mandatory if <code>com_ssl_certificate</code> is set. Defines CA private key filename.
<code>com_ssl_peer_verify</code>	Optional boolean (default is false). If <code>true</code> , server will check client certificate.
<code>com_ssl_ca</code>	Optional string. Set CA certificate. In case of peer verification with self signed certificate, you should use the client certificate.
<code>com_log_lock</code>	Optional boolean (default is false). If <code>true</code> , logs configuration can't be changed via <code>config_set</code> remote API.
<code>com_log_us</code>	Optional boolean (default is false). If <code>true</code> , logs sent by <code>log_get</code> remote API response will have a <code>timestamp_us</code> parameters instead of <code>timestamp</code>
<code>com_auth</code>	Optional object. If set, remote API access will require authentication. Authentication mechanism is describe in [Remote API Startup], page 69, section.
<code>passfile</code>	Optional string. Defines filename where password is stored (plaintext). If not set, <code>password</code> must be set
<code>password</code>	Optional string. Defines password. If not set, <code>passfile</code> must be set.
<code>unsecure</code>	Optional boolean (default false). If set, allow password to be sent plaintext. NB: you should set it to true if you access it from a Web Browser (Ex: Amarisoft GUI) without SSL (https) as your Web Browser may prevent secure access to work.
<code>com_log_count</code>	Optional number (Default = 8192). Defines number of logs to keep in memory before dropping them. Must be between 4096 and 2097152).
<code>sim_events</code>	Array of objects. Each element defines a remote API request ([Remote API], page 67) except that <code>message</code> field is replaced by <code>event</code> .
<code>sim_events_loop_count</code>	If set, will define <code>loop_count</code> for each event of <code>sim_events</code> , See [loop_count], page 68.

sim_events_loop_delay

If set, will define `loop_delay` for each event of `sim_events`, See [loop-delay], page 68.

license_server

Configuration of the Amarisoft license server to use.

Object with following properties:

server_addr

String. IP address of the license server.

name

Optional string. Text to be displayed inside server monitor or remote API.

tag

Optional string. If set, server will only allow license with same tag.

Example:

```
license_server: {
    server_addr: "192.168.0.20",
    name: "My license",
}
```

5.2.1 PDN options

Note that the options are also applicable to 5GS DNN.

ignore_initial_apn

Optional boolean (default = false). If false, UE will be rejected if its desired APN is unknown. Any unknown APN requested outside of the initial PDN connectivity request sent during the EPS NAS attach request procedure will be rejected. This parameter does not apply to 5GS.

explicit_apn_required

Optional boolean (default = false). If true, the UE must explicitly request an APN/DNN otherwise the PDN/PDU session establishment request will be rejected.

allow_apn_in_attach_req

Optional boolean (default = false). If true, the EPC accepts an attach request containing an APN even if it is strictly forbidden in 3GPP requirement. This is required for some specific operator requirement.

pdn_list

Array of objects. Configure the available EPS Packet Data Networks and 5GS Data Network Names. The first entry of each type (IP, unstructured, ethernet) is the default APN or DNN used in case the UE does not explicitly request it.

Each object contains the following properties:

access_point_name

String. Set the Access Point Name or Data Network Name network identifier. Use dots (.) to separate the APN or DNN elements.

Array of string. You can use array to define aliases.

pdn_type

Optional enumeration: `ipv4`, `ipv6`, `ipv4v6`, `unstructured`, `ethernet` (default = `ipv4`). Select the PDN or PDU session type.

first_ip_addr

Optional string. First available IPv4 address. Required if `pdn_type` is set to `ipv4` or `ipv4v6`.

last_ip_addr

Optional string. Last available IPv4 address. Required if **pdn_type** is set to **ipv4** or **ipv4v6**.

ipv4_auto_increment

Optional boolean (default = false). If set to false, the same IPv4 address is allocated for successive activation / deactivation of the PDN or PDU session. If set to true, the IPv4 address is incremented for successive activation / deactivation of the PDN or PDU session.

gateway Optional string. If set, forces the address of the gateway used for this PDN or PDU session and sent to mme-ifup script. With default config, it will be used to provide a IP address to the tun interface. If not set, the first IP of the subnet will be used.

ip_addr_shift

Optional integer (default = 0). The allocated IPv4 addresses are allocated starting from **first_ip_addr** with a difference of $2^{\text{ip_addr_shift}}$. Hence **last_ip_addr - first_ip_addr** must be a multiple of $2^{\text{ip_addr_shift}}$. This option can be useful in case of inter-UE communication to ensure that the IPv4 address of a given UE is the only one in its netmask.

ip_addr_config

Optional string. If set, this parameter defines the Access Point Name of a PDN or PDU session that will be used for IPv4 allocation. In such case, both PDNs or PDU sessions will share the same IPv4 range and thus, **first_ip_addr**, **last_ip_addr**, **ipv4_auto_increment**, **gateway**, **mtu_ipv4** and **ip_addr_shift** will be skipped.

first_ipv6_prefix

Optional string. First available global IPv6 prefix used in Router Advertisement message sent to the UE (only the high 8 bytes of the IPv6 address are meaningful). Note that the selected prefix will also be used as the interface identifier sent in NAS signalling. Required if **pdn_type** is set to **ipv6** or **ipv4v6**.

last_ipv6_prefix

Optional string. Last available global IPv6 prefix used in Router Advertisement message sent to the UE (only the high 8 bytes of the IPv6 address are meaningful). Note that the selected prefix will also be used as the interface identifier sent in NAS signalling. Required if **pdn_type** is set to **ipv6** or **ipv4v6**.

ipv6_auto_increment

Optional boolean (default = false). If set to false, the same IPv6 prefix is allocated for successive activation / deactivation of the PDN or PDU session. If set to true, the IPv6 prefix is incremented for successive activation / deactivation of the PDN or PDU session.

ipv6_interface_identifier

Optional string. IPv6 link local address interface identifier for the MME network interface of this PDN or PDU session (only the low 8 bytes of the IPv6 address are meaningful).

- ipv6_interface_addr**
Optional string. IPv6 global address for the MME network interface of this PDN or PDU session. If not present, the address is **first_ipv6_prefix** with a **::0** interface identifier.
- ipv6_prefix_config**
Optional string. If set, this parameter defines the Access Point Name of a PDN or PDU session that will be used for IPv6 prefixes allocation. In such case, both PDNs or PDU sessions will share the same IPv6 prefix range and thus, **first_ipv6_prefix**, **last_ipv6_prefix**, **ipv6_auto_increment**, **ipv6_interface_identifier**, **ipv6_prefix_delegation_count** and **ipv6_mtu** will be skipped.
- ipv6_router_lifetime**
Optional integer (range 0 to 65535, default is 65535). IPv6 Router Advertisement router lifetime in seconds.
- ipv6_valid_lifetime**
Optional integer (default is infinity - 0xffffffff). IPv6 Router Advertisement valid lifetime in seconds.
- ipv6_pref_lifetime**
Optional integer (default is **ipv6_valid_lifetime** value). IPv6 Router Advertisement preferred lifetime in seconds.
Must not be greater than **ipv6_valid_lifetime**.
- ipv6_onlink_flag**
Optional boolean (default is true). Defines IPv6 Router Advertisement on-link flag state.
- ipv6_managed_addr_config_flag**
Optional boolean (default is false). Defines IPv6 Router Advertisement managed address configuration flag state.
- ipv6_other_config_flag**
Optional boolean (default is false). Defines IPv6 Router Advertisement other configuration flag state.
- ipv6_mtu** Optional integer (default is 0). Defines the MTU sent in the IPv6 Router Advertisement message. If set to 0, the MTU option is not sent.
- ipv6_ra_transmission_interval**
Optional integer (range -1 to 1800, default is 0). Time in seconds between 2 periodical multicast Router Advertisement transmission, once the initial 3 transmissions have been performed after opening the PDN or PDU session. The value -1 means that no multicast transmission is done at all (including the 3 initial ones). The value 0 means that periodical transmission is deactivated.
- ipv6_drop_rs**
Optional boolean (default is false). Defines whether the incoming Router Solicitation messages should be dropped by the MME and UPF or not.
- ipv6_send_dns_in_ra**
Optional boolean (default is false). Defines whether Router Advertisement message should contain the configured IPv6 DNS servers address or not.

ipv6_prefix_delegation_count	Optional integer (2, 4, 8, 16, 32). Defines the number of prefixes delegated by DHCPv6-PD (including the one allocated by the Router Advertisement message). Only the first IA_PD option in the DHCPv6 Solicit message is considered.								
dhcpv6_t1	Optional integer (default = 0xffffffff). DHCPv6 T1 option in seconds (Cf rfc3633) used for prefix delegation replies.								
dhcpv6_t2	Optional integer (default = 0xffffffff). DHCPv6 T2 option in seconds (Cf rfc3633) used for prefix delegation replies.								
dns_addr	Optional string or array of strings. IPv4 or IPv6 addresses of the DNS servers.								
p_cscf_addr	Optional string or array of strings. IPv4 or IPv6 addresses of the P-CSCF servers (VoLTE).								
mtu_ipv4	Optional integer. Set MTU size (0 means disabled).								
mtu_non_ip	Optional integer. Set MTU size for non-IP PDN (0 means disabled, the minimum valid value is 128).								
mtu_unstructured_link	Optional integer (default value set to mtu_non_ip value). Set MTU size for unstructured PDU session (0 means disabled).								
mtu_ethernet_frame_payload	Optional integer. Set MTU size for ethernet PDN or PDU session (0 means disabled).								
operator	Optional array of objects. Each element defines an operator reserved container in protocol configuration. Properties of each element: <table> <tr> <td>id</td><td>Integer. Container identifier, must be between 0xff00 and 0xffff as defined in 3GPP TS 24.008.</td></tr> <tr> <td>plmn</td><td>String. PLMN info of container.</td></tr> <tr> <td>value</td><td>String. Value to send in hexadecimal string format.</td></tr> <tr> <td>force</td><td>Optional boolean. If true, container will be sent event without request (false by default).</td></tr> </table>	id	Integer. Container identifier, must be between 0xff00 and 0xffff as defined in 3GPP TS 24.008.	plmn	String. PLMN info of container.	value	String. Value to send in hexadecimal string format.	force	Optional boolean. If true, container will be sent event without request (false by default).
id	Integer. Container identifier, must be between 0xff00 and 0xffff as defined in 3GPP TS 24.008.								
plmn	String. PLMN info of container.								
value	String. Value to send in hexadecimal string format.								
force	Optional boolean. If true, container will be sent event without request (false by default).								
5gs_authentication	Optional enumeration: none , pap , chap , eap-md5 , eap-tls or eap-aka (default set to authentication). Defines the authentication mechanism used for this DNN in 5GS.								
authentication	Optional enumeration: none , pap or chap (default set to none). Defines the authentication mechanism used for this APN in EPS.								

- username** Optional string (up to 100 characters) containing the user name used for **pap**, **chap** or **eap-md5** authentication.
- password** Optional string (up to 100 characters) containing the password used for **pap**, **chap** or **eap-md5** authentication.
- apn_aggregate_max_bitrate_dl**
Optional integer (default = -1). EPS APN or 5GS PDU session aggregate maximum bitrate for downlink (in bits/s). If set to -1, no APN-AMBR or PDU session AMBR is configured and UE-AMBR is used instead.
- apn_aggregate_max_bitrate_ul**
Optional integer (default = -1). EPS APN or 5GS PDU session aggregate maximum bitrate for uplink (in bits/s). If set to -1, no APN-AMBR or PDU session AMBR is configured and UE-AMBR is used instead.
- emergency**
Optional boolean (default = false). If set, PDN will be selected for emergency calls.
- serving_plmn_rate_control**
Optional integer (range 0 to 65535, default = 0). Defines the serving PLMN rate control IE content when PDN is used with control plane CIoT optimization only. If the value configured is less than 10, the IE is not transmitted.
- apn_rate_control_params**
Optional object. If defined, and if the UE indicates APN rate control parameters support in its protocol configuration options, the following parameters will be sent in Core Network protocol configuration options:
- additional_exception_report**
Boolean. Indicates if exception reports are allowed once the limit is reached.
 - ul_time_unit**
Enumeration: **unrestricted**, **minute**, **hour**, **day** or **week**.
 - max_ul_rate**
Integer (range from 0 to 16777215). Number of messages allowed to be sent per **ul_time_unit**.
- additional_apn_rate_control_exception_data_params**
Optional object. If defined, and if the UE indicates additional APN rate control for exception data parameters support in its protocol configuration options, the following parameters will be sent in Core Network protocol configuration options:
- ul_time_unit**
Enumeration: **unrestricted**, **minute**, **hour**, **day** or **week**.
 - max_ul_rate**
Integer (range from 0 to 65535). Number of messages allowed to be sent per **ul_time_unit**.
- backoff_timer**
Optional integer (default = -1). Value in seconds of the T3396/T3584/T3585 timers. The timer is transmitted in the ESM and

5GSM reject messages if the value is not -1. -2 means that the timer is deactivated.

`re_attempt_ind`

Optional integer (range -1 to 255, default = -1). Value of octet 3 of the Re-attempt indicator information element, as specified in 3GPP TS 24.301 chapter 9.9.4.13A and 3GPP TS 24.501 chapter 9.11.4.17. The value -1 means that the information element is not sent.

`automatic_release`

Optional boolean (default = false). If set, when the last associated dedicated EPS bearer is released the MME releases the PDN connection. With 5GS, when the last non default QoS flow is released, the SMF releases the PDU session.

`automatic_release_timeout`

Optional integer (default = 0). Delay in ms before releasing the PDN connection or PDU session when `automatic_release` is set to true.

`allow_multiple_pdn_connections`

Optional boolean (default = false). If set, a UE can create multiple PDN connections to this APN, or multiple PDU sessions to this DNN for the same slice.

`single_address_bearers_only`

Optional boolean (default = false). Only applicable when `pdn_type` is set to "ipv4v6". If set to true, a UE trying to connect to the APN/DNN with IPv4v6 will only have a `single_address_bearers_first_address` address allocated and will receive the indication to create a new PDN connectivity/PDU session for the other IP address type.

`single_address_bearers_first_address`

Optional enumeration (ipv4, ipv6, default = ipv4). Only applicable when `single_address_bearers_only` is set to true. Defines the address type allocated when the UE tries to connect to the APN/DNN with IPv4v6.

`ue_initiated_modification`

Optional boolean (default = false). If set, the UE can request the modification of a bearer, otherwise the request is rejected.

`ip_src_violation_limit`

Optional integer (default = -1). If greater than -1, the MME or UPF checks the IP source address of uplink packets. When `ip_src_violation_limit` packets are received, the PDN or PDU session is released. The value 0 means that the packets are dropped without triggering a release.

`integrity_protection`

Optional enumeration (disabled, preferred, required, default = disabled). Defines whether integrity should be used for the PDN connection / PDU session or not.

For EPC, if the value is set to **preferred**, the EPC will activate integrity protection based on the UE capabilities. If set to **required**, and if the UE does not support integrity protection, the request will be rejected with ESM cause 30.

For 5GC, if the value is set to **preferred**, the 5GC will activate integrity

protection based on the UE capabilities and the configured PDU session AMBR. If set to **required**, and if the UE does not support integrity protection for the bitrate configured in the PDU session AMBR, the request will be rejected with 5GSM error cause #82.

tun_setup_script

Overrides [tun_setup_script], page 14, for this PDN or PDU session.

tun_ifname

Optional string. If set, use this tun device instead of creating it. Useful when LTEMME has no root privileges.

ifname

Optional string. If set, instead of creating a tun interface, all IP traffic will go through the designated network interface. In that case, **gateway** parameter is not used and local machine network (localhost and other interfaces) are not reachable from UEs.

This mode is incompatible with Generic Receive Offload, which has to be disabled on NIC.

```
ethtool -K <ifname> generic-receive-offload off
```

erabs

Array of objects. Each element defines an E-RAB (E-UTRAN Radio Access Bearer) associated to the PDN or a QoS flow associated to the PDU session. The first E-RAB or QoS flow is the default radio bearer and must always be present. The additional E-RABs and QoS flows are dedicated radio bearers and must include a Traffic Flow Template (TFT) unless they are defined as UE initiated.

Property of each element:

qci Range: 1 to 255. QoS Class Identifier of the E-RAB or 5G QoS Identifier of the QoS flow.

priority_level

Optional integer (range: 1 to 15, default 15). ARP priority level.

pre_emption_capability

Optional enumeration: **shall_not_trigger_pre_emption** or **may_trigger_pre_emption** (default **shall_not_trigger_pre_emption**).

pre_emption_vulnerability

Optional enumeration: **not_pre_emptable** or **pre_emptable** (default **not_pre_emptable**).

setup_type

Optional enumeration: **automatic**, **on_demand**, **ue_initiated** (default = **automatic**).

- If set to **automatic**, the dedicated bearer is created with the default bearer.
- If set to **on_demand**, the dedicated bearer is created when there is traffic matching the TFT filters. This option is useful to automatically create a dedicated bearer for IMS RTP voice traffic.
- If set to **ue_initiated**, the dedicated bearer is created when receiving a ESM bearer resource allocation request or PDU session modification request message. In

that case, the `gbr` object defines the maximum values allowed (MME will use the minimum between configured values and the ones sent by the UE) and `tft` object is not required (the core network will use the filters sent by the UE).

<code>5qi_qos</code>	Optional object. 5QI QoS characteristics. List of properties:
<code>priority_level</code>	Optional integer (range 0 to 127). 0 means that the field is absent. Its presence is mandatory if <code>packet_delay_budget</code> is present.
<code>packet_delay_budget</code>	Optional integer (range -1 to 1023) in 0.5 ms unit. -1 means that the field is absent.
<code>extended_packet_delay_budget</code>	Optional integer (range -1 to 109999) in 0.01 ms unit. -1 means that the field is absent.
<code>packet_error_rate</code>	Optional string. It should be set to "xE-y" where x is the scalar value (0 to 9) and y is the exponent value (0 to 9). Its presence is mandatory if <code>packet_delay_budget</code> is present.
<code>averaging_window</code>	Optional integer (range -1 to 4095) in ms unit. Only applicable to GBR bearers. -1 means that the field is absent.
<code>maximum_data_burst_volume</code>	Optional integer (range -1 to 2000000) in byte unit. Only applicable to delay critical GBR bearers. -1 means that the field is absent.
<code>cn_packet_delay_budget_dl</code>	Optional integer (range -1 to 1099990) in 0.01 ms unit. Only applicable to delay critical GBR bearers. -1 means that the field is absent.
<code>cn_packet_delay_budget_ul</code>	Optional integer (range -1 to 1099990) in 0.01 ms unit. Only applicable to delay critical GBR bearers. -1 means that the field is absent.
<code>gbr</code>	Optional object. Guaranteed Bitrate information. List of properties:
<code>maximum_bitrate_dl</code>	Integer. Bearer maximum bitrate for downlink (in bits/s).
<code>maximum_bitrate_ul</code>	Integer. Bearer maximum bitrate for uplink (in bits/s).

	guaranteed_bitrate_dl	Integer. Bearer guaranteed bitrate for downlink (in bits/s).
	guaranteed_bitrate_ul	Integer. Bearer guaranteed bitrate for uplink (in bits/s).
filters	Optional array of objects. List of TFT filters or QoS rules. Required for dedicated bearers with setup_type different from ue_initiated . Each filter has the following properties:	
	direction	Enumeration: dl , ul or both . Set the filter direction.
	id	Range: 1 to 15. Set the filter identifier.
	precedence	Range: 0 to 254. Set the filter precedence. All the filters must have different precedence. 0 is the highest precedence. Note that precedence 80 is reserved for derived QoS rules in 5GS and thus will be rejected if configured.
	reflective_qos	Optional integer (default = false). If set to true, this QoS rule in 5GS can use reflective QoS if supported by the UE. Is it only applicable if direction is set to dl .
	components	Array of objects. Each component contains one of the following properties as described in 3GPP TS 23.060 chapter 15.3.2:
	ipv4_remote_addr	String. Match a remote (external network entity) IPv4 address with the additional mask property.
	ipv4_local_addr	String. Match a local IPv4 address with the additional mask property. Note that not all UEs support it (they must indicate the support of the Local address in TFT in PCO/ePCO).
	ipv6_remote_addr	String. Match a remote (external network entity) IPv6 address with the additional mask property.
	ipv6_remote_addr_prefix	String. Match a remote (external network entity) IPv6 address with

the additional `prefix_len` property. Note that not all UEs support it (they must indicate the support of the Local address in TFT in PCO/ePCO).

ipv6_local_addr_prefix

String. Match a local IPv6 address with the additional `prefix_len` property. Note that not all UEs support it (they must indicate the support of the Local address in TFT in PCO/ePCO).

proto_id Range: 0 to 255. Match against the IP protocol identifier.

local_port

Range: 0 to 65535. Match against the local (UE) port.

local_port_range

Array of 2 integers. Match against a local (UE) port range.

remote_port

Range: 0 to 65535. Match against the remote (external network entity) port.

remote_port_range

Array of 2 integers. Match against a remote (external network entity) port range.

security_parameter_index

32 bit integer. Match the ESP or AH security parameter index.

type_of_service

Range: 0 to 255. Match the type of service (IPv4) or the traffic class (IPv6) field. The additional `mask` property is the corresponding mask.

mask

Depends on TFT component.

If `ipv4_remote_addr` is set, string representing IPv4 address used as a mask to apply on packet remote address.

If `ipv6_remote_addr` is set, string representing IPv6 address used as a mask to apply on packet remote address.

If `type_of_service` is set, integer between 0 and 255 used as a mask to apply on packet tos.

<code>flow_label</code>	20 bit integer. Match the IPv6 flow label.
<code>prefix_len</code>	Range: 1 to 128. IPv6 address prefix length.
<code>destination_mac_addr</code>	String. Match the destination MAC address.
<code>source_mac_addr</code>	String. Match the source MAC address.
<code>802.1q_ctag_vid</code>	Range: 0 to 4095. Match the 802.1Q C-TAG VID.
<code>802.1q_stag_vid</code>	Range: 0 to 4095. Match the 802.1Q S-TAG VID.
<code>802.1q_ctag_pcp_dei</code>	Range: 0 to 15. Match the 802.1Q C-TAG PCP and DEI.
<code>802.1q_stag_pcp_dei</code>	Range: 0 to 15. Match the 802.1Q S-TAG PCP and DEI.
<code>ethertype</code>	Range: 0 to 65535. Match the ethertype.
<code>destination_mac_addr_range</code>	Array of 2 strings. Match the destination MAC address range. Only applicable to 5GC.
<code>source_mac_addr_range</code>	Array of 2 strings. Match the source MAC address range. Only applicable to 5GC.
<code>on_demand_timeout</code>	Optional integer. When <code>setup_type</code> is <code>on_demand</code> , set the duration (in ms) after which the dedicated bearer is released when there is no downlink or uplink traffic.
<code>on_demand_ul_trigger</code>	Optional boolean (default = false). When <code>setup_type</code> is <code>on_demand</code> , if set to true an UL packet matching one of the TFT filters triggers the dedicated E-RAB or QoS flow establishment.
<code>transaction_identifier</code>	Optional integer (range 0 to 127). If present, the transaction identifier IE is put in the EPS bearer activation message.

- llc_sapi** Optional integer (range 0 to 15). If present, the LLC service access point identifier IE is put in the EPS bearer activation message.
- radio_priority** Optional integer (range 0 to 7). If present, the radio priority IE is put in the EPS bearer activation message.
- packet_flow_identifier** Optional integer (range 0 to 127). If present, the packet flow identifier IE is put in the EPS bearer activation message.
- sm_qos** Optional string. If present, the quality of service IE is put in the EPS bearer activation message. The string must contain the hexadecimal representation of the IE without its IEI and length.

The following parameters are applicable to EPC only:

esm_procedure_filter

Optional object. Allows to define the MME behavior for a list of ESM procedures.

Each property name represents an ESM procedure. The ones currently supported are **pdn_connectivity**, **pdn_disconnect**, **bearer_resource_allocation** and **bearer_resource_modification**.

Each property value is an object containing the following fields:

action Enumeration (**treat** (UE message is processed), **ignore** (UE message is ignored) or **reject** (UE message is rejected)).

t1 Optional integer. If set, the **reject** or **ignore** filter is applied **t1** times. If not set, the filter is applied until it is modified.

send_status_on_reject

Optional boolean. It set and if **action** is set to reject an ESM status message is sent.

ignore does not apply to procedure **pdn_connectivity** performed during the attach procedure.

By default all procedures are treated.

Example:

```
esm_procedure_filter: {
  pdn_connectivity: {
    action: "treat"
  },
  bearer_resource_allocation: {
    action: "reject",
    t1: 1
  }
}
```

The following parameters are applicable to 5GC only:

5gsm_procedure_filter

Optional object. Allows to define the SMF behavior for a list of 5GSM procedures.

Each property name represents a 5GSM procedure. The ones currently supported are `pdu_session_establishment`, `pdu_session_release` and `pdu_session_modification`.

Each property value is an object containing the following fields:

action Enumeration (`treat` (UE message is processed), `ignore` (UE message is ignored) or `reject` (UE message is rejected)).

ttl Optional integer. If set, the `reject` or `ignore` filter is applied `ttl` times. If not set, the filter is applied until it is modified.

send_status_on_reject

Optional boolean. It set and if `action` is set to `reject` a 5GSM status message is sent.

By default all procedures are treated.

Example:

```
"5gsm_procedure_filter": {
  pdu_session_establishment: {
    action: "treat"
  },
  pdu_session_modification: {
    action: "reject",
    ttl: 1
  }
}
```

always_on

Optional enumeration (`auto`, `required`, `not_allowed`, `default = auto`). Defines the always-on behavior for the PDU session. If the value is set to `auto`, the 5GC will follow whatever is requested by the UE. If the value is set to `required`, the 5GC will always set the PDU session as always-on required. If the value is set to `not_allowed`, the 5GC will always set the PDU session as always-on not allowed.

confidentiality_protection

Optional enumeration (`disabled`, `preferred`, `required`, `default = required`). Defines if confidentiality must be used for the PDU session or not.

apply_nas_transport_n1_sm_filter

Optional boolean (`default = true`). Indicates whether the 5GMM procedure filter `nas_transport_n1_sm` should apply to this DNN or not.

eps_5gs_interworking

Optional boolean (`default = true`). If set to `true`, interworking between EPS and 5GS is allowed for this APN/DNN. Otherwise it is forbidden.

5gsm_congestion_re_attempt_ind

Optional integer (range -1 to 255, `default = -1`). Value of octet 3 of the Re-attempt indicator information element, as specified in 3GPP TS 24.501 chapter 9.11.4.21. The value -1 means that the information element is not sent.

rq_timer	Optional integer (default = -1). Value in seconds of the RQ timer for the PDU session, sent to the UE if it supports reflective QoS feature. -2 means that the timer is deactivated.
slices	Optional array. Defines the QoS flows by S-NSSAI. If a supported S-NSSAI is not present in the array, the QoS flows defined in [erabs], page 35, applies. Each entry will set specific QoS flows for a slice as defined below:
snssai	S-NSSAI value.
sst	Integer (range 0-255). Slice Service Type.
sd	Optional integer (range 0-0xFFFFFE). Slice Differentiator.
qos_flows	Array of QoS flows. Each element of the array has the same structure as an element in [erabs], page 35, except that "5qi" shall be used instead of "qci".

5.2.2 User database options

ue_db

Array of objects. Configure the user database. Each element is an entry for one user. The following properties are available:

imsi	Optional string. Shall be present if nai is absent. Set the IMSI.
nai	Optional string applicable to 5G only. Shall be present if imsi is not set. Set the Network specific identifier-based SUPI.
sim_algo	Optional enumeration. xor, milenage or tuak (default = xor). Set the USIM authentication algorithm. Note: test USIM cards use the XOR algorithm.
sqn	Optional string (6 byte hexadecimal string). Default = "000000000000". Set the initial sequence number. For the XOR algorithm, the actual value does not matter. For the Milenage or TUAK algorithm, a sequence number resynchronization is initiated if the sequence number does not match the one stored in the USIM.
force_sqn_resync	Optional boolean (default = false). If set to true, the core network will reuse the last SQN value to force a SQN resynchronization procedure. Only applicable to Milenage or TUAK algorithm.
K	String. Set the user secret key (as a 16 bytes hexadecimal string, or eventually 32 bytes hexadecimal string for TUAK).
op	Optional string. Operator key (as a 16 byte hexadecimal string). When the Milenage authentication algorithm is used, either op or opc must be set.
opc	Optional string. Operator key preprocessed with the user secret key (as a 16 byte hexadecimal string). When the Milenage authentication algorithm is used, either op or opc must be set.

r	Optional array of 5 integers (range: 0 to 127). Allows to customize the r1 to r5 parameters when Milenage authentication algorithm is used. If the array is not present, the default values (as defined in 3GPP TS 35.206) are used.						
c	Optional array of 5 strings. Each value contains a 16 byte hexadecimal string. Allows to customize the c1 to c5 parameters when Milenage authentication algorithm is used. If the array is not present, the default values (as defined in 3GPP TS 35.206) are used.						
top	Optional string. Operator key (as a 32 byte hexadecimal string). When the TUAK authentication algorithm is used, either top or topc must be set.						
topc	Optional string. Operator key preprocessed with the user secret key (as a 32 byte hexadecimal string). When the TUAK authentication algorithm is used, either top or topc must be set.						
keccak_iter	Optional integer (range: 1 to MAX_INT). Allows to customize the number of Keccak permutations performed when using the TUAK authentication algorithm. If the item is not present, the default value 1 (as defined in 3GPP TS 35.231) is used.						
cag_info_list	Optional array. Subscribed CAG information list. Each element of the array contains: <table> <tr> <td>plmn</td><td>String (5 or 6 digits).</td></tr> <tr> <td>cag_id_list</td><td>Array of 1 to 12 integers (range 0 to 4294967295) giving the list of the allowed CAG-Identifiers.</td></tr> <tr> <td>cag_only_ind</td><td>Optional boolean (default = FALSE). Indication that the UE is only allowed to access 5GS via CAG cells.</td></tr> </table>	plmn	String (5 or 6 digits).	cag_id_list	Array of 1 to 12 integers (range 0 to 4294967295) giving the list of the allowed CAG-Identifiers.	cag_only_ind	Optional boolean (default = FALSE). Indication that the UE is only allowed to access 5GS via CAG cells.
plmn	String (5 or 6 digits).						
cag_id_list	Array of 1 to 12 integers (range 0 to 4294967295) giving the list of the allowed CAG-Identifiers.						
cag_only_ind	Optional boolean (default = FALSE). Indication that the UE is only allowed to access 5GS via CAG cells.						
csg_info_list	Optional array of objects. Subscribed CSG information. Each element of the array contains: <table> <tr> <td>plmn</td><td>String (5 or 6 digits).</td></tr> <tr> <td>csg_id_list</td><td>Array of integers in range 0 to 0x7FFFFFFF. Allowed CSG id list in the PLMN.</td></tr> </table>	plmn	String (5 or 6 digits).	csg_id_list	Array of integers in range 0 to 0x7FFFFFFF. Allowed CSG id list in the PLMN.		
plmn	String (5 or 6 digits).						
csg_id_list	Array of integers in range 0 to 0x7FFFFFFF. Allowed CSG id list in the PLMN.						
amf	Range: 0 to 65535. Set the Authentication Management Field.						
5gs_auth_type	Applicable to 5GC only. Optional enumeration: 5g_aka, eap_aka_prime, eap_tls (default = 5g_aka). 5GMM authentication method.						
at_result_ind	Applicable to 5GC only. Optional boolean (default = false). Indicates if the AUSF shall include the AT_RESULT_IND attribute in message EAP-request/AKA'-Challenge.						

res_len	Optional integer (default = 8). Defines length of response in bytes during authentication. For TUAK authentication algorithm, the value must be 4, 8 or 16 bytes long.
multi_sim	Optional boolean (default = false). If true, allow several UEs to have the same IMSI (useful when using several identical test SIM cards in different UEs at the same time). They are distinguished with their IMEI. Note: it is only allowed with the XOR authentication algorithm.
isim_auth	Optional object. If present, the object allows to configure some specific authentication parameters for the ISIM. Otherwise it uses the same parameters as those defined for the USIM. It contains the following configuration parameters: sim_algo , K , op , opc , r , c , top , topc , keccak_iter and res_len .
msisdn	Optional string. Sets the UE MSISDN (that will be sent in the NAS PCO message if requested by the UE for example).
ue_aggregate_max_bitrate_dl	Optional integer (default = 10e9). UE aggregate maximum bitrate for downlink (in bits/s).
ue_aggregate_max_bitrate_ul	Optional integer (default = 3e9). UE aggregate maximum bitrate for uplink (in bits/s).
t3412	Optional integer. Value in seconds of the T3412 (TAU update) or T3512 timer for this IMSI. If not present, the MME or AMF will use the value coming from HSS or configured locally. It is sent to the UE in NAS signalling, unless the UE is requesting the use of a longer timeout with T3412 extended value or requested T3512 value information element.
n3gpp_dereg_timer	Optional integer. Applicable to 5GC only. Value in seconds of the non-3GPP de-registration timer.
count	Optional integer (default = 1). Create n user entries by incrementing the IMSI and K.
restrict_nr_as_2nd_rat	Optional boolean (default = false). If set to true, the user is not allowed to use NR as secondary RAT (no DCNR).
restrict_5gc_access	Optional boolean (default = false). If set to true, the user is not allowed to access 5GC when coming from EPC (no handover or cell redirection).
restrict_epc_access	Optional boolean (default = false). If set to true, the user is not allowed to access EPC when coming from 5GC (no handover or cell redirection).
subscriber_profile_id	Optional integer (range 1 to 256). User subscriber profile ID for RAT/frequency priority or index to RAT/frequency selection priority.
restrict_pdn_list	Optional boolean (default = false). If set to true, only the PDNs or PDU sessions listed in the pdn_list object are allowed for the user.

pdn_list	Optional array. Each entry will set specific parameters for a PDN or PDU session as defined below:
access_point_name	String. Used to define what PDN or PDU session to configure.
default	Optional boolean (default = false). If true and UE does not specify the APN or DNN it wants to connect to, this PDN or PDU session will be used.
pdn_type	Optional enumeration: ipv4, ipv6, ipv4v6. Restrict the PDN type for this specific IMSI. The PDN or PDU session must be configured with a matching IP version.
ipv4_addr	Optional string. If set, the UE will always use this IPv4 address.
ipv6_prefix	Optional string. If set, the UE will always use this IPv6 prefix.
imei	Optional string (14 or 15 digits). If set, this configuration only applies to UE with matching IMEI. Only supported for EPS, not 5GS.
multicast	Optional boolean (default = false). If set, IPv4 multicast traffic will be forwarded to this PDN or PDU session.
ipv6_multicast	Optional boolean (default = false). If set, IPv6 multicast traffic will be forwarded to this PDN or PDU session.
broadcast	Optional boolean (default = false). If set, IPv4 broadcast traffic will be forwarded to this PDN or PDU session.
routes	Optional array. Each entry of array represent a list of filters. See [TFT components], page 37, for filters syntax except that remote refers to UE and local to network. When a packet enters MME or UPF, if it matches one of the filter list, it will be sent to associated UE. Ex: <pre> routes: [[{ ipv4_remote_addr: "10.0.0.0", mask: "255.255.255.0" }]] </pre> Means that all packets addressed to 10.0.0.0/24 network will be sent to UE.
nssai	Applicable to 5GC only. Optional array. List of subscribed S-NSSAIs per DNN.

If not present, the list of the S-NSSAIs served by the AMF applies.

See [nssai], page 25.

nssai_subject_to_nssaa

Applicable to 5GC only.

Optional array. List of the subscribed S-NSSAIs subject to NSSAA. Each entry of the array contains the following objects:

snssai S-NSSAI subject to NSSAA.

sst Integer (range 0-255). Slice Service Type.

sd Optional integer (range 0-0xFFFFFE). Slice Differentiator.

auth_type

Optional enumeration: **eap_md5**, **eap_tls** or **eap_aka** (default set to **eap_md5**). Defines the authentication mechanism used for this S-NSSAI.

username String (up to 100 characters) containing the user name used for algo **eap_md5**.

password String (up to 100 characters) containing the password used for algo **eap_md5**.

ue_slice_max_bitrate

Applicable to 5GC only.

Optional array. Each entry of the array gives the UE maximum bitrate for uplink and downlink in a slice subject to NSAC and contains the following objects:

snssai S-NSSAI value. The S-NSSAI shall be contained in the list of the S-NSSAIs subject to NSAC given by **nssai_subject_to_nsac**.

sst Integer (range 0-255). Slice Service Type.

sd Optional integer (range 0-0xFFFFFE). Slice Differentiator.

ue_slice_max_bitrate_ul

Integer. UE maximum bitrate in the S-NSSAI for uplink (in bits/s).

ue_slice_max_bitrate_dl

Integer. UE maximum bitrate in the S-NSSAI for downlink (in bits/s).

sms_over_nas_allowed

Optional boolean (default = true). Indicates if the subscription of the UE allows SMS over NAS.

allowed_5gs_tais

Optional object. Allowed or not allowed TAIs in 5GS as defined in 3GPP TS 29.571 chapter 5.4.4.18. If not present, all TAs in the registration area are allowed. Each object must contain the following parameters:

restriction_type

Optional enumeration: **allowed**, **not_allowed**. Default value is **allowed**. Only applicable to 5GS. Gives the type of the

	service area restriction. If set to allowed, the areas defined in tais are allowed.								
tais	Array of objects (up to 65535). Only applicable to 5GS. Each object must contain the following parameters: <table> <tr> <td>plmn</td><td>String (5 or 6 digits).</td></tr> <tr> <td>areas</td><td>Array of objects (up to 65535). Each object describes an area defined by a specific operator code or a list of tracking areas codes as defined in 3GPP TS 29.571 chapter 5.4.4.17. The same areas list must be configured for equivalent PLMNs. Each object contains the following parameters: <table> <tr> <td>code</td><td>Integer in range 0 to 0xFFFF. Area code as defined by the operator (See [areas_list_5gs], page 27). Must be present only if tacs is absent.</td></tr> <tr> <td>tacs</td><td>Array of up to 16 integers in range 1 to 0xFFFFFE, except 0xFFFFFE. Each element defines a 3 bytes long tracking area code. Must be present only if code is absent.</td></tr> </table> </td></tr> </table>	plmn	String (5 or 6 digits).	areas	Array of objects (up to 65535). Each object describes an area defined by a specific operator code or a list of tracking areas codes as defined in 3GPP TS 29.571 chapter 5.4.4.17. The same areas list must be configured for equivalent PLMNs. Each object contains the following parameters: <table> <tr> <td>code</td><td>Integer in range 0 to 0xFFFF. Area code as defined by the operator (See [areas_list_5gs], page 27). Must be present only if tacs is absent.</td></tr> <tr> <td>tacs</td><td>Array of up to 16 integers in range 1 to 0xFFFFFE, except 0xFFFFFE. Each element defines a 3 bytes long tracking area code. Must be present only if code is absent.</td></tr> </table>	code	Integer in range 0 to 0xFFFF. Area code as defined by the operator (See [areas_list_5gs], page 27). Must be present only if tacs is absent.	tacs	Array of up to 16 integers in range 1 to 0xFFFFFE, except 0xFFFFFE. Each element defines a 3 bytes long tracking area code. Must be present only if code is absent.
plmn	String (5 or 6 digits).								
areas	Array of objects (up to 65535). Each object describes an area defined by a specific operator code or a list of tracking areas codes as defined in 3GPP TS 29.571 chapter 5.4.4.17. The same areas list must be configured for equivalent PLMNs. Each object contains the following parameters: <table> <tr> <td>code</td><td>Integer in range 0 to 0xFFFF. Area code as defined by the operator (See [areas_list_5gs], page 27). Must be present only if tacs is absent.</td></tr> <tr> <td>tacs</td><td>Array of up to 16 integers in range 1 to 0xFFFFFE, except 0xFFFFFE. Each element defines a 3 bytes long tracking area code. Must be present only if code is absent.</td></tr> </table>	code	Integer in range 0 to 0xFFFF. Area code as defined by the operator (See [areas_list_5gs], page 27). Must be present only if tacs is absent.	tacs	Array of up to 16 integers in range 1 to 0xFFFFFE, except 0xFFFFFE. Each element defines a 3 bytes long tracking area code. Must be present only if code is absent.				
code	Integer in range 0 to 0xFFFF. Area code as defined by the operator (See [areas_list_5gs], page 27). Must be present only if tacs is absent.								
tacs	Array of up to 16 integers in range 1 to 0xFFFFFE, except 0xFFFFFE. Each element defines a 3 bytes long tracking area code. Must be present only if code is absent.								
forbidden_5gs_tais	Optional array of objects. Forbidden areas in 5GS. See [tais], page 47.								
mps_priority	Optional boolean (default = false). Indicates whether the UE is subscribed to multimedia priority service.								
mcs_priority	Optional boolean (default = false). Indicates whether the UE is subscribed to mission critical service.								
routing_indicator	Optional string (default = "0"). Routing Indicator consisting of 1 to 4 decimal digits assigned by the home network operator as defined in 3GPP 23.003 2.2B Subscription Concealed Identifier (SUCI)).								
akma	Optional boolean (default = false). Indicates if AKMA keys need to be generated for the UE.								

5.2.3 Public Warning System (ETWS/CMAS) options

pws_msgs	Optional array of objects. Define a list of ETWS/CMAS messages which can be sent to the connected eNodeBs with the pws_write monitor command. Check 3GPP TS 23.041 to have the exact definition of each field. Each message contains the following properties: <table> <tr> <td>local_identifier</td><td>Range: 0 to 65535. Local message identifier. Used as argument to the monitor commands pws_write or pws_kill.</td></tr> </table>	local_identifier	Range: 0 to 65535. Local message identifier. Used as argument to the monitor commands pws_write or pws_kill .
local_identifier	Range: 0 to 65535. Local message identifier. Used as argument to the monitor commands pws_write or pws_kill .		

- message_identifier**
Range: 0 to 65535. Message Identifier.
- serial_number**
Range: 0 to 65535. Serial Number.
- repetition_period**
Optional integer, range: 0 to 4095 for EPC, 131071 for 5GC (default = 10). Periodicity of the warning message to be broadcast.
- number_of_broadcasts_requested**
Optional integer, range: 0 to 65535 (default = 65535). Number of times a message is to be broadcast.
- warning_type**
Optional integer. Range: 0 to 65535. Warning type (ETWS only).
- warning_security_info**
Optional 50 byte hexadecimal string. Warning security information (ETWS optional).
- warning_message**
Optional array of string. Message content (ETWS: optional, CMAS: mandatory). Each string is a message page and contains at most 93 GSM 7 bit or 41 UCS2 characters. At most 15 pages are allowed. If a non GSM 7 bit character is found, the string is encoded as UCS2 otherwise it is encoded as GSM 7 bit.
- warning_message_hex**
Optional array of hexadecimal string. Message content (ETWS: optional, CMAS: mandatory). Each hexadecimal string is a message page and contains at most 164 characters. At most 15 pages are allowed. May be present only if **warning_message** is absent.
- data_coding_scheme**
Optional integer. Range 0 to 255. Data coding scheme. Must be present if **warning_message_hex** is present. If **warning_message** is used, its default value is set to 0x0f for GSM 7 bit encoding and 0x48 for UCS2 encoding. Note that this parameter does not override the encoding of the strings found in **warning_message** so setting an inconsistent value can lead to a garbled decoding in the UE.
- concurrent_warning_message_ind**
Optional boolean (default = false). Indicates that the warning message is a new message to be scheduled for concurrent broadcast with any other ongoing broadcast of warning messages.
- send_warning_indication**
Optional boolean (default = false). SBCAP interface: Gives the presence of Send Write Replace Warning Indication IE in the SBCAP message WRITE-REPLACE WARNING REQUEST. N50 interface: Gives the presence of sendRanResponse attribute the N50 message POST ../non-ue-n2-messages/transfer(N2InformationTransferReqData).
- warning_area_list**
Optional object. If present, the Warning Area List IE will be sent in the message WRITE-REPLACE WARNING REQUEST. It should contain one of the following objects:

cell_id_list	Optional array of objects (up to 65535). Each object must contain the following parameters:
plmn	String (5 or 6 digits).
cell_id	Integer. 28 bits long LTE cell identifier.
tai_list	Optional array of objects (up to 65535). Each object must contain the following parameters:
plmn	String (5 or 6 digits).
tac	Integer. 2 bytes long tracking area code.
emergency_area_id_list	Optional array of integers (up to 65535). 3 bytes long emergency area identifier.
warning_area_coordinates	Optional hexadecimal string. Maximum length 1024 bytes. Warning Area Coordinates octet string (CMAS only).
omc_id	Optional string. Maximum length 20 bytes. Identity of an Operation and Maintenance Centre.
enb	Optional object. Global eNB ID to send in the message WRITE-REPLACE WARNING REQUEST.
plmn	String (5 or 6 digits).
enb_type	Optional string (macro, home, short_macro or long_macro). Default value is "macro". Type of the global eNB ID.
enb_id	Integer. eNB ID.
tai_list	Optional array of objects (up to 65535). TAI List to send in the message WRITE-REPLACE WARNING REQUEST. See [tai_list], page 49.
warning_area_list_5gs	Optional object. 5GS Warning Area List to send in the message WRITE-REPLACE WARNING REQUEST. It should contain one of the following objects:
nr_cell_id_list	Optional array of objects (up to 65535). Each object must contain the following parameters:
plmn	String (5 or 6 digits).
cell_id	Integer. 36 bits long NR cell identifier.
tai_list	Optional array of objects (up to 65535). See [tai_list_5gs], page 49.
emergency_area_id_list	Optional array of integers (up to 65535). 3 bytes long emergency area identifier.
tai_list_5gs	Optional array of objects (up to 65535). List of 5GS TAIs to send in the SBCAP message WRITE-REPLACE

WARNING REQUEST or the N50 message POST `../non-ue-n2-messages/transfer(N2InformationTransferReqData)`. Each object must contain the following parameters:

plmn String (5 or 6 digits).
tac Integer. 3 bytes long tracking area code.

ran_node_id

Optional integer. Applicable to SBCAP interface only. Value of the global RAN node ID to send in the SBCAP message WRITE-REPLACE WARNING REQUEST. It should contain one of the following objects:

gnb gNB identifier.
 plmn String (5 or 6 digits).
 gnb_id_bits
 Integer. Range 22 to 32. gNB ID length in bits.
 gnb_id Integer. The gNB global identifier.
ng_enb ngENB identifier. See [enb], page 49.

rat_selector_5gs

Optional boolean. Default value is false. Applicable to SBCAP interface only. Indicates the presence of RAT Selector 5GS IE in the message WRITE-REPLACE WARNING REQUEST.

n50_rat_selector

Optional enumeration: nr, eutra, both. Default value is both. Applicable to N50 interface only. Gives the value of ratSelector attribute in N2InformationTransferReqDataTransfer.

n50_ran_node_id_list

Optional array of objects. Applicable to N50 interface only. See [ran_node_id], page 50. List of the global RAN node ID to send in the N50 message POST `../non-ue-n2-messages/transfer(N2InformationTransferReqData)`.

5.2.4 NAS special conformance testing options

The MME or AMF can automatically activate UE test mode during attachment and configure test loop mode A, B or G (see 3GPP TS 36.509 and 38.509 for details). Once the loop is closed, the user can transmit downlink IP packets to the UE that will be loopbacked in UL.

nas_test_procedure

Optional object allowing to configure the test procedure. It must contain the following objects:

test_loop_mode

Enumeration: none, a, b, g. Defines which test loop will be activated.

lb_setup_list

Optional array used for test loop mode A if UL PDCP SDU scaling is required. Each element of the array must contain the following 2 objects:

ul_pdcp_sdu_size

Integer (range 0 to 1520). UL PDCP SDU size in bytes.

<code>drb_id</code>	Integer (range 1 to 32). Data Radio Bearer identity on which the UL PDCP SDU scaling is applied.
<code>ip_pdu_delay</code>	Integer (range 0 to 255). Transmission delay in seconds of the EUTRA UL PDCP SDUs or NR UL SDAP SDUs when operating in test loop mode B.
<code>operation_mode</code>	Enumeration (<code>upper</code> or <code>rlc</code>). <code>upper</code> means that data is returned in uplink at the EMM entity. <code>rlc</code> means that data is returned in uplink at the RLC AM-SAP of SRB1bis for NB-IoT UE or at the RLC AM-SAP of SRB2 for E-UTRA UE. Used in test loop mode G.
<code>repetitions</code>	Integer (0 to 127). Number of repetitions of received content of received user data in downlink in uplink. Used in test loop mode G.
<code>ul_data_delay</code>	Integer (0 to 255). Uplink data delay in seconds. Used in test loop mode G.

5.2.5 Rx options

`rx`

Optional object allowing to configure the Rx options. It can contain the following objects:

<code>bind_addr</code>	Optional string. IP address and optional port on which the Rx SCTP connection is bound. The default address is the same as the first S1AP SCTP connection and the default port is 3868.				
<code>qos</code>	Optional object. It can contain 7 object properties: audio, video, application, data, control, text and message. Default QCI/5QI is 1 for audio, 2 for video and application, 6 for data and control, 8 for text and message. Each property contains the following fields: <table> <tr> <td><code>qci</code></td><td>Integer (range 1 to 255). QoS Class Identifier of the E-RAB, or 5QI of the QoS flow.</td></tr> <tr> <td><code>5qi_qos</code></td><td>Optional object. See [5QI QoS], page 36.</td></tr> </table>	<code>qci</code>	Integer (range 1 to 255). QoS Class Identifier of the E-RAB, or 5QI of the QoS flow.	<code>5qi_qos</code>	Optional object. See [5QI QoS], page 36.
<code>qci</code>	Integer (range 1 to 255). QoS Class Identifier of the E-RAB, or 5QI of the QoS flow.				
<code>5qi_qos</code>	Optional object. See [5QI QoS], page 36.				
<code>origin_realm</code>	Optional string. Defines the string sent in the Origin-Realm AVP for Rx messages. Default is set to <code>mnc<MNC>.mcc<MCC>.3gppnetwork.org</code> .				
<code>origin_host</code>	Optional string. Defines the string sent in the Origin-Host AVP for Rx messages. Default is set to <code>epc.mnc<MNC>.mcc<MCC>.3gppnetwork.org</code> .				
<code>reservation_priority</code>	Optional array of 16 elements defining the S1AP ARP (Allocation and Retention Priority) parameters to be used for each Rx reservation priority level. If not present, <code>priority_level</code> is set to 15 (no priority), <code>pre_emption_capability</code> is set to <code>shall_not_trigger_pre_emption</code> and <code>pre_emption_vulnerability</code> is set to <code>not_pre_emptable</code> . If present				

the array must be ordered by increasing Rx priority level (from 0 to 15) and must contain the following fields:

priority_level

Range: 1 to 15. ARP priority level.

pre_emption_capability

Enumeration: `shall_not_trigger_pre_emption` or `may_trigger_pre_emption`.

pre_emption_vulnerability

Enumeration: `not_pre_emptable` or `pre_emptable`.

emergency

Optional object defining the QCI and ARP parameters to be used for the emergency dedicated EPS bearer context. If not present, `qci` is set to 1, `priority_level` is set to 1 (highest priority), `pre_emption_capability` is set to `may_trigger_pre_emption` and `pre_emption_vulnerability` is set to `not_pre_emptable`.

qci Range: 1 to 255.

5qi_qos Optional object. See [5QI QoS], page 36.

priority_level

Range: 1 to 15. ARP priority level.

pre_emption_capability

Enumeration: `shall_not_trigger_pre_emption` or `may_trigger_pre_emption`.

pre_emption_vulnerability

Enumeration: `not_pre_emptable` or `pre_emptable`.

5.2.6 S6a options

s6

Optional object allowing to configure the S6a options. It can contain the following objects:

server_addr

String. IP address and optional port of the HSS used for S6a interface. The default port is 3868.

bind_addr

Optional string. IP address and optional port on which the S6a SCTP connection is bound. The default address is the same as the first S1AP SCTP connection.

origin_realm

Optional string. Defines the string sent in the Origin-Realm AVP for S6 messages. Default is set to `mnc<MNC>.mcc<MCC>.3gppnetwork.org`.

origin_host

Optional string. Defines the string sent in the Origin-Host AVP for S6 messages. Default is set to `epc.mnc<MNC>.mcc<MCC>.3gppnetwork.org`.

transaction_timeout

Optional integer (range 1 to 15000, default = 2000). Defines the timeout in milliseconds for a transaction with the HSS.

watchdog_duration

Optional integer (range 0 to 36000000, default = 30000). The watchdog timer in milliseconds to send the Diameter Device Watchdog Request message. The value 0 deactivates the watchdog.

5.2.7 EIR/S13 options**me_db**

Optional object allowing to define a list of IMEI (14 digits without the last Check Digit one) or IMEISV (16 digits), and their status (whitelisted, blacklisted, greylisted). If not present, all devices are considered as whitelisted.

It can contain the following objects:

default_status

Enumeration (whitelisted, blacklisted, greylisted). Defines the default status for devices not explicitly defined in the next objects.

whitelist

Optional array. It contains a list of IMEI or IMEISV whitelisted.

blacklist

Optional array. It contains a list of IMEI or IMEISV blacklisted.

greylist Optional array. It contains a list of IMEI or IMEISV greylisted.

Example:

```
me_db: {
  default_status: "blacklisted",
  whitelist: [
    "01234567100000",
    "0123456700000001"
  ]
}
```

s13

Optional object allowing to configure the S13 options. It can contain the following objects:

server_addr

String. IP address and optional port of the EIR used for S13 interface. The default port is 3868.

bind_addr

Optional string. IP address and optional port on which the S13 SCTP connection is bound. The default address is the same as the first S1AP SCTP connection.

origin_realm

Optional string. Defines the string sent in the Origin-Realm AVP for S13 messages. Default is set to `mnc<MNC>.mcc<MCC>.3gppnetwork.org`.

origin_host

Optional string. Defines the string sent in the Origin-Host AVP for S13 messages. Default is set to `epc.mnc<MNC>.mcc<MCC>.3gppnetwork.org`.

transaction_timeout

Optional integer (range 1 to 15000, default = 2000). Defines the timeout in milliseconds for a transaction with the EIR.

watchdog_duration

Optional integer (range 0 to 36000000, default = 30000). Tw watchdog timer in milliseconds to send the Diameter Device Watchdog Request message. The value 0 deactivates the watchdog.

5.2.8 SGs options**sgs**

Optional object allowing to configure the SGs options. It can contain the following objects:

csfb_allowed

Optional boolean (default = false). If set to true, Circuit Switched Fall back procedures are accepted, otherwise they are rejected.

lac

Optional integer (default = 0x001). Defines the Location Area Identifier of the MSC/VLR to connect to.

server_addr

String. IP address and optional port of the MSC/VLR used for SGs interface. The default port is 29118.

bind_addr

Optional string. IP address and optional port on which the SGs SCTP connection is bound. The default address is the same as the first S1AP SCTP connection.

5.2.9 SBc options**sbcap_bind_addr**

Optional string. IP address and optional port on which the SBc SCTP connection is bound. The default address is the same as the first S1AP SCTP connection.

5.2.10 LCS options**lcs**

Optional object allowing to configure the LCSAP interface options. It can contain the following objects:

server_addr

Optional string. IP address and optional port of the e-SMLC location server. The default port is 9082. If not present, the test e-SMLC located in the MME will be used.

bind_addr

Optional string. IP address and optional port on which the LCSAP SCTP connection is bound. The default address is the same as the first S1AP SCTP connection.

local_e_smlc

Optional object used to configure an internal test e-SMLC located in the MME. The internal e-SMLC is not a real location server. It provides some apis useful for testing. It can contain the following objects:

lpp_test

Optional boolean (default = false). If set to true, the e-SMLC will use the LPP protocol. Otherwise, LPPa protocol will be used.

e_smlc_id

Optional integer in range 0 to 255 (default = 0). e-SMLC identifier.

- transaction_id**
Optional integer in range 0 to 32767 (default = 0). Transaction identifier to use in the message LPPA E-CIDMeasurementInitiationRequest.
- cell_radius**
Optional integer (default = 5000). Cell radius in meters. This parameter is used to calculate the RSTD uncertainty in the LPP assistance data.
- periodic_meas**
Optional boolean (default = false). Indicates if periodic measurements will be requested in the message LPPA E-CIDMeasurementInitiationRequest.
- meas_period**
Optional integer in range 0 to 12 (default = 0), corresponding to [ms120, ms240, ms480, ms640, ms1024, ms2048, ms5120, ms10240, min1, min6, min12, min30, min60] according to the definition of MeasurementPeriodicity in 3GPP TS 36.455 MeasurementPeriodicity. Gives the measurement period to set in the message LPPA E-CIDMeasurementInitiationRequest in case of periodic measurements.
- meas_id**
Optional integer in range 1 to 15 (default = 1). Gives the value to use in the parameter E-SMLC-UE-Measurement-ID in the message LPPA E-CIDMeasurementInitiationRequest.
- otdoa_assistance_data**
Optional nullable property. Gives the name of the file containing the ASN.1 description of the OTDOA-ProvideAssistanceData LPP IE. If present, it will be used in LPP ProvideAssistanceData. To remove an existing configuration, use: otdoa_assistance_data: null.
See 3GPP TS 37.355 chapter 6.5.1.1 OTDOA Assistance Data.
- nr_tdoa_assistance_data**
Optional nullable property. Gives the name of the file containing the ASN.1 description of the NR-DL-TDOA-ProvideAssistanceData-r16 LPP IE. If present, it will be used in the message LPP Provide Assistance Data. To remove an existing configuration, use: nr_tdoa_assistance_data: null.
See 3GPP TS 37.355 chapter 6.5.10.1 NR DL-TDOA Assistance Data.
- gnss_assistance_data**
Optional nullable property. Gives the name of the file containing the ASN.1 description of the A-GNSS-ProvideAssistanceData LPP IE. If present, it will be used in LPP ProvideAssistanceData. To remove an existing configuration, use: gnss_assistance_data: null.
See 3GPP TS 37.355 chapter 6.5.2.1 GNSS Assistance Data.
- lpp_methods**
Optional array. Gives the list of the LPP methods that shall be used by the eSMLC during the LPP location procedure. If not present, the eSMLC shall use all the LPP methods supported by the UE. An element of the array is an enumeration: "ecid", "otdoa", "nr_ecid", "nr_tdoa", "gnss".
- autonomous_mode**
Optional boolean (default = FALSE). On receipt of a location request (See [location_req], page 118):

- if `autonomous_mode` is set to `TRUE`, the e-SMLC will autonomously initiate the LPPa and LPP procedures necessary to obtain or estimate the position of the UE (See [location_req], page 118). `enb` must be configured.
 - if it is set to `FALSE`, the e-SMLC will send LPP requestCapabilities or LPPa E-CIDMeasurementInitiationRequest (See [location_req], page 118), and the APIs described in the sections LPPa and LPP must be invoked to continue the localization procedure.
- enb** Object. Mandatory if `autonomous_mode` is set to `TRUE`. Contains the following properties:
- enb_plmn** String (5 or 6 digits). eNB PLMN.
 - enb_type** Optional enumeration: `macro`, `short_macro`, `long_macro` or `home` (default = `macro`).
 - enb_id** Integer. eNodeB global identifier. `enb_plmn`, `enb_type` and `enb_id` are used to identify the eNB to which the message LPPa OTDOAInformationRequest will be sent.

5.2.11 N12 options

n12

Optional object allowing to configure the N12 interface options. It can contain the following objects:

- api_root** Optional string. According to the definition in 3GPP TS 29.501, `api_root` is in the form: `<scheme>://<host>:<port>`, where `<scheme>` is "http" or "https". This field shall be present if an external AUSF is used.
- transaction_timeout** Optional integer (range 1 to 15000, default = 3000). Defines the timeout in milliseconds for a transaction with the AUSF.
- bind_addr** Optional string. IP address and optional port on which the N12 TCP connection is bound. The default address is the same as the first GTP-U connection.

5.2.12 N13 options

n13

Optional object allowing to configure the N13 interface options. It can contain the following objects:

- api_root** Optional string. According to the definition in 3GPP TS 29.501, `api_root` is in the form: `<scheme>://<host>:<port>`, where `<scheme>` is "http" or "https". This field shall be present if an internal AUSF is used with an external UDM.
- transaction_timeout** Optional integer (range 1 to 15000, default = 3000). Defines the timeout in milliseconds for a transaction between the AUSF and UDM.
- bind_addr** Optional string. IP address and optional port on which the N13 TCP connection is bound. The default address is the same as the first GTP-U connection.

5.2.13 N8 options

n8

Optional object allowing to configure the N8 interface options. It can contain the following objects:

api_root Optional string. According to the definition in 3GPP TS 29.501, `api_root` is in the form: `<scheme>://<host>:<port>`, where `<scheme>` is "http" or "https". This field shall be present if an external UDM is used.

transaction_timeout Optional integer (range 1 to 15000, default = 3000). Defines the timeout in milliseconds for a transaction with the UDM.

bind_addr Optional string. IP address and optional port on which the N8 TCP connection is bound. The default address is the same as the first GTP-U connection.

5.2.14 N17 options

n17

Optional object allowing to configure the N17 interface options. It can contain the following objects:

api_root Optional string. According to the definition in 3GPP TS 29.501, `api_root` is in the form: `<scheme>://<host>:<port>`, where `<scheme>` is "http" or "https". This field shall be present if an external 5G-EIR is used.

transaction_timeout Optional integer (range 1 to 15000, default = 3000). Defines the timeout in milliseconds for a transaction with the 5G-EIR.

bind_addr Optional string. IP address and optional port on which the N17 TCP connection is bound. The default address is the same as the first GTP-U connection.

The list of IMEIs is configured in the EIR options. See [me-db], page 53.

5.2.15 N20 options

n20

Optional object allowing to configure the N20 interface options. It can contain the following objects:

api_root Optional string. According to the definition in 3GPP TS 29.501, `api_root` is in the form: `<scheme>://<host>:<port>`, where `<scheme>` is "http" or "https". This field shall be present if an external SMSF is used.

transaction_timeout Optional integer (range 1 to 15000, default = 3000). Defines the timeout in milliseconds for a transaction with the SMSF.

`bind_addr`

Optional string. IP address and optional port on which the N20 TCP connection is bound. The default address is the same as the first GTP-U connection.

5.2.16 N50 options

`n50`

Optional object allowing to configure the N50 interface options. It can contain the following objects:

`transaction_timeout`

Optional integer (range 1 to 15000, default = 3000). Defines the timeout in milliseconds for a transaction with the CBC.

`server_bind_addr`

Optional string. IP address and optional port on which the N50 TCP connection is bound. The default address is the same as the first GTP-U connection and the default TCP port is 5558.

5.2.17 NL1 options

`nl1`

Optional object allowing to configure the NL1 interface options. It can contain the following objects:

`api_root` Optional string. According to the definition in 3GPP TS 29.501, `api_root` is in the form: `<scheme>://<host>:<port>`, where `<scheme>` is "http" or "https". This field shall be present if an external LMF is used.

`transaction_timeout`

Optional integer (range 1 to 1800000, default = 20000). Defines the timeout in milliseconds for a transaction with the LMF.

`server_bind_addr`

Optional string. IP address and optional port on which the NL1 TCP connection is bound. The default address is the same as the first GTP-U connection and the default TCP port is 5560.

`lmf_cfg` Optional object used to configure an internal test LMF located in the AMF. The internal LMF is not a real location server. It provides some apis useful for testing. It can contain the following objects:

`lpp_test` Optional boolean (default = false). If set to true, the LMF will use the LPP protocol. Otherwise, NRPPa protocol will be used.

`transaction_id`

Optional integer in range 0 to 32767 (default = 0). Transaction identifier to use in the message NRPPa E-CIDMeasurementInitiationRequest.

`tai_list_5gs`

Optional array of objects (up to 65535). 5GS TAI List to set in the parameter `taiList` as defined in 3GPP TS 29.518 chapter 6.1.6.2.9. See [tai_list_5gs], page 49.

`tai_list` Optional array of objects (up to 65535). TAI List to set in the parameter `taiList` as defined in 3GPP TS 29.518 chapter 6.1.6.2.9. See [tai_list], page 49.

rat_selector

Optional enumeration: nr, eutra, both (default = both). Value of the parameter ratSelector as defined in 3GPP TS 29.518 chapter 6.1.6.2.9.

cell_radius

Optional integer (default = 5000). Cell radius in meters. This parameter is used to calculate the RSTD uncertainty in the LPP assistance data.

periodic_meas

Optional boolean (default = false). Indicates if periodic measurements will be requested in the message NRPPa E-CIDMeasurementInitiationRequest.

meas_period

Optional integer in range 0 to 14 (default = 0), corresponding to [ms120, ms240, ms480, ms640, ms1024, ms2048, ms5120, ms10240, min1, min6, min12, min30, min60, ms20480, ms40960] according to the definition on MeasurementPeriodicity in 3GPP TS 38.455.

meas_id Optional integer in range 1 to 256 (default = 1). LMF-UE-Measurement-ID in the message NRPPa E-CIDMeasurementInitiationRequest.

otdoa_assistance_data

Optional nullable property. Gives the name of the file containing the ASN.1 description of the OTDOA-ProvideAssistanceData LPP IE. If present, it will be used in LPP ProvideAssistanceData. To remove an existing configuration, use: otdoa_assistance_data: null.
See 3GPP TS 37.355 chapter 6.5.1.1 OTDOA Assistance Data.

nr_tdoa_assistance_data

Optional nullable property. Gives the name of the file containing the ASN.1 description of the NR-DL-TDOA-ProvideAssistanceData-r16 LPP IE. If present, it will be used in the message LPP Provide Assistance Data. To remove an existing configuration, use: nr_tdoa_assistance_data: null.
See 3GPP TS 37.355 chapter 6.5.10.1 NR DL-TDOA Assistance Data.

gnss_assistance_data

Optional nullable property. Gives the name of the file containing the ASN.1 description of the A-GNSS-ProvideAssistanceData LPP IE. If present, it will be used in LPP ProvideAssistanceData. To remove an existing configuration, use: gnss_assistance_data: null.
See 3GPP TS 37.355 chapter 6.5.2.1 GNSS Assistance Data.

lpp_methods

Optional array. Gives the list of the LPP methods that shall be used by the LMF server during the LPP location procedure. If not present, the LMF shall use all the LPP methods supported by the UE. An element of the array is an enumeration: "ecid", "otdoa", "nr_ecid", "nr_tdoa", "gnss".

autonomous_mode

Optional boolean (default = FALSE). On receipt of a location request (See [nr_location_req], page 121):

- if `autonomous_modes` set to `TRUE`, the LMF will autonomously initiate the NRPPa and LPP procedures necessary to obtain or estimate the position of the UE (See `[nr_location_req]`, page 121). See `[ran_node_id_list]`, page 60, must be configured.
- if it is set to `FALSE`, the LMF will send LPP requestCapabilities or NRPPa E-CIDMeasurementInitiationRequest (See `[nr_location_req]`, page 121), and the APIs described in the sections NRPPa and LPP must be invoked to continue the localization procedure.

`ran_node_id_list`

Optional array of objects. List of the global RAN node ID to send in the HTTP POST request as defined in 3GPP TS 29.518 chapter 6.1.6.2.9. See `[ran_node_id]`, page 50.

5.2.18 N5 options

The 5QI QoS settings are configured in the `[Rx options]`, page 51.

`n5`

Optional object allowing to configure the address and port of the PCF server. It can contain the following objects:

`server_bind_addr`

Optional string. IP address and optional port on which the N5 TCP connection is bound. The default address is the same as the first GTP-U connection and the default TCP port is 5561.

5.2.19 N62 options

`n62`

Optional object allowing to configure the address and port of the AAnF server. It can contain the following objects:

`server_bind_addr`

Optional string. IP address and optional port on which the N62 TCP connection is bound. The default address is the same as the first GTP-U connection and the default TCP port is 5563.

5.2.20 CP-EDT options

`cp_edt` Optional object allowing to configure CP-EDT options. It can contain the following objects:

`mode` Optional enumeration: disabled, forced, automatic. Default value is automatic. If disabled is set: CP-EDT feature is disabled in the core network. If forced is set: CP-EDT is processed by the core network whatever the NAS RAI received with UL data. If automatic is set: if NAS RAI indicates that downlink data is expected, CP-EDT is processed by the core network. Otherwise connection establishment is requested by the core network.

`max_dl_len_nb`

Optional integer. Default value is 85. Largest DL transport block (including user payload and MAC/RLC/RRC/NAS overhead) allowed without fallback to RRC connection establishment in NB-IoT.

5.2.21 ePDG options

epdg Optional object allowing to configure ePDG options. It shall contain the following objects:

bind_addr

IP address on which the SWu connection is bound.

private_key

String. Defines the ePDG private key filename.

certificate

String. Defines the ePDG certificate filename. The default files `epdg_private_key.pem` and `epdg_cert.pem` are built for ePDG FQDN `"epdg.epc.mnc001.mcc001.pub.3gppnetwork.org"` following the procedure described below. For another ePDG FQDN, these files shall be re-built by setting the FQDN in `subjectAltName` field. Procedure to generate and check the private key file `epdg_private_key.pem` and the certificate file `epdg_cert.pem`:

```
openssl genrsa -out ca.key 2048
openssl req -new -x509 -days 3650 -key ca.key -out ca.crt
openssl req -newkey rsa:2048 -nodes -keyout epdg_private_key.pem
openssl x509 -req -extfile <(printf "subjectAltName=DNS:epdg.epc.mnc001.mcc001.pub.3gppnetwork.org") -key epdg_private_key.pem -out epdg_cert.pem
openssl x509 -in epdg_cert.pem -text
openssl rsa -in epdg_private_key.pem -text
```

esp_duration

Optional integer in range 10 to 5*3600 (default = 300). Gives the duration in seconds of the ESP-Sa.

ike_duration

Optional integer in range 20 to 48*3600 (default = 24*3600). Gives the duration in seconds of the IKE-Sa.

omit_auth_in_first_auth_rsp

Optional boolean (default = false). If set, configures the EPDG to not send the AUTH payload in the first IKE_AUTH exchange.

p_cscf_ipv4_address_attribute_type

Optional integer in range 16384-32767.

Allows to define a private type value for the attribute P-CSCF IPv4 address.

p_cscf_ipv6_address_attribute_type

Optional integer in range 16384-32767.

Allows to define a private type value for the attribute P-CSCF IPv6 address.

eap_aka_fast_reauth

Optional object. If present, indicates that EAP-AKA fast re-authentication is supported. It contains the following parameter:

max_count

Integer in range 1 to 65535. Gives the maximum count of fast re-authentication procedures before performing a full EAP-AKA authentication.

ike_encryption_algo_list

Optional list of IKE-Sa supported encryption algorithms "aes-cbc-128" (AES CBC 128 bits key length), "aes-cbc-192" (AES CBC 192 bits key length), "aes-cbc-256" (AES CBC 256 bits key length), "aes-gcm-128-16" (AES GCM 128 bits key length and 16 bytes ICV), "aes-gcm-256-16" (AES GCM 256 bits key length and 16 bytes ICV), "3des", "blowfish", "aes-ctr-128" (AES CTR 128 bits key length), "aes-ctr-192" (AES CTR 192 bits key length), and "aes-ctr-256" (AES CTR 256 bits key length) ordered from most preferred to least preferred.

Default value is ["aes-cbc-128", "aes-cbc-192", "aes-cbc-256", "aes-gcm-128-16", "aes-gcm-256-16", "3des", "blowfish", "aes-ctr-128", "aes-ctr-192", "aes-ctr-256"].

ike_integrity_algo_list

Optional list of IKE-Sa supported integrity algorithms "hmac-sha-1-96", "hmac-sha-1-160", "hmac-sha-256-128", "hmac-sha-384-192", "hmac-sha-512-256", "hmac-md5-96", "hmac-md5-128" and "aes-cmac-96" ordered from most preferred to least preferred.

Default value is ["hmac-sha-1-96", "hmac-sha-1-160", "hmac-sha-256-128", "hmac-sha-384-192", "hmac-sha-512-256", "hmac-md5-96", "hmac-md5-128", "aes-cmac-96"];

ike_prf_list

Optional list of IKE-Sa supported pseudo-random functions "prf-hmac-sha1", "prf-hmac-sha2-256", "prf-hmac-sha2-384", "prf-hmac-sha2-512", "prf-hmac-md5" and "prf-aes128-xcbc" ordered from most preferred to least preferred.

Default value is ["prf-hmac-sha1", "prf-hmac-sha2-256", "prf-hmac-sha2-384", "prf-hmac-sha2-512", "prf-hmac-md5", "prf-aes128-xcbc"].

ike_dh_group_list

Optional list of IKE-Sa supported Diffie-Hellman groups "group_1", "group_2", "group_5", "group_14", "group_15", "group_16", "group_17", "group_18", "group_19", "group_22", "group_23" and "group_24" ordered from most preferred to least preferred.

Default value is ["group_5", "group_14", "group_15", "group_16", "group_17", "group_18", "group_19", "group_22", "group_23", "group_24"].

esp_encryption_algo_list

Optional list of ESP-Sa supported encryption algorithms "null", "aes-cbc-128" (AES CBC 128 bits key length), "aes-cbc-192" (AES CBC 192 bits key length), "aes-cbc-256" (AES CBC 256 bits key length), "aes-gcm-128-16" (AES GCM 128 bits key length and 16 bytes ICV), "aes-gcm-192-16" (AES GCM 192 bits key length and 16 bytes ICV), "aes-gcm-256-16" (AES GCM 256 bits key length and 16 bytes ICV), "3des", "blowfish", "aes-ctr-128" (AES CTR 128 bits key length), "aes-ctr-192" (AES CTR 192 bits key length), "aes-ctr-256" (AES CTR 256 bits key length), "encr-null-auth-aes-gmac-128" (ENCR_NULL_AUTH_AES_GMAC 128 bits key length), "encr-null-auth-aes-gmac-192" (ENCR_NULL_AUTH_AES_GMAC 192 bits key length) and "encr-null-auth-aes-gmac-256" (ENCR_NULL_AUTH_AES_GMAC 256 bits key length) ordered from

most preferred to least preferred.

Default value is ["null", "aes-cbc-128", "aes-cbc-192", "aes-cbc-256", "aes-gcm-128-16", "aes-gcm-192-16", "aes-gcm-256-16", "3des", "blowfish", "aes-ctr-128", "aes-ctr-192", "aes-ctr-256", "encr-null-auth-aes-gmac-128", "encr-null-auth-aes-gmac-192", "encr-null-auth-aes-gmac-256"].

esp_integrity_algo_list

Optional list of ESP-Sa supported integrity algorithms "null", "hmac-sha-1-96", "hmac-sha-1-160", "hmac-sha-256-128", "hmac-sha-384-192", "hmac-sha-512-256", "hmac-md5-96", "hmac-md5-128" and "aes-cmac-96" ordered from most preferred to least preferred.

Default value is ["null", "hmac-sha-1-96", "hmac-sha-1-160", "hmac-sha-256-128", "hmac-sha-384-192", "hmac-sha-512-256", "hmac-md5-96", "hmac-md5-128", "aes-cmac-96"].

esp_dh_group_list

Optional list of ESP-Sa supported Diffie-Hellman groups "none", "group_1", "group_2", "group_5", "group_14", "group_15", "group_16", "group_17", "group_18", "group_19", "group_22", "group_23" and "group_24" ordered from most preferred to least preferred.

This list is used for rekeying ESP-Sa. Default value is ["none", "group_5", "group_14", "group_15", "group_16", "group_17", "group_18", "group_19", "group_22", "group_23" and "group_24"].

dpd_timer_value

Optional integer in range 5 to 300 (default = 300). Gives the "dead peer detection" timer value in seconds.

mobike Optional boolean (default = true). Indicates MOBIKE support.

ike_generate_error

Optional object. Allows to ignore a message or generate an error during an IKE exchange.

It contains the following objects:

exchange String. Gives the exchange to ignore or on which the error must be sent. Possible values are "none", "ike_sa_init", "ike_auth_step1", "ike_auth_step2", "ike_auth_step3", "dpd", "mobike".

reject_notify_list

Optional array. If absent, the message received during the exchange will be ignored.

If present, the message received during the exchange will be rejected, and each element of the array describes a Notify payload to insert in the response. Each element of the array contains:

error Optional integer. Gives the value of 'Notify Message Type' to send in the Notify payload rejecting the exchange.

data Optional hexadecimal string. May be present if error is present.

Gives the Notification Data to send in the Notify payload.

- ttnl** Optional integer. If set, `ike_generate_error` is applied `ttnl` times. If not set, `ike_generate_error` is applied until it is modified.
- apn_list** Optional array of 0 to 16 strings. Gives the list of the APNs for which `ike_generate_error` applies. If `apn_list` is empty or not present, `ike_generate_error` applies to any APN.

Example:

```
ike_generate_error: {
  exchange: "ike_auth_step1"
  reject_notify_list: [
    {
      error: 9002,
      data: "1234",
    },
    {
      error: 9003,
      data: "5678",
    }
  ],
  ttnl: 1
}
```

idr_for_emergency Optional string (default = "EMERGENCY"). Name of the APN for emergency.

dont_fragment Optional boolean (default = true) used to enable/disable the fragmentation of the ESP packets.

additional_ue_auth_type Optional enumeration: `none`, `pap`, `chap` (default = `none`). Defines the authentication mechanism used for the additional UE authentication as described in 3GPP 33.402 chapters 6.5.2 and 6.5.3.

If present and not set to `none`:

MULTIPLE_AUTH_SUPPORTED notify payload will be sent by the ePDG in the message IKE_SA_INIT response (see 3GPP 33.402 chapter 6.5), and username (See [username], page 32) and password (See [password], page 33) must be configured in `pdn_list` to perform the authentication and authorization for a given APN.

5.2.22 MBS options

mbs Optional object allowing to configure MBS parameters. It contains the following parameters:

nmbsmf Object allowing to configure the MB-SMF server on the NMBSMF interface. It contains the following parameters:

server_bind_addr

Optional string. IP address and optional port on which the nmbsmf TCP connection is bound. The default address is

the same as the first GTP-U connection and the default TCP port is 5581.

gtp_addr	String or array of 2 strings. IP address (and optional port) to which the GTP-U packets are sent. It is normally a multicast address. Several sessions can share the same IP address and have a different gtp_teid . Use an array to configure both an IPv4 (first entry in the array) and an IPv6 (second entry in the array) address if you use both IPv4 and IPv6 addresses in the ltemme gtp_addr top level parameter.														
ingress_tunnel_addr	Optional string or array of 2 strings. IP address used by the MB-UPF to receive the MBS data packets when unicast transfer is used over N6mb. The same address and different ports are used for different MBS sessions.														
tmgi_lifetime	Optional integer (range: 1 to MAX_INT, default value: 300). Lifetime of an allocated TMGI in seconds.														
service_area	Optional object. Describes the service area supported by the MBS server. it contains the following parameters: <table> <tr> <td>cell_list</td><td>Optional array of objects. Service area cell list. Each element contains the following parameters: <table> <tr> <td>plmn</td><td>String. PLMN.</td></tr> <tr> <td>cell_id</td><td>Integer. 36 bits NR cell identity.</td></tr> </table> </td></tr> <tr> <td>tai_list</td><td>Optional array of objects. Service area tracking area list. Each element contains the following parameters: <table> <tr> <td>plmn</td><td>String. PLMN.</td></tr> <tr> <td>tac</td><td>Integer. Tracking area code.</td></tr> </table> </td></tr> </table>	cell_list	Optional array of objects. Service area cell list. Each element contains the following parameters: <table> <tr> <td>plmn</td><td>String. PLMN.</td></tr> <tr> <td>cell_id</td><td>Integer. 36 bits NR cell identity.</td></tr> </table>	plmn	String. PLMN.	cell_id	Integer. 36 bits NR cell identity.	tai_list	Optional array of objects. Service area tracking area list. Each element contains the following parameters: <table> <tr> <td>plmn</td><td>String. PLMN.</td></tr> <tr> <td>tac</td><td>Integer. Tracking area code.</td></tr> </table>	plmn	String. PLMN.	tac	Integer. Tracking area code.		
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plmn	String. PLMN.														
tac	Integer. Tracking area code.														
broadcast_sessions	Optional array of objects allowing to configure the MBS sessions at the client side on the NMBSMF interface. Each element describes a broadcast session and contains the following parameters: <table> <tr> <td>tmgi</td><td>Optional object. Session TMGI. Used to configure a session context for which the TMGI is already allocated. It contains the following parameters: <table> <tr> <td>plmn</td><td>String. PLMN.</td></tr> <tr> <td>service_id</td><td>Integer. 24 bits service identity.</td></tr> </table> </td></tr> <tr> <td>nid</td><td>Optional object. See [nid], page 12.</td></tr> <tr> <td>snssai</td><td>Object. S-NSSAI value. it contains the following parameters: <table> <tr> <td>sst</td><td>Integer (range 0-255). Slice Service Type.</td></tr> <tr> <td>sd</td><td>Optional integer (range 0-0xFFFFFE). Slice Differentiator.</td></tr> </table> </td></tr> </table>	tmgi	Optional object. Session TMGI. Used to configure a session context for which the TMGI is already allocated. It contains the following parameters: <table> <tr> <td>plmn</td><td>String. PLMN.</td></tr> <tr> <td>service_id</td><td>Integer. 24 bits service identity.</td></tr> </table>	plmn	String. PLMN.	service_id	Integer. 24 bits service identity.	nid	Optional object. See [nid], page 12.	snssai	Object. S-NSSAI value. it contains the following parameters: <table> <tr> <td>sst</td><td>Integer (range 0-255). Slice Service Type.</td></tr> <tr> <td>sd</td><td>Optional integer (range 0-0xFFFFFE). Slice Differentiator.</td></tr> </table>	sst	Integer (range 0-255). Slice Service Type.	sd	Optional integer (range 0-0xFFFFFE). Slice Differentiator.
tmgi	Optional object. Session TMGI. Used to configure a session context for which the TMGI is already allocated. It contains the following parameters: <table> <tr> <td>plmn</td><td>String. PLMN.</td></tr> <tr> <td>service_id</td><td>Integer. 24 bits service identity.</td></tr> </table>	plmn	String. PLMN.	service_id	Integer. 24 bits service identity.										
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sst	Integer (range 0-255). Slice Service Type.														
sd	Optional integer (range 0-0xFFFFFE). Slice Differentiator.														

service_area	Optional object. It describes the service area in case of location dependent MBS session. See [service_area], page 65.
associated_session_id	Optional hexadecimal string. Associated session identity.
autostart	Optional boolean (default = true). Indicates if the session is started automatically or if it must be started manually.
user_plane_desc	Optional object allowing to configure the parameters used for the transfer of the data to the MBS server.
use_n6mb_multicast	Optional boolean (default = FALSE). Indicates if multicast transport is used over N6mb.
multicast_dest_addr	Optional string. Shall be present if sim is TRUE (to build the RTP packets headers), or use_n6mb_multicast is FALSE (to fill the ssm field in the MBS session create request over the Nmbsmf interface). Source-specific multicast (SSM) destination address of the RTP flow sent by the MBS client (used for both unicast and multicast transport over N6mb). See in RFC 4607 the addresses range reserved for use by source-specific applications and protocols: IANA assigned 232/8 for IPv4 and FF3x::/96 for IPv6.
source_addr	Optional string. Shall be present if sim is TRUE (to build the RTP packets headers), or use_n6mb_multicast is FALSE (to fill the ssm field in the MBS session create request over the Nmbsmf interface). IP unicast address used as source address of the RTP flow sent by the MBS client (used for both unicast and multicast transport over N6mb).
source_bind_addr	Optional string. IP source address and port used by the MBS client as bind address for the sending of MBS packets. If not present, the first address in the ltemme gtp_addr is used.
sim	Optional boolean (default = false). If true, RTP packets are generated and sent to the MBS server using a RTP payload of rtp_payload_len bytes and a bitrate of bitrate .

rtp_payload_len	Optional integer. RTP payload length in bytes (default = 1460 for IPv4 or 1440 for IPv6). RTP payload length of the RTP packets sent to the MBS server.
bitrate	Integer. Bitrate in bit/s of the generated RTP stream. The bitrate includes the size of the IP, UDP and RTP headers.
qos_flows	Array of objects. Each element contains the following parameters:
5qi	Integer. 5G QoS Identifier of the QoS flow.
priority_level	Optional integer (range: 1 to 15, default 15). ARP priority level.
pre_emption_capability	Optional enumeration: <code>shall_not_trigger_pre_emption</code> or <code>may_trigger_pre_emption</code> (default <code>shall_not_trigger_pre_emption</code>).
pre_emption_vulnerability	Optional enumeration: <code>not_pre_emptable</code> or <code>pre_emptable</code> (default <code>not_pre_emptable</code>).
5qi_qos	Optional object. See [5QI QoS], page 36.
gbr	Optional object. See [GBR], page 36.
flow_descs	Optional array of strings. Contains the flow description for the MBS Downlink IP flow. Each element contains an <code>IpFilterRule</code> according to clause 4.3 of IETF RFC 6733. Refer to clause 5.3.8 of 3GPP TS 29.214 for encoding.

6 Remote API

You can access LTEMME via a remote API.

Protocol used is WebSocket as defined in RFC 6455 (<https://tools.ietf.org/html/rfc6455>).

Note that Origin header is mandatory for the server to accept connections. This behavior is determined by the use of `noopll` library. Any value will be accepted.

To learn how to use it, you can refer to our the following tutorial (<https://tech-academy.amarisoft.com/RemoteAPI.html>).

6.1 Messages

Messages exchanged between client and LTEMME server are in strict JSON format.

Each message is represented by an object. Multiple message can be sent to server using an array of message objects.

Time and delay values are floating number in seconds.

There are 4 types of messages:

- Request

Message sent by client.

Common definition:

message String. Represent type of message. This parameter is mandatory and depending on its value, other parameters will apply.

message_id

Optional any type. If set, response sent by the server to this message will have same message_id. This is used to identify response as WebSocket does not provide such a concept.

start_time

Optional float. Represent the delay before executing the message. If not set, the message is executed when received.

absolute_time

Optional boolean (default = false). If set, **start_time** is interpreted as absolute.

You can get current clock of system using **time** member of any response.

standalone

Optional boolean (default = false). If set, message will survive WebSocket disconnection, else, if socket is disconnected before end of processing, the message will be cancelled.

loop_count

Optional integer (default = 0, max = 1000000). If set, message will be repeated **loop_count** time(s) after **loop_delay** (From message beginning of event). Response will have a **loop_index** to indicate iteration number.

loop_delay

Optional number (min = 0.1, max = 86400). Delay in seconds to repeat message from its **start_time**. Mandatory when **loop_count** is set > 0.

group_id Optional integer. If set, this number may be used with **cancel** message

- Notification

For some API, intermediate message may be sent by server before reception of response. Common definition:

message String. Same as request.

message_id

Optional any type. Same as in request.

time Number representing time in seconds of the message start, relative to the beginning of the process.
Useful to send command with absolute time.

notification

String. Notification purpose

utc Number representing UTC seconds (local clock) when the response has been generated.

- Response

Message sent by server after any request message has been processed. Common definition:

message String. Same as request.

message_id

Optional any type. Same as in request.

time Number representing time in seconds of the message start, relative to the beginning of the process.
Useful to send command with absolute time.

utc Number representing UTC seconds (local clock) when the response has been generated.

absolute_time

Optional string. If **absolute_time** has been set and message is reaching LTEMME too late, this field is present and set to **late**.

cancel Optional boolean. Set to true if request has been cancelled by **cancel** remote API.

- Events

Message sent by server on its own initiative. Common definition:

message String. Event name.

time Number representing time in seconds.
Useful to send command with absolute time.

6.2 Startup

When WebSocket connections is setup, LTEMME will send a first message with name set to `com_name` and type set to `MME`.

If authentication is not set, message will be `ready`:

```
{
  "message": "ready",
  "type": "MME",
  "name": <com_name>,
  "version": <software version>,
  "product": <Amarisoft product name (optional)>
}
```

If authentication is set, message will be `authenticate` :

```
{
  "message": "authenticate",
  "type": "MME",
  "name": <com_name>,
  "challenge": <random challenge>
}
```

To authenticate, the client must answer with a `authenticate` message and a `res` parameter where:

```
res = HMAC-SHA256( "<type>:<password>:<name>", "<challenge>" )
```

`res` is a string and HMAC-SHA256 refers to the standard algorithm (<https://en.wikipedia.org/wiki/HMAC>)

If the authentication succeeds, the response will have a `ready` field set to `true`.

```
{
  "message": "authenticate",
  "message_id": <message id>,
  "ready": true
}
```

If authentication fails, the response will have an `error` field and will provide a new challenge.

```
{
  "message": "authenticate",
  "message_id": <message id>,
  "error": <error message>,
  "type": "MME",
  "name": <name>,
  "challenge": <new random challenge>
}
```

If any other message is sent before authentication succeeds, the error `"Authentication not done"` will be sent as a response.

6.3 Errors

If a message produces an error, response will have an error string field representing the error.

6.4 Sample nodejs program

You will find in this documentation a sample program: `ws.js`.

It is located in `doc` subdirectory.

This is a nodejs program that allow to send message to LTEMME.

It requires nodejs to be installed:

```
dnf install nodejs npm
npm install nodejs-websocket
```

Use relevant package manager instead of NPM depending on your Linux distribution.

Then simply start it with server name and message you want to send:

```
./ws.js 127.0.0.1:9000 '{"message": "config_get"}'
```

6.5 Common messages

`config_get`

Retrieve current config.

Response definition:

type	Always "MME"																														
name	String representing server name.																														
logs	Object representing log configuration. With following elements: <table> <tr> <td>layers</td><td>Object. Each member of the object represent a log layer configuration: <table> <tr> <td>layer name</td><td>Object. The member name represent log layer name and parameters are: <table> <tr> <td>level</td><td>See [log-options], page 10,</td></tr> <tr> <td>max_size</td><td>See [log-options], page 10,</td></tr> <tr> <td>key</td><td>See [log-options], page 10,</td></tr> <tr> <td>crypto</td><td>See [log-options], page 10,</td></tr> <tr> <td>payload</td><td>See [log-options], page 10,</td></tr> <tr> <td>verbose</td><td>Optional boolean. See [log-options], page 10,</td></tr> </table> </td></tr> </table> </td></tr> <tr> <td>count</td><td>Number. Number of bufferizer logs.</td></tr> <tr> <td>rotate</td><td>Optional number. Max log file size before rotation.</td></tr> <tr> <td>rotate_count</td><td>Optional number. Max log count before rotation.</td></tr> <tr> <td>path</td><td>Optional string. Log rotation path.</td></tr> <tr> <td>bcch</td><td>Boolean. True if BCCH dump is enabled (eNB only).</td></tr> <tr> <td>mib</td><td>Boolean. True if MIB dump is enabled (eNB only).</td></tr> <tr> <td>locked</td><td>Optional boolean. If <code>true</code>, logs configuration can't be changed with <code>config_set</code> API.</td></tr> </table>	layers	Object. Each member of the object represent a log layer configuration: <table> <tr> <td>layer name</td><td>Object. The member name represent log layer name and parameters are: <table> <tr> <td>level</td><td>See [log-options], page 10,</td></tr> <tr> <td>max_size</td><td>See [log-options], page 10,</td></tr> <tr> <td>key</td><td>See [log-options], page 10,</td></tr> <tr> <td>crypto</td><td>See [log-options], page 10,</td></tr> <tr> <td>payload</td><td>See [log-options], page 10,</td></tr> <tr> <td>verbose</td><td>Optional boolean. See [log-options], page 10,</td></tr> </table> </td></tr> </table>	layer name	Object. The member name represent log layer name and parameters are: <table> <tr> <td>level</td><td>See [log-options], page 10,</td></tr> <tr> <td>max_size</td><td>See [log-options], page 10,</td></tr> <tr> <td>key</td><td>See [log-options], page 10,</td></tr> <tr> <td>crypto</td><td>See [log-options], page 10,</td></tr> <tr> <td>payload</td><td>See [log-options], page 10,</td></tr> <tr> <td>verbose</td><td>Optional boolean. See [log-options], page 10,</td></tr> </table>	level	See [log-options], page 10,	max_size	See [log-options], page 10,	key	See [log-options], page 10,	crypto	See [log-options], page 10,	payload	See [log-options], page 10,	verbose	Optional boolean. See [log-options], page 10,	count	Number. Number of bufferizer logs.	rotate	Optional number. Max log file size before rotation.	rotate_count	Optional number. Max log count before rotation.	path	Optional string. Log rotation path.	bcch	Boolean. True if BCCH dump is enabled (eNB only).	mib	Boolean. True if MIB dump is enabled (eNB only).	locked	Optional boolean. If <code>true</code> , logs configuration can't be changed with <code>config_set</code> API.
layers	Object. Each member of the object represent a log layer configuration: <table> <tr> <td>layer name</td><td>Object. The member name represent log layer name and parameters are: <table> <tr> <td>level</td><td>See [log-options], page 10,</td></tr> <tr> <td>max_size</td><td>See [log-options], page 10,</td></tr> <tr> <td>key</td><td>See [log-options], page 10,</td></tr> <tr> <td>crypto</td><td>See [log-options], page 10,</td></tr> <tr> <td>payload</td><td>See [log-options], page 10,</td></tr> <tr> <td>verbose</td><td>Optional boolean. See [log-options], page 10,</td></tr> </table> </td></tr> </table>	layer name	Object. The member name represent log layer name and parameters are: <table> <tr> <td>level</td><td>See [log-options], page 10,</td></tr> <tr> <td>max_size</td><td>See [log-options], page 10,</td></tr> <tr> <td>key</td><td>See [log-options], page 10,</td></tr> <tr> <td>crypto</td><td>See [log-options], page 10,</td></tr> <tr> <td>payload</td><td>See [log-options], page 10,</td></tr> <tr> <td>verbose</td><td>Optional boolean. See [log-options], page 10,</td></tr> </table>	level	See [log-options], page 10,	max_size	See [log-options], page 10,	key	See [log-options], page 10,	crypto	See [log-options], page 10,	payload	See [log-options], page 10,	verbose	Optional boolean. See [log-options], page 10,																
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locked	Optional boolean. If <code>true</code> , logs configuration can't be changed with <code>config_set</code> API.																														

config_set

Change current config.
Each member is optional.
Message definition:

logs Optional object. Represent logs configuration. Same structure as config-get (See [config-get logs member], page 71).
All elements are optional.
Layer name can be set to **all** to set same configuration for all layers.
If set and logs are locked, response will have **logs** property set to **locked**.

relative_capacity

Optional integer (range 0 to 255). Set the MME or AMF relative capacity value used for MME or AMF load balancing in S1AP S1 Setup Response, MME Configuration Update, NGAP NG Setup Response and NGAP AMF Configuration Update messages.

attach_reject_error

Optional integer (range 0 to 255).
Force value of EMM reject cause in NAS attach reject message.

tracking_area_update_reject_error

Optional integer (range 0 to 255).
Force value of EMM reject cause in NAS tracking area update reject message.

service_reject_error

Optional integer (range 0 to 255).
Force value of EMM reject cause in NAS service reject message.

pdn_connect_reject_error

Optional integer (range 0 to 255).
Force value of ESM reject cause in NAS PDN connectivity reject message.

pdn_disconnect_reject_error

Optional integer (range 0 to 255).
Force value of ESM reject cause in NAS PDN disconnect reject message.

bearer_resource_allocation_reject_error

Optional integer (range 0 to 255).
Force value of ESM reject cause in NAS bearer resource allocation reject message.

bearer_resource_modification_reject_error

Optional integer (range 0 to 255).
Force value of ESM reject cause in NAS bearer resource modification reject message.

registration_initial_reject_error

Optional integer (range 0 to 255).

Force value of 5GMM reject cause in NAS registration reject message (for 5GS registration type 1 or 4).

registration_mobility_periodic_error

Optional integer (range 0 to 255).

Force value of 5GMM reject cause in NAS registration reject message (for 5GS registration type 2 or 3).

5gs_service_reject_error

Optional integer (range 0 to 255).

Force value of 5GMM reject cause in NAS service reject message.

pdu_session_establishment_reject_error

Optional integer (range 0 to 255).

Force value of 5GSM reject cause in NAS PDU session establishment reject message.

pdu_session_release_reject_error

Optional integer (range 0 to 255).

Force value of 5GSM reject cause in NAS PDU session release reject message.

pdu_session_modification_reject_error

Optional integer (range 0 to 255).

Force value of 5GSM reject cause in NAS PDU session modification reject message.

5gmm_dl_nas_transport_error

Optional integer (range 0 to 255).

Force value of 5GMM reject cause in NAS DL NAS transport message.

eps_user_unknown_reject_cause

Optional integer (range 0 to 255).

EMM cause sent in the NAS attach reject message when the IMSI is unknown in the HSS.

5gs_user_unknown_reject_cause

Optional integer (range 0 to 255).

5GMM cause sent in the NAS registration reject message when the SUPI is unknown in the UDM.

attach_reject_filter

Optional object. Represent UE to reject when trying to attach.

Each property name represent IMSI. Use of wildcard "*" with an IMSI prefix is allowed to match IMSI range (Ex: 0010112456*). If multiple

filters are matching, the one with the longest prefix will be used.
Each property value may be:

null	Removes redirection matching IMSI
integer	Defines redirection type as described in <i>rrc_redirect</i> eNB configuration.
string	Defines PLMN to redirect to
t3402	Optional integer. Value in seconds of the T3402 or T3502 timer. -1 means that the timer value is not transmitted in attach accept or TAU accept or registration accept so that the UE uses the default value (12 minutes).
t3412	Optional integer. Value in seconds of the T3412 (TAU update) timer. -1 means that the timer is deactivated.
t3412_low_priority	Optional integer. Value in seconds of the T3412 (TAU update) timer if the UE indicates NAS signalling low priority. -1 means that the timer is deactivated.
t3512	Optional integer. Value in seconds of the T3512 (periodic registration) timer. -1 means that the timer is deactivated. This is the value sent to the UE in NAS signalling, unless the UE is requesting the use of a longer timeout with requested T3512 value information element.
t3501	Optional integer in range 1-30 (default = 5). Value in seconds of the MANAGE UE POLICY COMMAND timer in the PCF.
n3gpp_dereg_timer	Optional integer. Value in seconds of the non-3GPP de-registration timer. This is the value sent to the UE in NAS signalling.
psm	Option boolean. If set to false, MME will ignore the PSM request sent by the UE.
mico_support	Optional boolean. If set to false, AMF will ignore the MICO request sent by the UE.
registration_area_alloc_ind	Optional integer. Sets the Registration Area Allocation Indication bit in the 5GMM MICO indication IE. 0 means 'all PLMN registration area not allocated' and 1 means 'all PLMN registration area allocated'.
sprt_support	Optional boolean. If set to true and if mico_support is set to true, the AMF will accept the use of the strictly periodic registration timer.
t3412_extended_forced	Optional integer. Value in seconds of the T3412 extended timer if UE uses PSM. If different from -1, the MME will ignore the value requested by the UE and will send this one instead.
force_t3412_extended_ie	Optional boolean. If set to false, the MME selects the greatest T3412 value between the one configured in the MME and the one requested by

the UE for PSM (unless `t3412_extended_forced` is set), and it does not send the T3412 extended IE if the value can be encoded as a GPRS timer IE. If set to true, the MME accepts a T3412 value requested by the UE smaller than the configured one, and the T3412 extended IE is always sent.

`requested_t3512_forced`

Optional integer. Value in seconds of the T3512 timer if UE uses MICO. If greater than -1, the AMF will ignore the value requested by the UE and will send this one instead. If set to -2, the AMF will accept a T3512 value requested by the UE smaller than the configured one.

`t3324_forced`

Optional integer. Value in seconds of the T3324 timer if UE uses PSM. If different from -1, the MME will ignore the value requested by the UE and will send this one instead. -2 means that the timer is deactivated.

`t3346` Optional integer. Value in seconds of the T3346 timer. The timer is transmitted in the reject messages if the EMM of 5GSM cause is #22 (congestion) and the value is not -1.

`t3442` Optional integer. Value in seconds of the T3442 timer.

`t3448` Optional integer. Value in seconds of the T3448 timer. The timer is transmitted if the value is different from -1 and the UE indicates its support in the UE network capability information element.

`t3460` Optional integer. Value in seconds of the T3460 or T3560 timer.

`t3460_wb_ce`

Optional integer. Value in seconds of the T3460 or T3560 timer for UE operating in WB-S1/CE or WB-N1/CE mode.

`lower_bound_timer`

Optional integer. The timer is transmitted in the reject messages if the EMM or 5GMM cause is #78 (PLMN not allowed to operate at the present UE location) and the value is not -1.

`t3560_ng_ran_sat`

Optional integer. Value in seconds of the T3560 timer for UE operating in NR(MEO) or NR(GEO) satellite RAT.

`5gmm_backoff_timer`

Optional integer. Value in seconds of the 5GMM DL NAS transport back-off timer. The timer is transmitted if the value is not -1. -2 means that the timer is deactivated.

`edrx` Option boolean. If set to false, MME will ignore the eDRX request sent by the UE.

`edrx_ptw_wb_s1`

Optional integer. 4 bits Paging Time Window length for WB-S1 UEs as defined in 3GPP TS 24.008 chapter 10.5.5.32.

`edrx_ptw_nb_s1`

Optional integer. 4 bits Paging Time Window length for NB-S1 UEs as defined in 3GPP TS 24.008 chapter 10.5.5.32.

<code>edrx_ptw_nr</code>	Optional integer (0 to 31, default = 3). 8 bits Paging Time Window length for NR connected to 5GCN UEs as defined in 3GPP TS 24.008 chapter 10.5.5.32.
<code>edrx_cycle_forced</code>	Optional integer. 4 bits E-UTRAN eDRX cycle length duration as defined in 3GPP TS 24.008 chapter 10.5.5.32. If different from -1, the MME will ignore the value requested by the UE and will send this one instead.
<code>gwus_support</code>	Optional boolean. Group WUS support.
<code>gwus_prob_forced</code>	Optional enumeration: -1, 0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100. Force group WUS paging probability for all the UEs. The MME will ignore the value requested by the UE and will send this one instead.
<code>ims_vops_eps</code>	Optional boolean. Set the IMS voice over PS session in S1 mode supported bit of the EPS network feature support field in the NAS attach accept message (VoLTE).
<code>ims_vops_5gs_3gpp</code>	Optional boolean. Set the IMS voice over PS session over 3GPP access indicator of the 5GS network feature support IE of the NAS registration access message. See 3GPP TS 24.501 table 9.11.3.5.1.
<code>ims_vops_5gs_n3gpp</code>	Optional boolean. Set the IMS voice over PS session over non-3GPP access indicator of the 5GS network feature support IE of the NAS registration access message. See 3GPP TS 24.501 table 9.11.3.5.1.
<code>emc_bs</code>	Optional boolean. Set the emergency bearer services in S1 mode supported bit of the EPS network feature support field in the NAS attach accept message (VoLTE, Release 9).
<code>emc</code>	Optional integer. Set the emergency service support indicator for 3GPP access bits of the 5GS network feature support IE in the NAS registration accept message.
<code>emc_n3gpp</code>	Optional boolean. Set the emergency service support indicator for non-3GPP access bits of the 5GS network feature support IE in the NAS registration accept message. See 3GPP TS 24.501 table 9.11.3.5.1.
<code>emf</code>	Optional integer. Set the emergency service fallback indicator for 3GPP access bits of the 5GS network feature support IE in the NAS registration accept message.
<code>epc_lcs</code>	Optional boolean. Set the Location services indicator via EPC supported bit of the EPS network feature support field in the NAS attach accept message.
<code>5gs_sms_over_nas</code>	Optional boolean. Defines if 5GC should indicate the support of SMS over NAS in the 5GMM registration accept message, if the UE indicated its support in the 5GMM registration request message.

emergency_number_list

Optional nullable array of objects. Defines a list of emergency numbers to be sent to the UE in the NAS Attach Accept, Tracking Area Update Accept or Registration Accept messages.

To clear the current emergency number list, the array must be empty. To configure an emergency number list, each object must contain the following parameters:

category Integer. Bitmask of the category bits as defined in 3GPP TS 24.008 table 10.5.135d (bit 1: police, bit 2: ambulance, bit 3: fire brigade, bit 4: marine guard, bit 5: mountain rescue, bit 6: manually initiated eCall, bit 7: automatically initiated eCall).

digits String. Emergency number.

The configuration may be removed using:
emergency_number_list: null

extended_emergency_number_list

Optional nullable object. Defines a list of extended emergency numbers to be sent to the UE in the NAS Attach Accept, Tracking Area Update Accept or Registration Accept messages.

The object must contain the following parameters:

validity Optional enumeration (country or plmn).
Validity of the extended emergency number list.
Must be present if emergency_numbers is not empty.

emergency_numbers

Array of objects. To clear the current extended emergency number list, the array must be empty.

To configure an extended emergency number list, each object must contain the following parameters:

digits String. Emergency number.

sub_services

Optional string. Emergency number sub-services.

The configuration may be removed using:
extended_emergency_number_list: null

cp_ciot_opt

Optional boolean. If true, enable control plane CIoT optimization (if supported by the UE).

attach_without_pdn

Optional boolean. If true, enable attach without PDN functionality (if supported by the UE).

fifteen_bearers

Optional boolean. If true, enable the use of 15 EPS radio bearers (if supported by the UE).

attach_result_mode

Optional string. Set attach result of attach accept message.
Can be:

auto This is standard LTE behavior.

- eps_only** If set and UE is sending combined EPS/IMSI attach, the MME will answer with EPS only in attach accept message (EMM cause will be CS domain not available).
- combined** If set and UE is sending EPS only attach, the MME will answer with combined in attach accept message.
- combined_sms_only**
If set the UE will only be attached for EPS and SMS.
- additional_update_result**
Optional integer (range -1 to 3). Set the value of additional update result in NAS attach accept and tracking area update accept messages. If set to -1, the additional update result won't be set.
- sms_services_status**
Optional integer (range -1 to 7). Set the value of the SMS service status information element in the NAS attach accept and tracking area update accept messages. It is sent if the value is different from -1 and the UE performed an EPS service with SMS only procedure and does not have SMS service allowed.
- network_policy**
Optional integer (range -1 to 15). Set the value of the network policy information element described in 3GPP TS 24.301 chapter 9.9.3.52. The value -1 means that the IE is not transmitted.
- authentication_mode**
Optional string. Set NAS authentication procedure behavior. Can be:
- auto** The MME or AMF performs authentication procedure unless the UE is already successfully authenticated.
 - force** The MME or AMF forces a new NAS authentication procedure even if the Attach Request or Registration Request was already successfully authenticated
 - skip** The MME or AMF skips the NAS authentication procedure and uses EIA0/EEA0 or 5G-IA0/5G-EA0 algorithms. This needs to be supported on UE side also.
- dummy_authentication_autn_mac**
Optional boolean. If set to true, the network will send an invalid AUTN MAC value in the NAS authentication request message.
- authenticate_known_emergency_supl**
Optional boolean. If set to true, the network will authenticate known IMSI/SUPI during an emergency registration procedure and reject the UE if it fails.
- secure_known_emergency_supl**
Optional boolean. If set to true and if **authenticate_known_emergency_supl** is set to true, known IMSI/SUPI during an emergency registration procedure will use the integrity and ciphering algorithm matching the network and UE configuration. Otherwise it will use a null integrity and ciphering algorithm.

restrict_ec_wb
Optional boolean. Sets restriction on enhanced coverage for WB-S1 and WB-N1 UEs.

restrict_ec_nb
Optional boolean. Sets restriction on enhanced coverage for NB-S1 and NB-N1 UEs.

skip_smc_proc
Optional boolean. If set to true, the MME or AMF will not perform a NAS security mode control procedure and will send all messages as plain. This needs to be supported on UE side also.

coarse_location_information_support
Optional boolean. If set to true, and if a NB-IoT UE accessing the core network via a NTN cell supports the feature, the MME will request the coarse location information in the NAS security mode command message.

force_identity_request
Optional boolean. If set to true, the network will perform a NAS identity request procedure even if the GUTI in the attach request or the 5G-GUTI in the initial registration request is already known.

allocate_new_guti_in_tau
Optional boolean. If set to true, a new GUTI is allocated during each Tracking Area Update procedure. If set to false, the UE keeps the GUTI allocated during the attach procedure.

force_guti_in_tau
Optional boolean. If set to true, GUTI IE will be systematically present in Tracking Area Update Accept message even if it does not change.

emm_procedure_filter
Optional object. Allows to define the MME behavior for a list of EMM procedures.
Each property name represents an EMM procedure. The ones currently supported are **attach**, **tracking_area Updating**, **detach**, **service_request**, **identity**, **authentication**, **security_mode_control** and **nas_transport**.
Each property value is an object containing the following fields:

action Enumeration (**treat** (UE message is processed), **ignore** (UE message is ignored) or **reject** (UE message is rejected)).

t1 Optional integer. If set, the **reject** or **ignore** filter is applied **t1** times. If not set, the filter is applied until it is modified.

send_status_on_reject
Optional boolean. If set and if **action** is set to **reject** an EMM status message is sent.

Example:

```
emm_procedure_filter: {
  attach: {
    action: "treat"
```

```

    },
    service_request: {
      action: "reject",
      ttl: 1
    }
  }
}

```

5gmm_procedure_filter

Optional object. Allows to define the AMF behavior for a list of 5GMM procedures.

Each property name represents a 5GMM procedure. The ones currently supported are `registration_initial`, `registration_initial_with_security_protection`, `registration_mobility_periodic`, `service_request`, `identity`, `authentication`, `security_mode_control`, `generic_ue_update_command`, `nas_transport_n1_sm`, `nas_transport_sms` and `deregistration`.

Each property value is an object containing the following fields:

action Enumeration (`treat` (UE message is processed), `ignore` (UE message is ignored) or `reject` (UE message is rejected)).

ttl Optional integer. If set, the `reject` or `ignore` filter is applied `ttl` times. If not set, the filter is applied until it is modified.

send_status_on_reject

Optional boolean. It set and if `action` is set to `reject` a 5GMM status message is sent.

Note that `nas_transport_n1_sm` filter must be used together with the `apply_nas_transport_n1_sm_filter` DNN parameter.

Example:

```

"5gmm_procedure_filter": {
  registration_initial: {
    action: "treat"
  },
  service_request: {
    action: "reject",
    ttl: 1
  }
}

```

nr_support

Optional boolean. Set it to true to enable Dual Connectivity with NR support.

dcnr_implicit_support

Optional boolean. If set to true, the MME will not send the 2nd byte of the EPS network feature support IE because of DCNR. Can be useful to test the UE behavior.

ecc_params

Optional object. Set the ECC network configuration for the SUPI protection and de-concealment of the SUCI. Applicable to 5GC only. It contains the following objects:

A	Optional array of objects. Set the home network private key for profile A protection scheme.
	home_nw_private_key String. Set the home network private key;
	home_nw_key_id Optional integer in range 0 to 255 (default = 1). Set the home network key identifier.
B	Optional array of objects. Set the home network private key for profile B protection scheme.
	home_nw_private_key String. Set the home network private key;
	home_nw_key_id Optional integer in range 0 to 255 (default = 2). Set the home network key identifier.
nssai_inclusion_mode	
Applicable to 5GC only. Optional enumeration (none, A, B, C, D). NSSAI inclusion mode value to send in message Registration accept.	
cp_edt	Optional object allowing to configure CP-EDT options. It can contain the following objects:
mode	Optional enumeration: disabled, forced, automatic. Default value is automatic. If disabled is set: CP-EDT feature is disabled in the core network. If forced is set: CP-EDT is processed by the core network whatever the NAS RAI received with UL data. If automatic is set: if NAS RAI indicates that downlink data is expected, CP-EDT is processed by the core network. Otherwise connection establishment is requested by the core network.
max_dl_len_nb	
Optional integer. Default value is 85. Largest DL transport block (including user payload and MAC/RLC/RRC/NAS overhead) allowed without fallback to RRC connection establishment in NB-IoT.	
epdg	Applicable to EPC only. Optional object allowing to configure ePDG options. It may contain the following object:
esp_duration	
Optional integer in range 10 to 5*3600). Gives the duration in seconds of the ESP-Sa.	
ike_duration	
Optional integer in range 20 to 48*3600. Gives the duration in seconds of the IKE-Sa.	
ike_generate_error	
Optional object. Allows to ignore a message or generate an error during an IKE exchange. See [ike_generate_error], page 63.	
mobike	Optional boolean. Indicates MOBIKE support.

dont_fragment	Optional boolean (default = TRUE) used to enable/disable the fragmentation of the ESP packets.
additional_ue_auth_type	Optional enumeration: none , pap , chap . Defines the authentication mechanism used for the additional UE authentication as described in 3GPP 33.402 chapters 6.5.2 and 6.5.3.
pdn_list	Optional array of objects. Each object can contain the following properties:
apn	String. APN allowing to identify the PDN or PDU session to be modified.
operator	Optional array of objects. Each element defines an operator reserved container in protocol configuration. Properties of each element:
id	Integer. Container identifier, must be between 0xff00 and 0xffff as defined in 3GPP TS 24.008.
plmn	String. PLMN info of container.
value	String. Value to send in hexadecimal string format.
force	Optional boolean. If true, container will be sent event without request (false by default).
serving_plmn_rate_control	Optional integer (range 0 to 65535). Defines the serving PLMN rate control IE content when PDN is used with control plane CIoT optimization only. If the value configured is less than 10, the IE is not transmitted.
apn_rate_control_params	Optional object. If defined, and if the UE indicates APN rate control parameters support in its protocol configuration options, the following parameters will be sent in Core Network protocol configuration options:
additional_exception_report	Boolean. Indicates if exception reports are allowed once the limit is reached.
ul_time_unit	Enumeration: unrestricted , minute , hour , day or week .
max_ul_rate	Integer (range from 0 to 16777215). Number of messages allowed to be sent per ul_time_unit .
additional_apn_rate_control_exception_data_params	Optional object. If defined, and if the UE indicates additional APN rate control for exception data parameters support in its protocol configuration options, the following

parameters will be sent in Core Network protocol configuration options:

`ul_time_unit`

Enumeration: `unrestricted`, `minute`, `hour`, `day` or `week`.

`max_ul_rate`

Integer (range from 0 to 65535). Number of messages allowed to be sent per `ul_time_unit`.

`backoff_timer`

Optional integer. Value in seconds of the T3396/T3584/T3585 timers. The timer is transmitted in the ESM and 5GSM reject messages if the value is not -1. -2 means that the timer is deactivated.

`re_attempt_ind`

Optional integer (range -1 to 255). Value of octet 3 of the Re-attempt indicator information element, as specified in 3GPP TS 24.301 chapter 9.9.4.13A and 3GPP TS 24.501 chapter 9.11.4.17. The value -1 means that the information element is not sent.

`ipv6_router_lifetime`

Optional integer (range 0 to 65535). IPv6 Router Advertisement router lifetime in seconds.

`ipv6_valid_lifetime`

Optional integer. IPv6 Router Advertisement valid lifetime in seconds.

`ipv6_pref_lifetime`

Optional integer (default is `ipv6_valid_lifetime` value). IPv6 Router Advertisement preferred lifetime in seconds. Must not be greater than `ipv6_valid_lifetime`.

`ipv6_onlink_flag`

Optional boolean. Defines IPv6 Router Advertisement on-link flag state.

`ipv6_managed_addr_config_flag`

Optional boolean. Defines IPv6 Router Advertisement managed address configuration flag state.

`ipv6_other_config_flag`

Optional boolean. Defines IPv6 Router Advertisement other configuration flag state.

`ipv6_ra_transmission_interval`

Optional integer (range -1 to 1800). Time in seconds between 2 periodical multicast Router Advertisement transmission, once the initial 3 transmissions have been performed after opening the PDN or PDU session. The value -1 means that no multicast transmission is done at all (including the 3 initial ones). The value 0 means that periodical transmission is deactivated.

- ipv6_send_dns_in_ra**
Optional boolean. Defines whether Router Advertisement message should contain the configured IPv6 DNS servers address or not.
- ipv6_drop_rs**
Optional boolean. Defines whether the incoming Router Solicitation messages should be dropped by the MME and UPF or not.
- automatic_release**
Optional boolean. If set, when the last associated dedicated EPS bearer is released the MME releases the PDN connection. With 5GS, when the last non default QoS flow is released, the SMF releases the PDU session.
- automatic_release_timeout**
Optional integer. Delay in ms before releasing the PDN connection or PDU session when **automatic_release** is set to true.
- allow_multiple_pdn_connections**
Optional boolean. If set, a UE can create multiple PDN connections to this APN.
- ue_initiated_modification**
Optional boolean. If set, the UE can request the modification of a bearer, otherwise the request is rejected.
- ip_src_violation_limit**
Optional integer. If greater than -1, the MME or UPF checks the IP source address of uplink packets. When **ip_src_violation_limit** packets are received, the PDN or PDU session is released. The value 0 means that the packets are dropped without triggering a release.
- integrity_protection**
Optional enumeration (disabled, preferred, required, default = disabled). Defines whether integrity should be used for the PDN connection / PDU session or not.
For EPC, if the value is set to **preferred**, the EPC will activate integrity protection based on the UE capabilities. If set to **required**, and if the UE does not support integrity protection, the request will be rejected with ESM cause 30. For 5GC, if the value is set to **preferred**, the 5GC will activate integrity protection based on the UE capabilities and the configured PDU session AMBR. If set to **required**, and if the UE does not support integrity protection for the bitrate configured in the PDU session AMBR, the request will be rejected with 5GSM error cause #82.
- dns_addr** Optional string or array of strings. IPv4 or IPv6 addresses of the DNS servers. Use an empty array to remove any previously configured DNS servers.

p_cscf_addr

Optional string or array of strings. IPv4 or IPv6 addresses of the P-CSCF servers (VoLTE). Use an empty array to remove any previously configured P-CSCF servers.

The following parameters are applicable to EPC only:

esm_procedure_filter

Optional object. Allows to define the MME behavior for a list of ESM procedures.

Each property name represents an ESM procedure. The ones currently supported are **pdn_connectivity**, **pdn_disconnect**, **bearer_resource_allocation** and **bearer_resource_modification**.

Each property value is an object containing the following fields:

action Enumeration (**treat** (UE message is processed), **ignore** (UE message is ignored) or **reject** (UE message is rejected)).

t1 Optional integer. If set, the **reject** of **ignore** filter is applied **t1** times. If not set, the filter is applied until it is modified.

send_status_on_reject

Optional boolean. It set and if **action** is set to **reject** an ESM status message is sent.

ignore does not apply to procedure **pdn_connectivity** performed during the attach procedure.

Example:

```
esm_procedure_filter: {
  pdn_connectivity: {
    action: "treat"
  },
  bearer_resource_allocation: {
    action: "reject",
    t1: 1
  }
}
```

The following parameters are applicable to 5GC only:

5gsm_procedure_filter

Optional object. Allows to define the SMF behavior for a list of 5GSM procedures.

Each property name represents a 5GSM procedure. The ones currently supported are **pdu_session_establishment**, **pdu_session_release** and **pdu_session_modification**.

Each property value is an object containing the following fields:

action Enumeration (**treat** (UE message is processed), **ignore** (UE message is ignored) or **reject** (UE message is rejected)).

ttl Optional integer. If set, the **reject** or **ignore** filter is applied **ttl** times. If not set, the filter is applied until it is modified.

send_status_on_reject
Optional boolean. If set and if **action** is set to **reject** a 5GSM status message is sent.

Example:

```
"5gsm_procedure_filter": {
  pdu_session_establishment: {
    action: "treat"
  },
  pdu_session_modification: {
    action: "reject",
    ttl: 1
  }
}
```

always_on

Optional enumeration (auto, required, not_allowed, default = auto). Defines the always-on behavior for the PDU session. If the value is set to **auto**, the 5GC will follow whatever is requested by the UE. If the value is set to **required**, the 5GC will always set the PDU session as always-on required. If the value is set to **not_allowed**, the 5GC will always set the PDU session as always-on not allowed.

confidentiality_protection

Optional enumeration (disabled, required). Defines if confidentiality must be used for the PDU session or not.

apply_nas_transport_n1_sm_filter

Optional boolean. Indicates whether the 5GMM procedure filter **nas_transport_n1_sm** should apply to this DNN or not.

eps_5gs_interworking

Optional boolean. If set to true, interworking between EPS and 5GS is allowed for this APN/DNN. Otherwise it is forbidden.

5gsm_congestion_re_attempt_ind

Optional integer (range -1 to 255). Value of octet 3 of the Re-attempt indicator information element, as specified in 3GPP TS 24.501 chapter 9.11.4.21. The value -1 means that the information element is not sent.

rq_timer

Optional integer. Value in seconds of the RQ timer for the PDU session, sent to the UE if it supports reflective QoS feature. -2 means that the timer is deactivated.

log_get Get logs.

This API has a per connection behavior. This means that the response will depend

on previous calls to this API within the same WebSocket connection.

In practice, logs that have been provided in a response won't be part of subsequent request unless connection is reestablished. To keep on receiving logs, client should send a new `log_get` request as soon as the previous response has been received.

If a request is sent before previous request has been replied, previous request will be replied right now without considering specific min/max/timeout conditions.

Message definition:

<code>min</code>	Optional number (default = 1). Minimum amount of logs to retrieve. Response won't be sent until this limit is reached (Unless timeout occurs).
<code>max</code>	Optional number (default = 4096). Maximum logs sent in a response.
<code>timeout</code>	Optional number (default = 1). If at least 1 log is available and no more logs have been generated for this time, response will be sent.
<code>allow_empty</code>	Optional boolean (default = false). If set, response will be sent after timeout, event if no logs are available.
<code>rnti</code>	Optional number. If set, send only logs matching rnti.
<code>ue_id</code>	Optional number. If set, send only logs with matching ue_id.
<code>layers</code>	Optional Object. Each member name represents a log layer and values must be string representing maximum level. See [log_options], page 10. If <i>layers</i> is not set, all layers level will be set to <i>debug</i> , else it will be set to <i>none</i> . Note also the logs is also limited by general log level. See [log_options], page 10.
<code>short</code>	Optional boolean (default = false). If set, only first line of logs will be dumped.
<code>headers</code>	Optional boolean. If set, send log file headers.
<code>start_timestamp</code>	Optional number. Is set, filter logs older than this value in milliseconds.
<code>end_timestamp</code>	Optional number. Is set, filter logs more recent than this value in milliseconds.
<code>max_size</code>	Optional number (default = 1048576, i.e. 1MB). Maximum size in bytes of the generated JSON message. If the response exceeds this size, the sending of logs will be forced independently from other parameters.

Response definition:

<code>logs</code>	Array. List of logs. Each item is a an object with following members:
<code>data</code>	Array. Each item is a string representing a line of log.
<code>timestamp</code>	Number. Milliseconds since January 1st 1970. Not present if <code>com_log_us</code> is set in configuration.

	timestamp_us	Number. Microseconds since January 1st 1970. Only present if com_log_us is set in configuration.
	layer	String. Log layer.
	level	String. Log level: <i>error</i> , <i>warn</i> , <i>info</i> or <i>debug</i> .
	dir	Optional string. Log direction: <i>UL</i> , <i>DL</i> , <i>FROM</i> or <i>TO</i> .
	ue_id	Optional number. UE_ID.
	cell	Optional number (only for PHY layer logs). Cell ID.
	rnti	Optional number (only for PHY layer logs). RNTI.
	frame	Optional number (only for PHY layer logs). Frame number (Subframe is decimal part).
	channel	Optional string (only for PHY layer logs). Channel name.
	src	String. Server name.
	idx	Integer. Log index.
	headers	Optional array. Array of strings.
	discontinuity	Optional number. If set, this means some logs have been discarded due to log buffer overflow.
	microseconds	Optional boolean. Present and set to true if com_log_us is set in configuration file.
log_set	Add log. Message definition:	
	log	Optional string. Log message to add. If set, <i>layer</i> and <i>level</i> are mandatory.
	layer	String. Layer name. Only mandatory if <i>log</i> is set.
	level	String. Log level: <i>error</i> , <i>warn</i> , <i>info</i> or <i>debug</i> . Only mandatory if <i>log</i> is set.
	dir	Optional string. Log direction: <i>UL</i> , <i>DL</i> , <i>FROM</i> or <i>TO</i> .
	ue_id	Optional number. UE_ID.
	flush	Optional boolean (default = false). If set, flushes fog file.
	rotate	Optional boolean (default = false). If set, forces log file rotation.
	cut	Optional boolean (default = false). If set, forces log file reset.
log_reset	Resets logs buffer.	
license	Retrieves license file information. Response definition:	
	products	String. List of products, separated by commas.
	user	String. License username.

validity	String. License end of validity date.
id	Optional string. License ID.
id_type	Optional string. License ID type. Can be <code>host_id</code> or <code>dongle_id</code>
uid	Optional string. License unique ID.
filename	Optional string. License filename.
server	Optional string. License server URL.
server_id	Optional string. License server ID.
quit	Terminates ltemme.
help	Provides list of available messages in <i>messages</i> array of strings and events to register in <i>events</i> array of strings.
stats	Report statistics for LTEMME. Every time this message is received by server, statistics are reset. Warning, calling this message from multiple connections simultaneously will modify the statistics sampling time. Response definition: cpu Object. Each member name defines a type and its value cpu load in % of one core. instance_id Number. Constant over process lifetime. Changes on process restart. counters Object. List of counters, with following sub members: messages Object. Each member name is the message name and its value is its occurrence. To get list of message, type <i>cevent help msg</i> in LTEMME monitor. errors Object. Each member name is the error name and its value is its occurrence. To get list of message, type <i>cevent help error</i> in LTEMME monitor. emm_registered_ue_count Integer. Number of UEs in EMM-REGISTERED or 5GMM-REGISTERED state. s1_connections Array of objects. List of S1AP connection between eNBs and MME. Each object contains the following fields: plmn String. PLMN of the Global eNB ID. enb_id_type String (macro, home, short_macro or long_macro). Type of identifier of the Global eNB ID. enb_id Integer. Identifier of the Global eNB ID. ip_addr String. IP address and port of the eNB.

ta_list	Array of objects. List of the Tracking Areas served by the eNB. Each object contains the following fields:
plmn	String. PLMN of Tracking Area.
tac	Integer. Tracking Area Code.
emm_connected_ue_count	Integer. Number of UEs in EMM-CONNECTED state for this S1AP connection.
ng_connections	Array of objects. List of NGAP connection between RANs and AMF. Each object contains the following fields:
plmn	String. PLMN of the Global RAN ID.
ran_id_type	String (gNB, ng-eNB or N3IWF). Type of identifier of the Global RAN ID.
ran_id	Integer. Identifier of the Global RAN ID.
ip_addr	String. IP address and port of the RAN.
ta_list	Array of objects. List of the Tracking Areas served by the RAN. Each object contains the following fields:
plmn	String. PLMN of Tracking Area.
tac	Integer. Tracking Area Code.
cn_connected_ue_count	Integer. Number of UEs in 5GMM-CONNECTED state for this NGAP connection.
pdn_list	Array of objects. List of the APNs/DNNs configured. Each object contains the following fields:
access_point_name	String. APN / DNN.
ul_bytes	Integer. Number of UL bytes received since the last call on the same socket.
dl_bytes	Integer. Number of DL bytes received since the last call on the same socket.
users	Object. It contains the following fields:
total	Integer. Total number of entries in the user database.
max_registered	Integer. Number of different users having registered.
gtp_tx_bitrate	Optional integer. This field will be filled when multiple calls on the same socket are done and represents the GTP payload bitrate (bits/seconds) sent to RAN and is equivalent to IP traffic. The bitrate is computed using the delay between two calls.
gtp_rx_bitrate	Optional integer. This field will be filled when multiple calls on the same socket are done and represents the GTP payload bitrate (bits/seconds)

received from RAN and is equivalent to IP traffic.

The bitrate is computed using the delay between two calls.

ip_tx_bitrate

Optional integer. This field will be filled when multiple calls on the same socket are done and represents the IP payload bitrate (bits/seconds) sent to tun interfaces.

The bitrate is computed using the delay between two calls.

ip_rx_bitrate

Optional integer. This field will be filled when multiple calls on the same socket are done and represents the IP payload bitrate (bits/seconds) received from tun interfaces.

The bitrate is computed using the delay between two calls.

register Register client for messages generated by LTEMME. Message definition:

register Optional string or array of string. List of messages to register to.
Can be `registration`, `registration_reject`, `non_ip_data`,
`generic_nas_transport`, `5gs_nas_transport`, `eps_bearer_`
`notification`, `qos_flow_notification`

unregister

Optional string or array of string. List of messages to unregister.

Can be `registration`, `registration_reject`, `non_ip_data`,
`generic_nas_transport`, `5gs_nas_transport`, `eps_bearer_`
`notification`, `qos_flow_notification`

ipsec Report ipsec SAs.
For ePDG

Response definition:

SAs Array. List of object representing a security association with following definition:

type String. IP version, can be IPv4 or IPv6.

dir String. Direction, can be `in` or `out`.

spi Number. SPI.

ue_id Number. Associated `ue_id`.

mode String. ESP type, can be `tunnel` or `transport`

src String. Source IP address.

dst String. Destination IP address.

tun_src Optional string. Tunnel source IP address.

tun_dst Optional string. Tunnel destination IP address.

src_prefix

Number. Source network prefix.

dst_prefix

Number. Destination network prefix.

authent_key

String. Authentication key in hexadecimal form (Empty string authentication is disabled).

	cipher_key	String. Ciphering key in hexadecimal form (Empty string ciphering is disabled).
cancel	Cancel pending remote API requests. Message definition:	
	group_id	Optional integer. If set, cancellation will be applied to all messages having same group_id , otherwise, all pending messages will be cancelled.

6.6 LTE messages

ue_get	Get UE informations. Message definition:	
	imsi	Optional string. If set, retrieve only information from UE with matching IMSI.
	nai	Optional string. Not applicable to 4G UEs. May be present only if imsi is absent. If set, retrieve only information from UE with matching NAI.
	imei	Optional string (14 or 15 digits). If set, retrieve only information from UE with matching IMEI.
	type	Optional enumeration (3gpp, n3gpp, both). Default value is both. Only display a UE connected to a RAN with matching type.
	mme_ue_id	Optional integer. If set, retrieve only information from UE connected to EPC with matching MME UE id. The imsi , nai , imei and type parameters are ignored.
	amf_ue_id	Optional integer. If set, retrieve only information from UE connected to 5CC with matching AMF UE id. The imsi , nai , imei and type parameters are ignored.
	radio_capabilities	Optional boolean. If set, provides radio_capabilities in response.
	Response definition:	
	ue_list	Array of current UEs. Each element has the following definition:
	rat_type	Enumeration (LTE, NB-IOT, NR or NON_3GPP). RAT currently used by the UE.
	imsi	Optional string. IMSI.
	nai	Optional string. Network specific identifier-based SUPI.
	imeisv	String. IMEISV.
	m_tmsi	Optional string. M-TMSI. Present for UEs connected to EPC.
	5g_tmsi	Optional string. 5G-TMSI. Present for UEs connected to 5GC.
	tac	Integer. Last known tracking area code.
	tac_plmn	String. Last known tracking area PLMN.

<code>ue_aggregate_max_bitrate_dl</code>	Number. UE aggregate maximum bitrate for downlink.				
<code>ue_aggregate_max_bitrate_ul</code>	Number. UE aggregate maximum bitrate for uplink.				
<code>registered</code>	Boolean. True if UE is currently registered to the network.				
<code>t3412</code>	Optional integer. T3412 timer in seconds. Only present if the UE connected to EPC is registered to the network.				
<code>t3324</code>	Optional integer. T3324 timer in seconds. Only present if the UE connected to EPC is registered to the network and PSM is activated, or if the UE connected to 5GC is registered to the network and MICO is activated.				
<code>edrx</code>	Optional object. eDRX configuration. Only present if the UE is registered to the network and eDRX is activated. The object has the following definition: <table> <tr> <td><code>paging_time_window</code></td><td>Integer. 4 bits or 8 bits Paging Time Window length as defined in 3GPP TS 24.008 chapter 10.5.5.32</td></tr> <tr> <td><code>cycle</code></td><td>Integer. 4 bits E-UTRAN or NR eDRX cycle length duration as defined in 3GPP TS 24.008 chapter 10.5.5.32.</td></tr> </table>	<code>paging_time_window</code>	Integer. 4 bits or 8 bits Paging Time Window length as defined in 3GPP TS 24.008 chapter 10.5.5.32	<code>cycle</code>	Integer. 4 bits E-UTRAN or NR eDRX cycle length duration as defined in 3GPP TS 24.008 chapter 10.5.5.32.
<code>paging_time_window</code>	Integer. 4 bits or 8 bits Paging Time Window length as defined in 3GPP TS 24.008 chapter 10.5.5.32				
<code>cycle</code>	Integer. 4 bits E-UTRAN or NR eDRX cycle length duration as defined in 3GPP TS 24.008 chapter 10.5.5.32.				
<code>t3512</code>	Optional integer. T3512 timer in seconds. Only present if the UE connected to 5GC is registered to the network.				
<code>enb_plmn</code>	Optional string. eNB PLMN. This field would only be present if the UE connected to EPC is still in connected mode.				
<code>enb_id</code>	Optional integer. eNB id. This field would only be present if the UE connected to EPC is still in connected mode.				
<code>enb_ue_id</code>	Optional integer. eNB UE id. This field would only be present if the UE connected to EPC is still in connected mode.				
<code>mme_ue_id</code>	Optional integer. MME UE id. This field would only be present if the UE connected to EPC is still in connected mode.				
<code>ran_plmn</code>	Optional string. RAN PLMN. This field would only be present if the UE connected to 5GC is still in connected mode.				
<code>ran_id</code>	Optional integer. RAN id. This field would only be present if the UE connected to 5GC is still in connected mode.				
<code>ran_ue_id</code>	Optional integer. RAN UE id. This field would only be present if the UE connected to 5GC is still in connected mode.				

amf_ue_id	Optional integer. AMF UE id. This field would only be present if the UE connected to 5GC is still in connected mode.																				
ecgi	Optional object. E-UTRAN Cell Global Identity. This field would only be present if the UE is connected via an eNB or ng-eNB. It contains the following fields: <table> <tr> <td>plmn</td><td>String. PLMN Identifier.</td></tr> <tr> <td>cell_id</td><td>Integer. E-UTRAN Cell Identity Cell Identity.</td></tr> </table>	plmn	String. PLMN Identifier.	cell_id	Integer. E-UTRAN Cell Identity Cell Identity.																
plmn	String. PLMN Identifier.																				
cell_id	Integer. E-UTRAN Cell Identity Cell Identity.																				
ncgi	Optional object. NR Cell Global Identity. This field would only be present if the UE is connected via a gNB. It contains the following fields: <table> <tr> <td>plmn</td><td>String. PLMN Identifier.</td></tr> <tr> <td>cell_id</td><td>Integer. NR Cell Identity Cell Identity.</td></tr> </table>	plmn	String. PLMN Identifier.	cell_id	Integer. NR Cell Identity Cell Identity.																
plmn	String. PLMN Identifier.																				
cell_id	Integer. NR Cell Identity Cell Identity.																				
nid	Optional integer. NID. This field would only be present if the UE is connected via a gNB and NID is used.																				
n3iwf	Optional object. NR Cell Global Identity. This field would only be present if the UE is connected via a N3IWF. It contains the following fields: <table> <tr> <td>ue_addr</td><td>String. UE IP address and UDP source port.</td></tr> </table>	ue_addr	String. UE IP address and UDP source port.																		
ue_addr	String. UE IP address and UDP source port.																				
bearers	Array. List of connected default bearers or PDU sessions. Each object has the following definition: <table> <tr> <td>erab_id</td><td>Optional integer. EPS Bearer ID. Present UEs connected to EPC.</td></tr> <tr> <td>pdu_session_id</td><td>Optional integer. 5GS PDU session ID. Present for UEs connected to 5GC.</td></tr> <tr> <td>sst</td><td>Optional integer. Slice Service Type. Present for UEs connected to 5GC.</td></tr> <tr> <td>sd</td><td>Optional integer. Slice Differentiator. Can be present for UEs connected to 5GC.</td></tr> <tr> <td>qos_flow_id</td><td>Optional integer. 5GS QoS flow ID. Present for UEs connected to 5GC.</td></tr> <tr> <td>ip</td><td>String. IPv4 address.</td></tr> <tr> <td>ipv6</td><td>String. Global IPv6 prefix.</td></tr> <tr> <td>ul_total_bytes</td><td>Number. Total uplink transferred bytes.</td></tr> <tr> <td>dl_total_bytes</td><td>Number. Total downlink transferred bytes.</td></tr> <tr> <td>apn</td><td>String. Access point name.</td></tr> </table>	erab_id	Optional integer. EPS Bearer ID. Present UEs connected to EPC.	pdu_session_id	Optional integer. 5GS PDU session ID. Present for UEs connected to 5GC.	sst	Optional integer. Slice Service Type. Present for UEs connected to 5GC.	sd	Optional integer. Slice Differentiator. Can be present for UEs connected to 5GC.	qos_flow_id	Optional integer. 5GS QoS flow ID. Present for UEs connected to 5GC.	ip	String. IPv4 address.	ipv6	String. Global IPv6 prefix.	ul_total_bytes	Number. Total uplink transferred bytes.	dl_total_bytes	Number. Total downlink transferred bytes.	apn	String. Access point name.
erab_id	Optional integer. EPS Bearer ID. Present UEs connected to EPC.																				
pdu_session_id	Optional integer. 5GS PDU session ID. Present for UEs connected to 5GC.																				
sst	Optional integer. Slice Service Type. Present for UEs connected to 5GC.																				
sd	Optional integer. Slice Differentiator. Can be present for UEs connected to 5GC.																				
qos_flow_id	Optional integer. 5GS QoS flow ID. Present for UEs connected to 5GC.																				
ip	String. IPv4 address.																				
ipv6	String. Global IPv6 prefix.																				
ul_total_bytes	Number. Total uplink transferred bytes.																				
dl_total_bytes	Number. Total downlink transferred bytes.																				
apn	String. Access point name.																				

	dedicated Array of objects. Each object represents a dedicated bearer or non default QoS flow defined as follow: erab_id Optional integer. EPS Bearer ID. Present for UEs connected to EPC. qos_flow_id Optional integer. 5GS QoS flow ID. Present for UEs conencted ot 5GC. ul_total_bytes Number. Total uplink transferred bytes. dl_total_bytes Number. Total downlink transferred bytes. radio_capabilities GSER string. UE radio access capabilities. Only present if radio_capabilities is set to true in request.
ue_set	Modify the UE configuration in database. imsi Optional string. nai Optional string. Not applicable to 4G UEs. May be present only if imsi is absent. pdn_list Array. Each entry will set specific parameters for a PDN or PDU session as defined below: access_point_name String. Used to define what PDN or PDU session to configure. routes Array of filters. See [routes], page 45.
ue_add	Add UE to UE database. Message definition: ue_db Array. List of UE configuration. See [ue.db], page 42.
ue_del	Remove UE from the UE database and performs a detach if necessary. Message definition: imsi Optional string. IMSI of the UE to delete. Shall be present if nai is absent. nai Optional string. Network specific identifier-based SUPI. Not applicable to 4G UEs. Shall be present if imsi is absent. local Optional boolean (default = false). If set to true, the UE is locally detached without any NAS signalling.
ue_detach	Force a detach from network. Message definition: imsi Optional string. IMSI of the UE to detach. Shall be present if nai is absent.

nai	Optional string. Network specific identifier-based SUPI. Not applicable to 4G UEs. Shall be present if imsi is absent.
imei	Optional string (14 or 15 digits). UE IMEI (with or without check digit), required if multi_sim is set to true.
n3gpp	Optional boolean (default = false). Set it to true for a non 3GPP UE connected to 5GC.
type	Optional number (EPS default = 2 / re-attach not required; 5GS default = 1 / re-registration not required, 3GPP access). Set NAS detach request type (according to 3GPP TS 24.301 9.9.3.7 Detach type) or de-registration type (according to 3GPP TS 24.501 9.11.3.20 De-registration type).
cause	Optional number (default = 3 / illegal UE). Set EMM or 5GMM cause. The value -1 means that the EMM cause IE is not sent in the NAS Detach Request message or the 5GMM cause is not sent in the NAS Deregistration Request message.
local	Optional boolean (default = false). If set to true, the UE is locally detached without any NAS signalling.

ue_identity_request

Force an identification procedure.

Message definition:

imsi	Optional string. IMSI of the UE. Shall be present if nai is absent.
nai	Optional string. Network specific identifier-based SUPI. Not applicable to 4G UEs. Shall be present if imsi is absent.
imei	Optional string (14 or 15 digits). UE IMEI (with or without check digit), required if multi_sim is set to true.
n3gpp	Optional boolean (default = false). Set it to true for a non 3GPP UE connected to 5GC.
type	Integer (range 1 to 5). Identity type.

me_add Add or update one or several devices in ME database.

Message definition:

default_status

Optional enumeration (whitelisted, blacklisted, greylisted). Defines the default status for devices not explicitly defined in the next objects.

whitelist

Optional array. It contains a list of IMEI (14 digits) or IMEISV (16 digits) whitelisted.

blacklist

Optional array. It contains a list of IMEI (14 digits) or IMEISV (16 digits) blacklisted.

greylist

Optional array. It contains a list of IMEI (14 digits) or IMEISV (16 digits) greylisted.

me_del	Remove one or several devices in ME database. Message definition:
list	Array of strings. Each entry must be an IMEI (14 digits) or IMEISV (16 digits).
pws_write	Start broadcasting Public Warning System message. Message definition:
local_id	Number. ID of the message as defined by local_identifier in MME configuration file
nf	Optional boolean (default = false). If not set, SBC interface is used. If set, N50 interface is used.
increment_serial_number	Optional boolean (default = true). If set to false, the serial_number is not incremented.
pws_kill	Stop broadcasting Public Warning System message. Message definition:
local_id	Number. ID of the message as defined by local_identifier in MME configuration file
stop_all	Optional boolean. Gives the presence of Stop-All-Indicator IE in the message STOP-WARNING-REQUEST.
send_warning_indication	Optional boolean. Default value is 0. Gives the presence of Send-Stop-Warning-Indication IE in the message STOP WARNING REQUEST.
nf	Optional boolean (default = false). If not set, SBC interface is used. If set, N50 interface is used.
cbc_notif_subscribe	CBC subscription to notification. Applicable to N50 interface only. Message definition:
notify_cbk_uri	String. Callback URI on which the N2 information shall be notified.
info_class	Optional enumeration: write-cancel, restart-failure (default = write-cancel). Class of N2 information to which the CBC wants to subscribe.
cbc_notif_unsubscribe	CBC unsubscription to notification. Applicable to N50 interface only. Message definition:
info_class	Optional enumeration: write-cancel, restart-failure (default = write-cancel). Class of N2 information to which the CBC wants to unsubscribe.
enb	Get list of eNB connections. Response definition:
enb_list	Array of objects. Each object represents an eNB connection:
plmn	String. PLMN.

	eNB_ID_type	String (macro, home, short_macro or long_macro). eNB type.
	eNB_ID	Integer. eNB ID.
	name	Optional string. eNB name.
	address	String. eNB IP address and port.
	ue_ctx	Number. Number of UE contexts.
ng_ran	Get list of NG-RAN node connections. Response definition:	
	ng_ran_list	Array of objects. Each object represents a RAN connection:
	plmn	String. PLMN.
	RAN_ID_type	String (gNB, ng-eNB or N3IWF). RAN type.
	RAN_ID	Integer. RAN ID.
	name	Optional string. RAN node name.
	address	String. RAN IP address and port.
	ue_ctx	Number. Number of UE contexts.
s6	Get information regarding the S6a connection. Response definition:	
	state	String. S6a connection state (disconnected, connecting, connected or inactive).
	address	String. HSS address and port.
	host	Optional string. HSS Diameter host identifier retrieved during Capabilities Exchange procedure.
	realm	Optional string. HSS Diameter realm identifier retrieved during Capabilities Exchange procedure.
s6connect	Force S6a connection establishment. Message definition:	
	addr	Optional string. If not set, the MME will try to connect to the previously configured address
s6disconnect	Force S6a connection release.	
s13	Get information regarding the S13 connection. Response definition:	
	state	String. S13 connection state (disconnected, connecting, connected or inactive).
	address	String. EIR address and port.
	host	Optional string. EIR Diameter host identifier retrieved during Capabilities Exchange procedure.

	realm	Optional string. EIR Diameter realm identifier retrieved during Capabilities Exchange procedure.
s13connect	Force S13 connection establishment. Message definition:	
	addr	Optional string. If not set, the MME will try to connect to the previously configured address
s13disconnect	Force S13 connection release.	
sgs	Get information regarding the SGs connection. Response definition:	
	state	String. SGs connection state (disconnected, connecting, connected or inactive).
	address	String. MSC/VLR address and port.
sgsconnect	Force SGs connection establishment. Message definition:	
	addr	Optional string. If not set, the MME will try to connect to the previously configured address
sgsdisconnect	Force SGs connection release.	
sbc	Get list of CBC connections. Response definition:	
	cbc_list	Array of objects. Each object represents a CBC connection:
	address	String. CBC address and port.
lcs	Get information regarding the LCS connection. Response definition:	
	state	String. LCS connection state (disconnected, connecting, connected or inactive).
	address	String. E-SMLC address and port.
lcsconnect	Force LCS connection establishment. Message definition:	
	addr	Optional string. If not set, the MME will try to connect to the previously configured address
n8	Get information regarding the N8 interface. Response definition:	
	server_address	String. UDM address and port.
n8connect	Force N8 connections establishment. Message definition:	
	api_root	Optional string. api_root of the UDM server in the form: <scheme>://<host>:<port>, where <scheme> is "http" or "https".

If not set, the AMF will try to connect to the previously configured address

n8disconnect

Disconnect the AMF client on the interface N8 from the UDM.

n12 Get information regarding the N12 interface.

Response definition:

server_address

String. AUSF address and port.

client_address

String. Address of the AMF client connected to the AUSF.

n12connect

Connect or reconnect the AMF client on the interface N12 to the AUSF.

Message definition:

api_root Optional string. api_root of the AUSF server in the form: <scheme>://<host>:<port>, where <scheme> is "http" or "https".
If not set, the AMF will try to connect to the previously configured address

n12disconnect

Disconnect the AMF client on the interface N12 from the AUSF.

n13 Available only in case of internal AUSF.

Get information regarding the N13 interface.

Response definition:

server_address

String. UDM address and port.

client_address

String. Address of the AUSF client connected to the UDM.

n13connect

Available only in case of internal AUSF.

Connect or reconnect the AUSF client on the interface N13 to the UDM.

Message definition:

addr

api_root Optional string. api_root of the UDM server in the form: <scheme>://<host>:<port>, where <scheme> is "http" or "https".
If not set, the AUSF will try to connect to the previously configured address.

n13disconnect

Available only in case of internal AUSF.

Disconnect the AUSF client on the interface N13 from the UDM.

n17 Get information regarding the N17 interface.

Response definition:

server_address

String. EIR address and port.

client_address

String. Address of the AMF client connected to the EIR.

n17connect

Connect or reconnect the AMF client on the interface N17 to the EIR.

Message definition:

api_root Optional string. api_root of the EIR server in the form: <scheme>://<host>:<port>, where <scheme> is "http" or "https".
If not set, the AMF will try to connect to the previously configured address.

n17disconnect

Disconnect the AMF client on the interface N17 from the EIR.

n11

Get information regarding the NL1 interface.

Response definition:

server_address

String. LMF address and port.

client_address

String. Address of the AMF client connected to the LMF.

nl1connect

Connect or reconnect the AMF client on the interface NL1 to the LMF.

Message definition:

api_root Optional string. api_root of the LMF server in the form: <scheme>://<host>:<port>, where <scheme> is "http" or "https".
If not set, the AMF will try to connect to the previously configured address.

nl1disconnect

Disconnect the AMF client on the interface NL1 from the LMF.

ue_activate_dedicated_bearer

Trigger a network initiated dedicated EPS bearer activation or a 5GS QoS flow activation.

Message definition:

imsi Optional string. UE IMSI.

Shall be present if **nai** is absent.

nai Optional string. Network specific identifier-based SUPI.

Not applicable to 4G UEs.

Shall be present if **imsi** is absent.

imei Optional string (14 or 15 digits). UE IMEI (with or without check digit), required if **multi_sim** is set to true.

apn Optional string. APN of the default EPS bearer associated to the dedicated one. Must be present if **pdu_session_id** or **linked_erab_id** are not present.

sst Optional integer. SST of the PDU session. Used for UEs connected to 5GC.

sd Optional integer. SD of the PDU session. Used for UEs connected to 5GC.

linked_erab_id

Optional integer. Identity of the default EPS bearer associated to the dedicated one. Used for UEs connected to EPC.

<code>pdu_session_id</code>	Optional integer. PDU session identity of the PDU session to select. Used for UEs connected to 5GC.
<code>qci</code>	Integer (range 1 to 255). QoS Class Identifier of the E-RAB, or 5QI of the QoS flow.
<code>5qi_qos</code>	Optional object. See [5QI QoS], page 36.
<code>priority_level</code>	Optional integer (1 to 15, default 15). ARP priority level.
<code>pre_emption_capability</code>	Optional enumeration (<code>shall_not_trigger_pre_emption</code> or <code>may_trigger_pre_emption</code> , default <code>shall_not_trigger_pre_emption</code>).
<code>pre_emption_vulnerability</code>	Optional enumeration (<code>not_pre_emptable</code> or <code>pre_emptable</code> , default <code>not_pre_emptable</code>).
<code>filters</code>	Array. See [TFT], page 37.
<code>gbr</code>	Optional object. See [GBR], page 36.
<code>transaction_identifier</code>	Optional integer (range 0 to 127). If present, the transaction identifier IE is put in the EPS bearer activation message.
<code>llc_sapi</code>	Optional integer (range 0 to 15). If present, the LLC service access point identifier IE is put in the EPS bearer activation message.
<code>radio_priority</code>	Optional integer (range 0 to 7). If present, the radio priority IE is put in the EPS bearer activation message.
<code>packet_flow_identifier</code>	Optional integer (range 0 to 127). If present, the packet flow identifier IE is put in the EPS bearer activation message.
<code>sm_qos</code>	Optional string. If present, the quality of service IE is put in the EPS bearer activation message. The string must contain the hexadecimal representation of the IE without its IEI and length.
Response definition:	
<code>erab_id</code>	Integer. Allocated ERAB identity for this dedicated EPS bearer. Sent if the procedure is for EPS.
<code>pdu_session_id</code>	Integer. PDU session identifier associated to the QoS flow identifier. Sent if the procedure is for 5GS.
<code>qos_flow_id</code>	Integer. Allocated QoS flow identifier for this bearer. Sent if the procedure is for 5GS.

ue_modify_bearer

Trigger a network initiated EPS bearer modification.

Message definition:

<code>imsi</code>	String. UE IMSI.
-------------------	------------------

<code>imei</code>	Optional string (14 or 15 digits). UE IMEI (with or without check digit), required if <code>multi_sim</code> is set to true.										
<code>erab_id</code>	Integer. ERAB identity of the bearer to be modified.										
<code>qos</code>	Optional object. If present a QoS modification is done. It should contain the following objects: <table> <tr> <td><code>qci</code></td><td>Integer (range 1 to 255). QoS Class Identifier of the E-RAB.</td></tr> <tr> <td><code>priority_level</code></td><td>Optional integer (1 to 15, default 15). ARP priority level.</td></tr> <tr> <td><code>pre_emption_capability</code></td><td>Optional enumeration (<code>shall_not_trigger_pre_emption</code> or <code>may_trigger_pre_emption</code>, default <code>shall_not_trigger_pre_emption</code>).</td></tr> <tr> <td><code>pre_emption_vulnerability</code></td><td>Optional enumeration (<code>not_pre_emptable</code> or <code>pre_emptable</code>, default <code>not_pre_emptable</code>).</td></tr> <tr> <td><code>gbr</code></td><td>Optional object. See [GBR], page 36.</td></tr> </table>	<code>qci</code>	Integer (range 1 to 255). QoS Class Identifier of the E-RAB.	<code>priority_level</code>	Optional integer (1 to 15, default 15). ARP priority level.	<code>pre_emption_capability</code>	Optional enumeration (<code>shall_not_trigger_pre_emption</code> or <code>may_trigger_pre_emption</code> , default <code>shall_not_trigger_pre_emption</code>).	<code>pre_emption_vulnerability</code>	Optional enumeration (<code>not_pre_emptable</code> or <code>pre_emptable</code> , default <code>not_pre_emptable</code>).	<code>gbr</code>	Optional object. See [GBR], page 36.
<code>qci</code>	Integer (range 1 to 255). QoS Class Identifier of the E-RAB.										
<code>priority_level</code>	Optional integer (1 to 15, default 15). ARP priority level.										
<code>pre_emption_capability</code>	Optional enumeration (<code>shall_not_trigger_pre_emption</code> or <code>may_trigger_pre_emption</code> , default <code>shall_not_trigger_pre_emption</code>).										
<code>pre_emption_vulnerability</code>	Optional enumeration (<code>not_pre_emptable</code> or <code>pre_emptable</code> , default <code>not_pre_emptable</code>).										
<code>gbr</code>	Optional object. See [GBR], page 36.										
<code>filters</code>	Array. Contains the new TFT after modification. See [TFT], page 37.										
<code>llc_sapi</code>	Optional integer (range 0 to 15). If present, the LLC service access point identifier IE is put in the EPS bearer activation message.										
<code>radio_priority</code>	Optional integer (range 0 to 7). If present, the radio priority IE is put in the EPS bearer activation message.										
<code>packet_flow_identifier</code>	Optional integer (range 0 to 127). If present, the packet flow identifier IE is put in the EPS bearer activation message.										
<code>sm_qos</code>	Optional string. If present, the quality of service IE is put in the EPS bearer activation message. The string must contain the hexadecimal representation of the IE without its IEI and length.										
<code>p_cscf</code>	Optional boolean. Adds the P-CSCF addresses to the PCO information element of the modify EPS bearer context request message.										
<code>dns</code>	Optional boolean. Adds the DNS addresses to the PCO information element of the modify EPS bearer context request message.										

Response definition:

<code>erab_id</code>	Integer. ERAB identity of the EPS bearer.
----------------------	---

`ue_modify_pdu_session`

Trigger a network initiated PDU session modification.

Message definition:

<code>imsi</code>	Optional string. UE IMSI. Shall be present if <code>nai</code> is absent.
<code>nai</code>	Optional string. Network specific identifier-based SUPI. Shall be present if <code>imsi</code> is absent.
<code>imei</code>	Optional string (14 or 15 digits). UE IMEI (with or without check digit), required if <code>multi_sim</code> is set to true.

n3gpp	Optional boolean (default = false). Set it to true for a non 3GPP UE connected to 5GC.														
pdu_session_id	Integer. PDU session identity of the PDU session to be modified.														
qos_rules	Optional array. List of the QoS rules after modification other than the default one. Each element of the array contains the followings properties: <table> <tr> <td>id</td><td>QoS rule identifier.</td></tr> <tr> <td>qfi</td><td>Range: 0 to 63. QoS flow identifier.</td></tr> <tr> <td>filters</td><td>Array of packet filters. See [TFT], page 37.</td></tr> </table>	id	QoS rule identifier.	qfi	Range: 0 to 63. QoS flow identifier.	filters	Array of packet filters. See [TFT], page 37.								
id	QoS rule identifier.														
qfi	Range: 0 to 63. QoS flow identifier.														
filters	Array of packet filters. See [TFT], page 37.														
qos_flow	Optional array. List of the QoS flows after modification other than the QoS flow associated to the default QoS rule. Each element of the array contains the following properties: <table> <tr> <td>qfi</td><td>Integer. Range: 0 to 63. QoS flow identifier.</td></tr> <tr> <td>5qi</td><td>Integer. Range: 1 to 254. 5QI of the QoS flow.</td></tr> <tr> <td>5qi_qos</td><td>Optional object. See [5QI QoS], page 36.</td></tr> <tr> <td>priority_level</td><td>Range: 1 to 15. ARP priority level.</td></tr> <tr> <td>pre_emption_capability</td><td>Enumeration: <code>shall_not_trigger_pre_emption</code> or <code>may_trigger_pre_emption</code>.</td></tr> <tr> <td>pre_emption_vulnerability</td><td>Enumeration: <code>not_pre_emptable</code> or <code>pre_emptable</code>.</td></tr> <tr> <td>gbr</td><td>Optional object. See [GBR], page 36. Must be present for a GBR 5QI.</td></tr> </table>	qfi	Integer. Range: 0 to 63. QoS flow identifier.	5qi	Integer. Range: 1 to 254. 5QI of the QoS flow.	5qi_qos	Optional object. See [5QI QoS], page 36.	priority_level	Range: 1 to 15. ARP priority level.	pre_emption_capability	Enumeration: <code>shall_not_trigger_pre_emption</code> or <code>may_trigger_pre_emption</code> .	pre_emption_vulnerability	Enumeration: <code>not_pre_emptable</code> or <code>pre_emptable</code> .	gbr	Optional object. See [GBR], page 36. Must be present for a GBR 5QI.
qfi	Integer. Range: 0 to 63. QoS flow identifier.														
5qi	Integer. Range: 1 to 254. 5QI of the QoS flow.														
5qi_qos	Optional object. See [5QI QoS], page 36.														
priority_level	Range: 1 to 15. ARP priority level.														
pre_emption_capability	Enumeration: <code>shall_not_trigger_pre_emption</code> or <code>may_trigger_pre_emption</code> .														
pre_emption_vulnerability	Enumeration: <code>not_pre_emptable</code> or <code>pre_emptable</code> .														
gbr	Optional object. See [GBR], page 36. Must be present for a GBR 5QI.														
p_cscf	Optional boolean. Adds the P-CSCF addresses to the ePCO information element of the PDU session modification command message.														
dns	Optional boolean. Adds the DNS addresses to the ePCO information element of the PDU session modification command message.														
mtu_ipv4	Optional boolean. Adds the IPv4 MTU to the ePCO information element of the PDU session modification command message.														
ue_modify_reflective_qos	Modify the reflective QoS indicator bit for an existing UE packet filter. Message definition: <table> <tr> <td>imsi</td><td>Optional string. UE IMSI. Shall be present if nai is absent.</td></tr> <tr> <td>nai</td><td>Optional string. Network specific identifier-based SUPI. Shall be present if imsi is absent.</td></tr> <tr> <td>imei</td><td>Optional string (14 or 15 digits). UE IMEI (with or without check digit), required if multi_sim is set to true.</td></tr> <tr> <td>n3gpp</td><td>Optional boolean (default = false). Set it to true for a non 3GPP UE connected to 5GC.</td></tr> </table>	imsi	Optional string. UE IMSI. Shall be present if nai is absent.	nai	Optional string. Network specific identifier-based SUPI. Shall be present if imsi is absent.	imei	Optional string (14 or 15 digits). UE IMEI (with or without check digit), required if multi_sim is set to true.	n3gpp	Optional boolean (default = false). Set it to true for a non 3GPP UE connected to 5GC.						
imsi	Optional string. UE IMSI. Shall be present if nai is absent.														
nai	Optional string. Network specific identifier-based SUPI. Shall be present if imsi is absent.														
imei	Optional string (14 or 15 digits). UE IMEI (with or without check digit), required if multi_sim is set to true.														
n3gpp	Optional boolean (default = false). Set it to true for a non 3GPP UE connected to 5GC.														

pdu_session_id
Integer. PDU session identity of the QoS flow to be modified.

qos_flow_id
Integer. QoS flow identity of the QoS flow to be modified.

filter_id
Integer. Packet filter identity of the DL only filter to be modified.

reflective_qos
Boolean. Reflective QoS indicator for this packet filter.

ue_deactivate_bearer

Trigger a network initiated default or dedicated EPS bearer deactivation, or a 5GS QoS flow deactivation. If the UE is in RRC idle state, the bearer will be locally released without any NAS signalling.

Message definition:

imsi Optional string. UE IMSI.
Shall be present if **nai** is absent.

nai Optional string. Network specific identifier-based SUPI.
Not applicable to 4G UEs.
Shall be present if **imsi** is absent.

imei Optional string (14 or 15 digits). UE IMEI (with or without check digit), required if **multi_sim** is set to true.

n3gpp Optional boolean (default = false). Set it to true for a non 3GPP UE connected to 5GC.

erab_id Optional integer. ERAB identity of the bearer to be released. Must be present for an EPS procedure.

esm_cause
Optional integer (default = 36). ESM cause for the message. Can be present for an EPS procedure.

pdu_session_id
Optional integer. PDU session identifier of the QoS flow to release.
Must be present for a 5GS procedure.

qos_flow_id
Optional integer. QoS flow identifier to release. Must be present for a 5GS procedure.

5gsm_cause
Optional integer (default = 36). 5GSM cause for the message. Can be present for a 5GS procedure.

non_ip_data

Send data over a non IP PDN or unstructured PDU session.

Message definition:

imsi Optional string. UE IMSI.
Shall be present if **nai** is absent.

nai Optional string. Network specific identifier-based SUPI.
Not applicable to 4G UEs.
Shall be present if **imsi** is absent.

imei	Optional string (14 or 15 digits). UE IMEI (with or without check digit), required if multi_sim is set to true.
apn	Optional string. APN of the non IP bearer. Used for UEs connected to EPC. Shall be present if erab_id is absent.
erab_id	Optional integer. ERAB identity of the non IP default bearer. Used for UEs connected to EPC. Shall be present if apn is absent.
dnn	Optional string. DNN of the non IP bearer. Used for UEs connected to 5GC. Shall be present if pdu_session_id is absent.
sst	Optional integer. SST of the non IP bearer. Used for UEs connected to 5GC. May be present if dnn is present.
sd	Optional integer. Optional SD of the non IP bearer. Used for UEs connected to 5GC. May be present if dnn is present.
pdu_session_id	Optional integer. PDU session ID of the non IP bearer. Used for UEs connected to 5GC. Shall be present if dnn is absent.
data	String. ASCII representation of the data hexadecimal dump.

generic_nas_transport

Send an EPS downlink generic NAS transport message.

Message definition:

imsi	String. UE IMSI.
imei	Optional string (14 or 15 digits). UE IMEI (with or without check digit), required if multi_sim is set to true.
type	Integer (range: 0 to 255). Generic message container type information element.
payload	String. ASCII representation of the generic message container hexadecimal dump.
add_info	Optional string. ASCII representation of the additional information hexadecimal dump.

5gs_nas_transport

Send an 5GS downlink NAS transport message for LPP, SOR, UE policy, UE parameters update or location services.

Message definition:

imsi	Optional string. UE IMSI. Shall be present if nai is absent.
nai	Optional string. Network specific identifier-based SUPI. Not applicable to 4G UEs. Shall be present if imsi is absent.
imei	Optional string (14 or 15 digits). UE IMEI (with or without check digit), required if multi_sim is set to true.
n3gpp	Optional boolean (default = false). Set it to true for a non 3GPP UE connected to 5GC.
type	Integer (range: 3 to 6). Payload container type information element.
payload	String. ASCII representation of the payload container hexadecimal dump.

add_info	Optional string. ASCII representation of the additional information hexadecimal dump for LPP or location services.
ursp_rules	Send URSP rules to the UE. Message definition:
imsi	Optional string. UE IMSI. Shall be present if nai is absent.
nai	Optional string. Network specific identifier-based SUPI. Not applicable to 4G UEs. Shall be present if imsi is absent.
imei	Optional string (14 or 15 digits). UE IMEI (with or without check digit), required if multi_sim is set to true.
n3gpp	Optional boolean (default = false). Set it to true for a non 3GPP UE connected to 5GC.
ue_policy_section_management_list	Array of objects. Contains the description of the UE policy section management list as defined in 3GPP TS 24.501 chapter D.6.2. Each element of the array has the following properties:
plmn	String. PLMN identity of the MME (5 or 6 digits).
instruction_list	Array of objects. Each element describes an instruction as defined in 3GPP TS 24.501 chapter D.6.2. Each element has the following definition:
upsc	Integer (range: 0 to 65 535). UE Policy Section Code.
ue_policy_part_list	Optional array. UE policy section contents as defined in 3GPP TS 24.501 chapter D.6.2 Figure D.6.2.6. Each element of the array contains a UE policy part and has the following properties:
ursp_rules	Optional array of objects. Each element describes a URSP rule as defined in 3GPP TS 24.526 chapter 5.2 Figure 5.2.2 and has the following properties:
precedence	Integer (range: 0 to 255). Precedence value of URSP rule.
traffic_descriptor_components	Array describing the Traffic descriptor as defined in 3GPP TS 24.526 chapter

5.2. Each element of the array contains one of the following properties:

`match_all` Optional boolean (default = false). Indicates the presence of `match_all` descriptor component type.

`os_id_os_app_id` Object. Match the OS Id and the OS application Id. Contains the following properties:

<code>os_id</code>	String. Hexadecimal representation of the OS Id.
<code>os_app_id</code>	String.


```

ipv4_remote_addr
    String.
    Match a
    remote
    (external
    network
    entity)
    IPv4
    address
    with the
    additional
    mask
    property.

ipv6_remote_addr_
prefix
    String.
    Match a
    remote
    (external
    network
    entity)
    IPv6
    address
    with the
    additional
    prefix_
    len
    property.

proto_id Range: 0
to 255.
Match
against
the IP
protocol
identifier.

remote_port
    Range: 0
    to 65535.
    Match
    against
    the
    remote
    (external
    network
    entity)
    port.

remote_port_range
    Array of 2
    integers.
    Match

```

against
a remote
(external
network
entity)
port
range.

`ip_3_tuple`

Array of
objects.
Each
element of
the array
contains
one of the
`ipv4_`
`remote_`
`addr`,
`ipv6_`
`remote_`
`addr`,
`prefix`,
`proto_id`,
`remote_`
`port` or
`remote_`
`port_`
`range`
proper-
ties.

`security_`

`parameter_index`

32 bit
integer.
Match
the ESP
or AH
security
parameter
index.

`type_of_service`

Range: 0
to 255.
Match the
type of
service
(IPv4) or
the traffic
class
(IPv6)
field. The

additional
mask
property
is the
corre-
sponding
mask.

flow_label
20 bit
integer.
Match the
IPv6 flow
label.

destination_mac_
addr
String.
Match the
destina-
tion MAC
address.

802.1q_ctag_vid
Range: 0
to 4095.
Match the
802.1Q
C-TAG
VID.

802.1q_stag_vid
Range: 0
to 4095.
Match the
802.1Q
S-TAG
VID.

802.1q_ctag_pcp_
dei
Range:
0 to 15.
Match the
802.1Q
C-TAG
PCP and
DEI.

802.1q_stag_pcp_
dei
Range:
0 to 15.
Match the
802.1Q

S-TAG
PCP and
DEI.

ethertype
Range: 0
to 65535.
Match the
ethertype.

dnn String.
Match
a DNN
value.

**connection_
capabilities**
Array of
enumer-
ation:
ims, mms,
supl,
internet.
Match
a con-
nection
capability
identifier
given in
the array.

destination_fqdn
String.
Match
the des-
tination
FQDN.

os_app_id
String.
Match the
OS app
id.

**destination_mac_
addr_range**
Array of
2 strings.
Match the
destina-
tion MAC
address
range.

route_list

Array of objects describing the Route selection descriptor list as defined in 3GPP TS 24.526 Figure 5.2.3. Each element contains a Route selection descriptor as defined in 3GPP TS 24.526 Figure 5.2.4 and contains a variable number (at least one) of route selection descriptor components as described below:

precedence

Integer (range 0-255). Precedence value of route selection descriptor.

components

Array of objects. Contains a variable number (at least one) of route selection descriptor components as defined below:

ssc_

mode Integer
In-
di-
cates
the
SSC
mode.

	Only value 1 is supported.
snssai	Indicates the S-NSSAI value.
sst	Indicates the S-NSSAI value.
sd	Indicates the S-NSSAI value.
dnn	String. Indicates the DNN value.
pdu_session_type	Enumeration. Indicates the PDU session type.

	eth-
	er-
	net.
	In-
	di-
	cates
	the
	PDU
	ses-
	sion
	type.
preferred_	
access_	
type	Enumeration
	3gpp,
	non-
	3gpp.
	In-
	di-
	cates
	the
	pre-
	ferred
	ac-
	cess
	type.
multi_	
access_	
preference	Boolean
	(de-
	fault
	=
	false).
	If
	set
	to
	true.
	in-
	di-
	cates
	the
	multi-
	access
	pref-
	er-
	ence.

reset_ue_pos_stored_info
Send a test procedure reset UE positioning stored information message.
Message definition:

imsi	String. UE IMSI.
imei	Optional string (14 or 15 digits). UE IMEI (with or without check digit), required if multi_sim is set to true.
techno	Integer (range: 0 to 255). UE positioning technology as specified in 3GPP TS 36.509 chapter 6.9.

mt_cs_paging
Trigger a CS paging for a UE connected to EPC.
Message definition:

imsi	String. UE IMSI.
-------------	------------------

guti_realloc
Initiate a LTE procedure GUTI reallocation. Message definition:

imsi	String. UE IMSI.
imei	Optional string (14 or 15 digits). UE IMEI (with or without check digit), required if multi_sim is set to true.

load_balancing_tau
Initiate a LTE load balancing TAU procedure. Message definition:

imsi	String. UE IMSI.
imei	Optional string (14 or 15 digits). UE IMEI (with or without check digit), required if multi_sim is set to true.

mbs_broadcast_session_setup
Starts a MBS broadcast session. Message definition:

index	Optional integer. Index (starting from 0) of the session to start in the broadcast_sessions array. If not present, the properties above are used to configure dynamically the session to start.
--------------	--

session_create
Optional object. Must be present if **index** is absent. Gives the json description of the message Nbmbsmf_MBSSession_Create Request (CreateReqData structure as defined in 3GPP 29.532).

user_plane_desc
Optional object. See [user_plane_desc], page 66. **multicast_dest_addr** and **source_addr** are not needed because **ssm** is already encoded in **session_create**.

Response definition:

cid	Integer. Unique connection ID of the client MBS session context.
------------	--

mbs_broadcast_session_release
Stops a MBS broadcast session. Message definition:

cid	Integer. Index (starting from 0) of the session to stop returned by mbs_broadcast_session_setup .
------------	--

mbs_tmgi_allocate
Requests TMGIs allocation. Message definition:

tmgi_allocate
Mandatory parameter in json format encoded as described in 3GPP 29.532 6.1.6.2.2 Type: TmgiAllocate.
For example: "tmgiNumber": 3,


```

"tmgiList": [
  {
    "mbsServiceId": "000000",
    "plmnId": {
      "mcc": "001",
      "mnc": "01"
    }
  }
],

```

mbs_tmgi_deallocate

Requests TMGIs deallocation. Has an optional parameter `tmgi_list` as described below. If no parameter is given, requests to deallocate all previously allocated TMGIs. Message definition:

tmgi_list

Optional parameter given as a json array of TMGIs as defined in 29.571 5.9.4.2 Type: Tmgi.

Example:

```

"tmgiList": [
  {
    "mbsServiceId": "000002",
    "plmnId": {
      "mcc": "001",
      "mnc": "01"
    }
  },
  {
    "mbsServiceId": "000001",
    "plmnId": {
      "mcc": "001",
      "mnc": "01"
    }
  }
],

```

mbs_session_info

Gets MBS session info at the server side. Response definition:

broadcast

Array of objects. Each object contains the following properties:

session_id

Object. MBS session identity. It contains the following properties:

tmgi Object. Temporary Mobile Group Identity. It contains the following properties:

plmn String. PLMN.

service_id

Integer. Service ID.

nid Optional integer. Network Identifier.

snssai Object. It contains the following properties:

sst Integer. Slice Service Type.

sd	Optional integer. Slice Differentiator.
qos_flows	Array of objects. Each object contains the following properties:
qfi	Integer. QoS Flow identifier.
packets	Integer. Number of packets transmitted.
bytes	Integer. Number of bytes transmitted.

6.7 Positioning messages

The messages in this section are related to the positioning procedures. The messages that are initiated by the location server are used when the test e-SMLC or test LMF is used. See [lcs], page 54, to configure the test e-SMLC. See [nl1], page 58, to configure the test LMF.

6.7.1 LCSAP

The messages in this section are applicable to EPC only. They are sent on the LCSAP interface between the LCS client located in the MME and the e-SMLC.

location_req

Start a location procedure by sending LCSAP Location-Request from the LCS client to the e-SMLC.

When the e-SMLC receives the message LCSAP Location-Request:

- if the e-SMLC is configured to run in autonomous mode (See [autonomous_mode], page 55), the e-SMLC will autonomously run the LPPa and LPP procedures necessary to obtain the location of the UE.

- else

- : if the e-SMLC is configured to use the LPP protocol (See [lpp_test], page 54), the e-SMLC will send the message LPP RequestCapability message to the UE. The command lpp_request_location (See [lpp_request_location], page 119) can then be used to trigger a LPP RequestLocationInformation message in the e-SMLC.

Else if the e-SMLC is configured to use the LPPa protocol (See [lpp_test], page 54), the e-SMLC will send the message LPPa E-CIDMeasurementInitiationRequest to the eNB using the measurement configuration in the e-SMLC (See [local_e_smlc], page 54).

The following commands can then be used in the e-SMLC:

ecid_periodic_meas_termination to send LPPa e-CIDMeasurementTermination (See [ecid_periodic_meas_termination], page 119).

otdoa_information_req to send LPPa OTDOAInformationRequest ([otdoa_information_req], page 119).

Message definition:

plmn	String (5 or 6 digits).
cell_id	Integer. 28 bits long LTE cell identifier.
session_id	Optional integer in range 0 to 255. Location session identifier.
type	Enumeration: geographic, assistance_info, last_known. Location type geographic information.
imsi	String. IMSI of the UE.
imei	Optional string (14 or 15 digits). UE IMEI (with or without check digit), required if multi_sim is set to true.

Response definition: If the autonomous mode is configured, the command will return the result of the location procedure. If the autonomous mode is not configured, the command returns the correlation ID:

correlation_id

Integer in range 0 to 0xFFFFFFFF. Identifier of the created location context. This value shall be used in the command `lpp_request_location`.

lcsap_reset_req

Send LCSAP Reset-Request from the test e-SMLC to the LCS client both located in the MME.

6.7.2 LPPa

The messages in this section are applicable to EPC only, and are useful if the e-SMLC is not configured in autonomous mode (See [autonomous_mode], page 55). They are sent by the test e-SMLC to the eNB via the MME. LCSAP interface is used between the e-SMLC and the MME.

ecid_periodic_meas_termination

Send LPPa e-CIDMeasurementTermination from the test e-SMLC.

Message definition:

session_id

Integer in range 0 to 255. Location session identifier.

transaction_id

Optional integer in range 0 to 32767 (default = 0). Transaction identifier in LPPa e-CIDMeasurementTermination.

e_smlc_meas_id

Integer in range 1 to 15. E-SMLC-UE-Measurement_ID in LPPa e-CIDMeasurementTermination.

enb_meas_id

Integer in range 1 to 15. eNB-UE-Measurement_ID in LPPa e-CIDMeasurementTermination.

otdoa_information_req

Send LPPa OTDOAInformationRequest from the test e-SMLC.

Message definition:

transaction_id

Optional integer in range 0 to 32767. Transaction identifier in LPPa OTDOAInformationRequest.

enb_plmn String (5 or 6 digits). eNB PLMN.

enb_type Optional enumeration: macro, short_macro, long_macro or home (default = macro).

enb_id Integer. eNodeB global identifier. enb_plmn, enb_type and enb_id are used to identify the eNB to which the message LPPa OTDOAInformationRequest will be sent.

6.7.3 LPP

The messages in this section are applicable to EPC and 5GC. They are sent by the test e-SMLC in EPC or the test LMF in 5GC to the UE.

lpp_request_location

Send LPP RequestLocationInformation from the test e-SMLC or the test LMF.

If the UE answers with a LPP RequestAssistanceData message, the location

server will use the assistance data in the e-SMLC or LMF configuration (See [otdoa_assistance_data], page 55, and/or See [nr_tdoa_assistance_data], page 55) to provide the OTDOA and NR-TDOA assistance data in LPP ProvideAssistance-Data.

If `otdoa_assistance_data` and/or `nr_tdoa_assistance_data` are not configured, the location server must have previously retrieved the information necessary to build the requested assistance data (see [otdoa_information_req], page 119, [nr_otdoa_information_req], page 122, and [trp_information_req], page 122).

Message definition:

<code>is_5gs</code>	Optional boolean (default = false). If set to true, send the LPP message from the test LMF using the NL1 interface. Otherwise send the LPP message from the test e-SMLC using the LCSAP interface.
<code>supi</code>	String in the form "imsi-" or "nai-" followed by the value of the IMSI or the NAI.
<code>imei</code>	Optional string (14 or 15 digits). UE IMEI (with or without check digit), required if <code>multi_sim</code> is set to true.
<code>correlation_id</code>	Integer in range 0 to 0xFFFFFFFF in EPC and to 0xFFFFFFFFFFFFFFFF in 5GC. Correlation ID as defined in 3GPP TS 29.171 and 3GPP TS 29.518.
<code>lpp_methods</code>	Optional array (default = ["ecid"]). An element of the array is an enumeration: "ecid", "otdoa", "nr_ecid", "nr_tdoa", "gnss". Gives the list of the LPP methods to request in the message LPP RequestLocationInformation. Note that "gnss" cannot be requested together with any other method, because only locationEstimate is supported for GNSS, and only locationMeasurements is supported for the other methods.
<code>report_type</code>	Enumeration: "once", "periodic", "triggered". Reporting configuration in the message LPP RequestLocationInformation.

6.7.4 NRPPa

The messages in this section are applicable to 5GC only and are useful if the LMF is not configured in autonomous mode (See [lmf_autonomous_mode], page 59). They are used to subscribe the LMF to N1/N2 notifications, trigger UE location, cancel UE location, and send NRPPa PDUs from the test LMF to the gNB via the AMF. The NL1 interface is used between the LMF and the AMF.

`lmf_non_ue_n2_subscribe`

LMF server subscription to the reception of non-UE NRPPa PDUs. See 3GPP TS 29.518 chapter 5.2.2.4.2 NonUeN2InfoSubscribe.

Message definition:

`notify_cbk_uri`

String. `n2NotifyCallbackUri` as defined in 3GPP TS 29.518 chapter 6.1.6.2.10 Type: NonUeN2InfoSubscriptionCreateData.

`lmf_non_ue_n2_unsubscribe`

LMF server unsubscription to the reception of non-UE NRPPa PDUs. See 3GPP TS 29.518 chapter 5.2.2.4.3 NonUeN2InfoUnSubscribe.

Message definition:

subscription_id

n2NotifySubscriptionId as defined in 3GPP TS 29.518 chapter 6.1.3.9
Resource: Non UE N2 Messages Subscriptions Collection.

nr_location_req

Start location procedure for a target UE in the AMF. The AMF will send to the LMF on the N1 interface an HTTP POST request to the resource URI associated with the "determine-location" operation as specified in 3GPP TS 29.572 chapter 5.2.2.2.2 Retrieve UE Location.

If the embedded test LMF is used:

- if the LMF is configured to run in autonomous mode (See [lmf_autonomous_mode], page 59), the LMF will autonomously initiate the NRPPa and LPP procedures necessary to obtain or estimate the position of the UE.
- if the autonomous mode (See [lmf_autonomous_mode], page 59) is not configured and if the LMF is configured with LPP (See [lpp_test], page 54), the LMF will send a LPP requestCapabilities message to the UE.
- otherwise, the LMF will send a NRPPa E-CIDMeasurementInitiationRequest using the parameters **transaction_id**, **meas_id**, **periodic_meas** and **meas_period** configured in **lmf_cfg**.

Message definition:

supi	String in the form "imsi-" or "nai-" followed by the value of the IMSI or the NAI.
pei	Optional string in the form "imei-" or "imeisv-" followed by the value of the IMEI or the IMEISV. Required if multi_sim is set to true.
lcs_qos	Optional object. QoS of the location request defined as specified in 3GPP 29.572 paragraph 6.1.6.2.13:
hAccuracy	Optional float. Horizontal accuracy.
vAccuracy	Optional float. Vertical accuracy.
vertRequested	Optional boolean. Vertical accuracy requested.
responseTime	Optional enumeration: LOW_DELAY, DELAY_TOLERANT, NO_DELAY.
lcsQosClass	Optional enumeration: BEST_EFFORT, ASSURED, MULTIPLE_QOS.

Response definition: if the LMF is configured to run in autonomous mode (See [lmf_autonomous_mode], page 59), the command returns the result of the location procedure. if the LMF is not configured to run in autonomous mode (See [lmf_autonomous_mode], page 59), the command returns the correlation ID:

correlation_id

Integer in range 0 to 0xFFFFFFFFFFFFFFFF. Identifier of the created location context. This value shall be used in the commands **lpp_request_location** and **nr_cancel_location**.

nr_cancel_location

Cancel periodic or triggered location procedure for a target UE.

Message definition:

- imsi** Optional string. UE IMSI.
Shall be present if **nai** is absent.
- nai** Optional string. Network specific identifier-based SUPI. Shall be present if **imsi** is absent.
- imei** Optional string (14 or 15 digits). UE IMEI (with or without check digit), required if **multi_sim** is set to true.
- correlation_id** Integer in range 0 to 0xFFFFFFFF in EPC and to 0xFFFFFFFFFFFFFFFF in 5GC. Correlation ID as defined in 3GPP TS 29.171 and 29.518.

nr_otdoa_information_req

Send an HTTP POST request for "NonUeN2MessageTransfer" operation containing a NRPPa OTDOAInformationRequest PDU from the test LMF to the AMF.

See 3GPP TS 38.455 chapter 8.2.5 OTDOA Information Exchange.

See 3GPP TS 29.518 chapter 5.2.2.4.1 NonUeN2MessageTransfer.

Message definition:

- transaction_id** Optional integer in range 0 to 32767 as defined in 3GPP TS 38.455 chapter 9.1.1.7. All other fields in the message NRPPa OTDOAInformationRequest are hardcoded.
- ran_node_id_list** Array of objects. List of the global RAN node ID to send in the HTTP POST request as defined in 3GPP TS 29.518 chapter 6.1.6.2.9. See [ran_node_id], page 50.
- tai_list_5gs** Optional array of objects (up to 65535). 5GS TAI List to set in the parameter taiList as defined in 3GPP TS 29.518 chapter 6.1.6.2.9. See [tai_list_5gs], page 49.
- tai_list** Optional array of objects (up to 65535). TAI List to set in the parameter taiList as defined in 3GPP TS 29.518 chapter 6.1.6.2.9. See [tai_list], page 49.
- rat_selector** Optional enumeration: nr, eutra, both (default = both). Value of the parameter ratSelector as defined in 3GPP TS 29.518 chapter 6.1.6.2.9.

trp_information_req

Send an HTTP POST request for "NonUeN2MessageTransfer" operation containing a NRPPa TRPInformationRequest PDU from the test LMF to the AMF.

See 3GPP TS 38.455 chapter 8.2.8 TRP Information Exchange.

See 3GPP TS 29.518 chapter 5.2.2.4.1 NonUeN2MessageTransfer.

Message definition:

- ran_node_id_list** Array of objects. List of the global RAN node ID to send in the HTTP POST request as defined in 3GPP TS 29.518 chapter 6.1.6.2.9. See [ran_node_id], page 50.

tai_list_5gs
Optional array of objects (up to 65535). 5GS TAI List to set in the parameter taiList as defined in 3GPP TS 29.518 chapter 6.1.6.2.9. See [tai_list_5gs], page 49.

tai_list Optional array of objects (up to 65535). TAI List to set in the parameter taiList as defined in 3GPP TS 29.518 chapter 6.1.6.2.9. See [tai_list], page 49.

rat_selector
Optional enumeration: nr, eutra, both (default = both). Value of the parameter ratSelector as defined in 3GPP TS 29.518 chapter 6.1.6.2.9.

transaction_id
Optional integer in range 0 to 32767 as defined in 3GPP TS 38.455 chapter 9.1.1.7.

meas_preconfig_req
Send an HTTP POST request for "N1N2MessageTransfer" operation containing a NRPPa MEASUREMENT PRECONFIGURATION REQUIRED from the test LMF to the AMF.
See 3GPP TS 38.455 chapter 8.2.12 Measurement Preconfiguration.
See 3GPP TS 29.518 chapter 5.2.2.3.1 N1N2MessageTransfer.
Message definition:

imsi Optional string. UE IMSI.
Shall be present if nai is absent.

nai Optional string. Network specific identifier-based SUPI. Shall be present if imsi is absent.

imei Optional string (14 or 15 digits). UE IMEI (with or without check digit), required if multi_sim is set to true.

correlation_id
Integer in range 0 to 0xFFFFFFFFFFFFFFFF. Correlation ID as defined in 3GPP TS 29.518.

transaction_id
Optional integer in range 0 to 32767 as defined in 3GPP TS 38.455 chapter 9.1.1.7.

nr_ecid_periodic_meas_termination_req
Send an HTTP POST request for "N1N2MessageTransfer" operation containing a NRPPa E-CID Measurement Termination PDU from the test LMF to the AMF.
See 3GPP TS 38.455 chapter 8.2.4 E-CID Measurement Termination.
See 3GPP TS 29.518 chapter 5.2.2.3.1 N1N2MessageTransfer.
Message definition:

imsi Optional string. UE IMSI.
Shall be present if nai is absent.

nai Optional string. Network specific identifier-based SUPI. Shall be present if imsi is absent.

imei Optional string (14 or 15 digits). UE IMEI (with or without check digit), required if multi_sim is set to true.

<code>correlation_id</code>	Integer in range 0 to 0xFFFFFFFFFFFFFFFF. Correlation ID as defined in 3GPP TS 29.518.
<code>transaction_id</code>	Optional integer in range 0 to 32767 as defined in 3GPP TS 38.455 chapter 9.1.1.7.
<code>lmf_meas_id</code>	Integer in range 1 to 256. Value of LMF-UE-Measurement-ID in the message NRPPa E-CID Measurement Termination.
<code>ran_meas_id</code>	Integer in range 1 to 256. Value of RAN-UE-Measurement-ID in the message NRPPa E-CID Measurement Termination.

6.7.5 NRPPa/LPP common

The messages in this section are common to LPP and NRPPa and are applicable to 5GC only. They are used to connect and configure the LMF on the N1 interface.

`lmf_ue_n1_n2_subscribe`

LMF server subscription to the reception of LPP messages and/or UE NRPPa PDUs. See 3GPP TS 29.518 chapter 5.2.2.3.3 N1N2MessageSubscribe.

Message definition:

<code>supi</code>	String in the form "imsi-" or "nai-" followed by the value of the IMSI or the NAI.
<code>imei.</code>	Optional string (14 or 15 digits). UE IMEI (with or without check digit), required if <code>multi_sim</code> is set to true.
<code>n1_notify_cbk_uri</code>	String. <code>n1NotifyCallbackUri</code> as defined in 3GPP TS 29.518 chapter 6.1.6.2.12 Type: <code>UeN1N2InfoSubscriptionCreateData</code> . This parameter shall be present if the LMF subscribes for LPP message notification. This IE represents the callback URI on which the N1 message shall be notified.
<code>n2_notify_cbk_uri</code>	String. <code>n2NotifyCallbackUri</code> as defined in 3GPP TS 29.518 chapter 6.1.6.2.12 Type: <code>UeN1N2InfoSubscriptionCreateData</code> . This parameter shall be present if the LMF subscribes for a UE NRPPa information notification. This IE represents the callback URI on which the N2 information shall be notified.

`lmf_ue_n1_n2_unsubscribe`

LMF server unsubscription to the reception of LPP messages and/or UE NRPPa PDUs. See 3GPP TS 29.518 chapter 5.2.2.3.4 N1N2MessageUnSubscribe.

Message definition:

<code>supi</code>	String in the form "imsi-" or "nai-" followed by the value of the IMSI or the NAI.
<code>imei.</code>	Optional string (14 or 15 digits). UE IMEI (with or without check digit), required if <code>multi_sim</code> is set to true.
<code>subscription_id</code>	String. <code>subscriptionId</code> as defined in 3GPP TS 29.518 chapter 6.1.3.3 Resource: N1N2 Subscriptions Collection for Individual UE Contexts.

6.8 ePDG

The messages in this section are initiated in the ePDG and are related to an active non-3GPP PDN.

epdg_send_p_cscf

Request to ePDG to send IKE_INFORMATIONAL request to the UE containing the P-CSCF addresses in the CONFIGURATION payload.

The P-CSCF addresses are taken from the list of addresses configured in the PDN See [p_cscf_addr], page 32. Message definition:

imsi	Optional string. UE IMSI. Shall be present if nai is absent.
nai	Optional string. UE network specific identifier-based SUPI. Not applicable to UEs connected to EPC.
imei	Optional string. UE IMEI, sent if available.
apn	String. Access point name.

epdg_pdn_reestab

Request to ePDG to send IKE_INFORMATIONAL request to the UE containing DELETE Payload and NOTIFY payload of type REACTIVATION_REQUESTED_CAUSE.

Message definition:

imsi	Optional string. UE IMSI. Shall be present if nai is absent.
nai	Optional string. UE network specific identifier-based SUPI. Not applicable to UEs connected to EPC.
imei	Optional string. UE IMEI, sent if available.
apn	String. Access point name.

6.9 Remote events

Some messages (events) may be sent by LTEMME without client solicitation.

To receive them, you need to register to those events via [remote event registration], page 91.

The received JSON will have a **message** property with the events name.

Ex:

Register to <event name> event:

```
{
  message: "register",
  register: "<event name>"
}
```

Message received:

```
{
  message: "<event name>",
  ...
}
```

Here is the list of events generated by LTEMME:

registration

Generated when the UE registers or deregisters from the network, or when its M-TMSI or 5G-TMSI changes.

Message definition:

imsi	Optional string. UE IMSI. Shall be present if nai is absent.
nai	Optional string. UE network specific identifier-based SUPI. Not applicable to UEs connected to EPC.
imei	Optional string. UE IMEI, sent if available.
m_tmsi	Optional string. M-TMSI. Present for UEs connected to EPC.
5g_tmsi	Optional string. 5G-TMSI. Present for UEs connected to 5GC.
registered	Boolean. True if UE is currently registered to the network.

registration_reject

Generated when an initial registration attempt is rejected.

Message definition:

imsi	Optional string. UE IMSI, sent if available.
nai	Optional string. UE network specific identifier-based SUPI, sent if available. Not applicable to UEs connected to EPC.
imei	Optional string. UE IMEI, sent if available.
emm_cause	Integer. EMM reject cause. Present for UEs connected to EPC.
5gmm_cause	Integer. 5GMM reject cause. Present for UEs connected to 5GC.

non_ip_data

Generated by data reception over a non IP PDN or unstructured PDU session.

Message definition:

imsi	Optional string. UE IMSI. Shall be present if nai is absent.
nai	Optional string. Network specific identifier-based SUPI. Not applicable to 4G UEs. Shall be present if imsi is absent.
imei	Optional string. UE IMEI, sent if multi_sim is set to true.
apn	Optional string. APN of the non IP bearer. Used for UEs connected to EPC.
erab_id	Optional integer. ERAB identity of the non IP default bearer. Used for UEs connected to EPC.
dnn	Optional string. DNN of the non IP bearer. Used for UEs connected to 5GC.
sst	Optional integer. SST of the non IP bearer. Used for UEs connected to 5GC.

<code>sd</code>	Optional integer. Optional SD of the non IP bearer. Used for UEs connected to 5GC.
<code>pdu_session_id</code>	Optional integer. PDU session ID of the non IP bearer. Used for UEs connected to 5GC.
<code>data</code>	String. ASCII representation of the data hexadecimal dump.

generic_nas_transport

Generated when receiving an EPS uplink generic NAS transport message.
Message definition:

<code>imsi</code>	String. UE IMSI.
<code>imei</code>	Optional string. UE IMEI, sent if <code>multi_sim</code> is set to true.
<code>type</code>	Integer. Generic message container type information element.
<code>payload</code>	String. ASCII representation of the generic message container hexadecimal dump.
<code>add_info</code>	Optional string. ASCII representation of the additional information hexadecimal dump.

5gs_nas_transport

Generated when receiving a 5GS uplink NAS transport message for LPP, SOR, UE policy or UE parameters update.
Message definition:

<code>imsi</code>	Optional string. UE IMSI. Shall be present if <code>nai</code> is absent.
<code>nai</code>	Optional string. Network specific identifier-based SUPI. Not applicable to 4G UEs. Shall be present if <code>imsi</code> is absent.
<code>imei</code>	Optional string. UE IMEI, sent if <code>multi_sim</code> is set to true.
<code>type</code>	Integer (range: 3 to 6). Payload container type information element.
<code>payload</code>	String. ASCII representation of the payload container hexadecimal dump.
<code>add_info</code>	Optional string. ASCII representation of the additional information hexadecimal dump for LPP.

eps_bearer_notification

Generated when an EPS bearer is opened or released.
Message definition:

<code>imsi</code>	Optional string. UE IMSI. Might not be present in case of emergency call.
<code>imei</code>	Optional string. UE IMEI, sent if <code>multi_sim</code> is set to true.
<code>apn</code>	String. Access point name.
<code>pdn_type</code>	Enumeration (ipv4, ipv6, ipv4v6, non-ip). PDN type.
<code>activated</code>	Boolean. True on EPS bearer establishment, false on EPS bearer release.

<code>ipv4_address</code>	Optional string. IPv4 address allocated to the UE.
<code>ipv6_prefix</code>	Optional string. IPv6 prefix allocated to the UE.
<code>erab_id</code>	Integer. ERAB identity.
<code>linked_erab_id</code>	Optional integer. ERAB identity of the default EPS bearer. Present when the EPS bearer opened is a dedicated bearer.
<code>dl_bytes</code>	Optional integer. Number of downlink bytes sent to the UE. Present when <code>activated</code> is set to false.
<code>ul_bytes</code>	Optional integer. Number of uplink bytes received from the UE. Present when <code>activated</code> is set to false.
<code>start_date</code>	Integer. Start date in seconds since 1970-01-01 00:00:00
<code>duration</code>	Optional number. Duration in seconds of bearer lifetime. Present when <code>activated</code> is set to false.
<code>qos_flow_notification</code>	Generated when a QoS flow is opened or released. Message definition:
<code>imsi</code>	Optional string. UE IMSI. Might not be present in case of emergency call.
<code>nai</code>	Optional string. Network specific identifier-based SUPI.
<code>imei</code>	Optional string. UE IMEI, sent if <code>multi_sim</code> is set to true.
<code>dnn</code>	String. Data network name.
<code>pdn_type</code>	Enumeration (ipv4, ipv6, ipv4v6, unstructured, ethernet). PDN type.
<code>activated</code>	Boolean. True on EPS bearer establishment, false on EPS bearer release.
<code>ipv4_address</code>	Optional string. IPv4 address allocated to the UE.
<code>ipv6_prefix</code>	Optional string. IPv6 prefix allocated to the UE.
<code>pdu_session_id</code>	Integer. PDU session identity.
<code>qos_flow_id</code>	Integer. QoS flow identity;
<code>dl_bytes</code>	Optional integer. Number of downlink bytes sent to the UE. Present when <code>activated</code> is set to false.
<code>ul_bytes</code>	Optional integer. Number of uplink bytes received from the UE. Present when <code>activated</code> is set to false.
<code>start_date</code>	Integer. Start date in seconds since 1970-01-01 00:00:00
<code>duration</code>	Optional number. Duration in seconds of bearer lifetime. Present when <code>activated</code> is set to false.

6.10 Examples

1. Config

1. Client sends

```
{
  "message": "config_get",
  "message_id": "foo"
}
```

2. Server replies

```
{
  "message_id": "foo",
  "message": "config_get",
  "name": "UE",
  "logs": {
    "phy": {
      "level": "error",
      "max_size": 0
    },
    ...
    "rrc": {
      "level": "debug",
      "max_size": 1
    }
  }
}
```

2. Error

1. Client sends

```
{
  "message": "bar",
  "message_id": "foo"
}
```

2. Server replies

```
{
  "message_id": "foo",
  "message": "bar",
  "error": "Unknown message: bar"
}
```

7 Command line monitor reference

The following commands are available:

- help** Display the help. Use **help *command*** to have a more detailed help about a command.
- log** [*log_options*] Display the current log state. If *log_options* are given, change the log options. The syntax is the same as the **log_options** configuration property.
- enb** List the connected eNodeBs.
- ng_ran** List the connected NG-RAN nodes.
- ue** [*reg*] List all the UE contexts (the UEs can be connected or not). If used with parameter *reg*, only registered UEs will be displayed.
- uctx** List all the active S1 or NG UE contexts.
- apn** List the configured APN/DNNs.
- pws_write** *local_id* Start broadcasting the ETWS/CMAS message identified by *local_id* on all connected eNodeBs.
- pws_kill** *local_id* Stop broadcasting the ETWS/CMAS message identified by *local_id* on all connected eNodeBs.
- n8** List the addresses of the AMF client and server and the UDM client and server on the N8 interface.
- n8disconnect** Disconnect the AMF client from the UDM on the N8 interface.
- n8connect** *api_root* Connect the AMF client to the UDM server on the N8 interface.
api_root is the api_root of the UDM server in the form: <scheme>://<host>:<port>, where <scheme> is "http" or "https".
 If *api_root* is not provided, the AMF will try to connect to the previously configured address.
- n12** List the AMF client address and the AUSF server address on the N12 interface.
- n12disconnect** Disconnect the AMF client from the AUSF on the N12 interface.
- n12connect** *api_root* Connect the AMF client to the AUSF server on the N12 interface.
api_root is the api_root of the AUSF server in the form: <scheme>://<host>:<port>, where <scheme> is "http" or "https".
 If *api_root* is not provided, the AMF will try to connect to the previously configured address.
- n17** List the AMF client address and the EIR server address on the N17 interface.
- n17disconnect** Disconnect the AMF client from the EIR on the N17 interface.
- n17connect** *api_root* Connect the AMF client to the EIR server on the N17 interface.
api_root is the api_root of the EIR server in the form: <scheme>://<host>:<port>.

where <scheme> is "http" or "https".

If *api_root* is not provided, the AMF will try to connect to the previously configured address.

n11 List the AMF client and server addresses and the LMF client and server addresses on the NL1 interface.

n11disconnect

Disconnect the AMF client from the LMF on the NL1 interface.

n11connect *api_root*

Connect the AMF client to the LMF server on the NL1 interface.

api_root is the api_root of the LMF server in the form: <scheme>://<host>:<port>, where <scheme> is "http" or "https".

If *api_root* is not provided, the AMF will try to connect to the previously configured address.

lmf_client_connect

Connect an LMF client to the AMF server on the NL1 interface.

lmf_non_ue_n2_unsubscribe *n2_callback_uri*

LMF server unsubscription to the reception of non-UE NRPPa PDUs.

See 3GPP TS 29.518 chapter 5.2.2.4.3 NonUeN2InfoUnSubscribe. *n2_callback_uri*: string. Non-UE N2 callback uri.

lmf_ue_n1_n2_subscribe *supi imei n1_callback_uri n2_callback_uri*

LMF server subscription to the reception of LPP messages and UE NRPPa PDUs.

See 3GPP TS 29.518 chapter 5.2.2.3.3 N1N2MessageSubscribe.

supi: string in the form "imsi-" or "nai-" followed followed by the value of the IMSI or the NAI.

imei: optional string. IMEI (14 digits).

n1_callback_uri: string. N1 callback uri.

n2_callback_uri: string. N2 callback uri.

mbs_broadcast_session_setup *index*

Starts a MBS broadcast session.

index: Integer. Index (starting from 0) of the session to start in the **broadcast_sessions** array.

mbs_broadcast_session_release *index*

Stops a MBS broadcast session.

index: Integer. Cid (starting from 0) of the session to stop, returned by **mbs_broadcast_session_setup**.

mbs_session_info

Displays MBS sessions info.

quit

Stop the program and exit.

8 Log file format

8.1 NAS layer

When a NAS message is dumped, the format is:

```
time layer - message
```

When a NAS data PDU is dumped (debug level), the format is:

```
time layer dir MME_UE_ID message_type
      long_content
```

time Time using the selected format

layer Indicate the layer ([NAS] here).

dir UL (uplink) or DL (downlink).

MME_UE_ID
MME S1AP UE identifier (hexadecimal).

message_type
NAS message type.

long_content
Full content of the NAS message if `nas.max_size > 0`.

8.2 IP layer

When a IP data PDU is dumped (debug level), the format is:

```
time layer dir short_content
      long_content
```

time Time using the selected format

layer Indicate the layer ([IP] here).

dir UL (uplink) or DL (downlink).

short_content
Single line content (at least the IP protocol and the source and destination address).

long_content
Optional hexadecimal dump of the PDU if `ip.max_size > 0`.

8.3 S1AP, NGAP, SBcAP, LCSAP and GTP-U layers

When a message is dumped, the format is:

```
time layer - message
```

When a data PDU is dumped (debug level), the format is:

```
time layer dir ip_address short_content
      long_content
```

time Time using the selected format.

layer Indicate the layer ([S1AP], [NGAP], [SBcAP], [LCSAP], or [GTPU] here).

dir Direction: TO or FROM.

ip_address
source or destination IP address, depending on the `dir` field.

`short_content`

Single line content.

`long_content`

- S1AP, NGAP, SBCAP, LCSAP: full ASN.1 content of the message if `layer.max_size > 0`.
- GTPU: hexadecimal dump of the message if `layer.max_size > 0`.

9 FAQ

9.1 Traffic control

I want to generate errors, limit bandwidth, introduce latency...

Easiest and most powerful way is to do this at IP level using the *tc* Linux command. There are various tutorials on the internet but it is not a piece of cake so here are some common commands to handle simple case.

First, *tc* will operate at Linux interface level, which means that for LTE we will control the *tun0* interface created by MME.

Note that this configuration will be dropped each time you restart the MME so if you want to set it automatically and keep it we recommend to place the commands inside *config/mme-ifup* (See [tun_setup_script], page 14).

- To limit overall bandwidth to 2mbps:


```
tc qdisc add dev tun0 root handle 1:0 htb default 1
tc class add dev tun0 parent 1:0 classid 1:1 htb rate 2000kbit
```
- To simulate 10% packet loss:


```
tc qdisc add dev tun0 root handle 1: netem loss 10%
```
- To change previous packet loss to 20%:


```
tc qdisc change dev tun0 root handle 1: netem loss 10%
```
- To add 100ms latency with more or less 10ms:


```
tc qdisc add dev tun0 root handle 1: netem delay 100ms 10ms
```
- Same as previous but with a normal distribution:


```
tc qdisc add dev tun0 root handle 1: netem delay 100ms 10ms distribution normal
```

tc is very powerful and you may also chain filters (qdisc), apply them on specific traffic...

10 Known limitations

We present here the known limitations of LTEMME:

- A single equivalent PLMNs list is supported.
- No interface with external SGW/SMF/UPF is implemented.

11 Change history

11.1 Version 2026-06-12

- added NSAG support
 - `nsag` parameter is added
- added NTN forbidden tracking areas support
- `sms_services_status` parameter is added
- `category` parameter is updated with new values in `emergency_number_list` configuration object
- `attach_result_mode` parameter has a new `combined_sms_only` value
- `eap_aka_fast_reauth` is added to `epdg` object
- `slap_time_to_wait` and `ngap_time_to_wait` parameters are added
- `cancel` remote API message is added
- `group_id` parameter is added to remote API request and cancel messages
- `cancel` parameter is added to remote API response message

11.2 Version 2026-03-13

- `secure_known_emergency_supl` parameter is added
- `ue_aggregate_max_bitrate_dl` and `ue_aggregate_max_bitrate_ul` default values are increased
- `emergency_number_list` and `extended_emergency_number_list` parameters can be set to null in `config_set` remote API
- `ifname` parameter is added to `pdn_list` object for direct interface PDN mode
- `ran_node_id_list`, `tai_list_5gs`, `tai_list` and `rat_selector` parameters are added to `lmf_cfg` object
- `enb` parameter is added to `local_e_smlc` object
- added NSAG support
 - `nsag` parameter is added

11.3 Version 2025-12-12

- OpenSSL library is upgraded to 3.5.4
- removed deprecated DES-CBC IKE and IPsec encryption algorithm
- added IMEI based emergency call support in ePDG
- EPS tracking area update procedure now always allocates a new GUTI
 - added `allocate_new_guti_in_tau` parameter in configuration file and `config_set` remote API
 - the previous behavior can be restored by setting `allocate_new_guti_in_tau` to false
- `automatic_release_timeout` parameter is added to `pdn_list` configuration object and `config_set` remote API
- `eps_assign_5gs_resources_without_n1_mode_capability` parameter is added
- added the Nmbsmf interface to create and delete MBS sessions from an external AF
 - `nmbsmf`, `gtp_addr`, `ingress_tunnel_addr`, `tmgi_lifetime` and `service_area` parameters are added in `mbs` object

- `broadcast_sessions` object parameters have changed
- `mbs_tmgi_deallocate` remote API is added
- no backward compatibility with previous configuration files is ensured
- `gnss_assistance_data` parameter is added to `lmf_cfg` and `local_e_smlc`
- remote APIs `epdg_send_p_cscf` and `epdg_pdn_reestab` are added
- `ecgi`, `ncgi`, `nid` and `n3iwf` parameters are added to `ue_get` remote API
- `lower_bound_timer` parameter is added

11.4 Version 2025-09-19

- `lpp_methods` parameter is added to `local_e_smlc` and `lmf_cfg` configuration objects
- `subscriber_profile_id` parameter is added to `ue_db` configuration object
- `id` parameter range in `filters` is changed from `[0;14]` to `[1;15]`
- `tac` parameter range is updated to forbid reserved values specified in 3GPP 23.003
- `p_cscf`, `dns` and `mtu_ipv4` parameters are added to `ue_modify_pdu_session` remote API
- `force_sqn_resync` parameter is added to user database options
- `coarse_location_information_support` parameter is added
- DCSP values for GTP-U tunnels are set to Expedited Forwarding for QCI/5QI 1 and 65 in the configuration files delivered

11.5 Version 2025-06-13

- NRPPa ASN.1 is updated to v18.5.0
- LPPa ASN.1 is updated to v18.1.0
- LPP ASN.1 is updated to v18.4.0
- MBS broadcast support is added
 - `mbs` parameter is added
 - `mbs_broadcast_session_setup`, `mbs_broadcast_session_release` and `mbs_session_info` remote API and monitor commands are added
 - added a `mme-mbs.cfg` sample configuration file
- `prf-aes128-xcbc` value is added to `ike_prf_list` in the `epdg` object
- `ip_3_tuple` parameter is added to the `traffic_descriptor_components` object of the `ursp_rules` remote API
- added an option to force EMM, ESM, 5GMM or 5GSM status message sending
 - `send_status_on_reject` parameter is added to `emm_procedure_filter`, `esm_procedure_filter`, `5gmm_procedure_filter` and `5gsm_procedure_filter` configuration objects
- `error` and `data` objects in `ike_generate_error` object are replaced by `reject_notify_list` object. The old syntax is still supported for backward compatibility
- `mme_ue_id` and `amf_ue_id` parameters are added to `ue_get` remote API
- `lcs_qos` object is added to the `nr_location_req` remote API
- `lmf_client_connect` remote API is removed. The embedded LMF connects automatically now
- `autonomous_mode` is added to `local_e_smlc` and `lmf_cfg` objects

11.6 Version 2025-03-14

- NGAP ASN.1 is updated to v18.4.0
- `restrict_ec_wb` and `restrict_ec_nb` parameters are added
- `gtp_addr`, `gtp_ext_addr`, `slap_bind_addr` and `ngap_bind_addr` parameters can be arrays
- `destination_mac_addr`, `802.1q_ctag_vid`, `802.1q_stag_vid`, `802.1q_ctag_pcp_dei`, `802.1q_stag_pcp_dei`, `ethertype` and `destination_mac_addr_range` parameters are added to the `traffic_descriptor_components` object of the `ursp_rules` remote API
- `ssc_mode` parameter is added to the `components` object of the `ursp_rules` remote API
- `p_cscf_ipv4_address_attribute_type` and `p_cscf_ipv6_address_attribute_type` parameters are added to the `epdg` object
- `apn_list` parameter is added to `ike_generate_error` configuration object

11.7 Version 2024-12-13

- S1AP ASN.1 is updated to v18.2.0
- NGAP ASN.1 is updated to v18.3.0
- `additional_ue_auth_type` parameter is added to `epdg` configuration object and `config_set` remote API
- `5gs_authentication` parameter is added
- `tta` parameter is added to `emm_procedure_filter`, `5gmm_procedure_filter`, `esm_procedure_filter` and `5gsm_procedure_filter` configuration objects. The previous syntax is still supported for backward compatibility
- `amf_nf_instance_id` parameter is added
- `data` and `tta` parameters are added to `ike_generate_error` configuration object
- `local` parameter is added to `ue_detach` and `ue_del` remote APIs
- `authenticate_known_emergency_supl` parameter is added

11.8 Version 2024-09-13

- N62 interface is added
- `eap` is now deprecated and `eap-md5`, `eap-tls` and `eap-aka` are added to authentication in `pdn_list`
- `auth_type` is added to `nssai_subject_to_nssaa`
- `5gmm_backoff_timer` parameter now accepts the value -2
- `akma` and `routing_indicator` are added to `ue_db`
- `akma_kaf_lifetime` parameter is added
- `ipv6_multicast` parameter is added in UE database
- `rq_timer` parameter is added to `pdn_list` array and `config_set` remote API
- `reflective_qos` parameter is added to `filters` array
- `ue_modify_reflective_qos` remote API is added
- `license` remote API is added
- `linked_erab_id` and `pdu_session_id` parameters are added to the `ue_activate_dedicated_bearer` remote API
- `mobike` parameter is added in `epdg` object and `config_set` remote API
- `ue_set` remote API is added

- use ARP priority 1 for emergency default bearer in mme-ims.cfg configuration file, like our default value for emergency dedicated bearers
- `dont_fragment` parameter is added to `epdg` configuration object and `config_set` remote API
- `nas_cipher_algo_null_allowed` and `nas_integ_algo_null_allowed` parameters are added
- `mobike` value is added to parameter exchange in `ike_generate_error`
- `com_logs_lock` parameter is renamed to `com_log_lock`. `com_logs_lock` is still supported for backward compatibility
- `com_log_us` parameter is added
- `encr-null-auth-aes-gmac-128`, `encr-null-auth-aes-gmac-192` and `encr-null-auth-aes-gmac-256` values are added to `esp_encryption_algo_list`

11.9 Version 2024-06-14

- OpenSSL library is upgraded to 1.1.1w
- nghttp2 library is upgraded to 1.61.0
- added indirect data forwarding support during intra RAT handover
- `correlation_id` parameter is added to remote APIs `location_req` and `nr_location_req` response message
- `mps_priority` and `mcs_priority` are added to `ue_db`
- `operator_defined_access_categories` is added
- `idr_for_emergency` parameter description is added
- `nr_ecid_periodic_meas_termination_req` remote API is added
- `t3324_forced` parameter now accepts the value -2
- added `apn monitor` command
- added `registration_reject` remote API event

11.10 Version 2024-03-15

- default APN or DNN is defined per type (IP, unstructured, ethernet)
- added MOBIKE support
- `emergency_number_list` and `extended_emergency_number_list` parameters can be changed with `config_set` remote API
- `5qi_qos` parameter is added to `erabs` array, `rx` object and `ue_activate_dedicated_bearer` and `ue_modify_pdu_session` remote APIs
- `qci` object in `rx` object is replaced by `qos` object. The old syntax is still supported for backward compatibility
- `type` input parameter and `rat_type` output parameter are added to `ue_get` remote API
- `local_e_smlc` and `lmf_cfg` parameters are added
- LPPa related `location_req`, `lcsap_reset_req`, `ecid_periodic_meas_termination` and `otdoa_information_req` remote apis are added
- NRPPa related `lmf_non_ue_n2_subscribe`, `lmf_non_ue_n2_unsubscribe`, `nr_location_req`, `nr_cancel_location`, `nr_otdoa_information_req` and `trp_information_req` remote apis are added
- LPP related `lpp_request_location` remote api is added

- common LPP/NRPPa `lmf_client_connect`, `lmf_ue_n1_n2_subscribe` and `lmf_ue_n1_n2_unsubscribe` remote apis are added
- `load_balancing_tau` remote API is added
- `sprrt_support` parameter is added
- `mme_name` parameter is added
- `name` parameter is added to `s1` and `ng_ran` remote APIs

11.11 Version 2023-12-15

- EPS user plane integrity protection is added
- `registration_mobility_periodic_error` parameter is renamed to `registration_mobility_periodic_reject_error`. `registration_mobility_periodic_error` is still supported for backward compatibility
- `value_preferred` is added to `confidentiality_protection` parameter
- `loop_count` and `loop_delay` are added to remote API messages
- `sim_events`, `sim_events_loop_count` and `sim_events_loop_delay` parameters are added
- `t3442` parameter is added
- `tai_lists` and `tai_lists_5gs` parameters are added
- `forbidden_eps_tacs` parameter is added
- `areas_list_5gs`, `allowed_5gs_tais` and `forbidden_5gs_tais` parameters are added
- `csg_info_list` is added to `ue_db`
- `com_ssl_ca` parameter is added for SSL verification
- `backoff_timer` parameter now accepts the value -2
- `always_on` parameter is added to `pdn_list`
- `registration_initial_with_security_protection` parameter is added to `5gmm_procedure_filter`

11.12 Version 2023-09-08

- NGAP ASN.1 is updated to v17.5.0
- N20 interface support is added
- `ursp_rules` remote API is added
- `t3501` parameter is added
- `attach_with_security_protection` parameter is added to `emm_procedure_filter` object
- `gtp_use_packet_bundling` parameter is added for GTP-U PDUs bundling support
- `ipsec` remote API added
- `ipv6_mtu` parameter can no longer be changed using the `config_set` remote API
- `ethernet` value is added to `pdn_type` parameter
- `mtu_ethernet_frame_payload` parameter is added
- `destination_mac_addr`, `source_mac_addr`, `802.1q_ctag_vid`, `802.1q_stag_vid`, `802.1q_ctag_pcp_dei`, `802.1q_stag_pcp_dei`, `ethertype`, `destination_mac_addr_range` and `source_mac_addr_range` parameters are added to `components` object
- `nssai_subject_to_nsac` parameter is added
- `ue_slice_max_bitrate` parameter is added

11.13 Version 2023-06-10

- N5 interface support is added
- NGAP ASN.1 is updated to v17.4.0
- `gtp_tx_bitrate`, `gtp_rx_bitrate`, `ip_tx_bitrate` and `ip_rx_bitrate` are added to the `stats remote API`
- `eplmn_list` parameter can no longer be changed with `config_set remote API`
- `nas_transport_lpp` parameter is added to `5gmm_procedure_filter`
- `pdn_type` parameter value `non-ip` in `pdn_list` array is renamed to `unstructured`. `non-ip` is still supported for backward compatibility
- `com_logs_lock` parameter added to disable logs configuration change via remote API

11.14 Version 2023-03-17

- `com_addr` parameter now uses `::` address instead of `0.0.0.0` in the delivered configuration files to allow IPv6 connection
- `cag_supported` parameter is renamed to `cag_support`. `cag_supported` is still supported for backward compatibility
- `requested_t3512_forced` parameter is added
- `gwus_support` and `gwus_prob_forced` parameters are added
- `purge_timer` parameter is added
- `nssai_subject_to_nssaa` parameter is added
- `network_slice_specific_authentication` parameter is added to `5gmm_procedure_filter`
- `ext_emm_cause` parameter is added
- `ipv6_send_dns_in_ra` parameter is added

11.15 Version 2022-12-16

- S1AP ASN.1 is updated to v17.2.0
- NGAP ASN.1 is updated to v17.2.0
- improved support for Equivalent PLMNs
- added `ca_certificate` parameter to `eap_tls`
- added `extended_emergency_number_list` parameter
- added `t3560_ng_ran_sat` parameter
- added S1AP/NGAP Masked IMEISV support
- added `sst` and `sd` parameters to the `ue_activate_dedicated_bearer remote API`
- added new ePDG IKE-Sa and ESP-Sa algorithms
- added `cag_supported`, `nid`, `cag_info_list`, `cag_id_list` and `cag_only_ind` parameters for NPN support
- added `edrx_ptw_nr` parameter
- added `single_address_bearers_only` and `single_address_bearers_first_address` parameters
- added `dpd` value to `exchange` parameter
- added `utc` parameter to remote API response messages

11.16 Version 2022-09-16

- "ipsec debug" monitor is now deprecated. Set ipsec.verbose to 1 in log configuration
- S1AP ASN.1 is updated to v17.1.0
- NGAP ASN.1 is updated to v17.1.1
- added missing n3gpp parameter to ue_detach, ue_identity_request, ue_deactivate_bearer, ue_modify_pdu_session and 5gs_nas_transport remote APIs
- added dhcpv6_t1 and dhcpv6_t2 parameters
- added dpd_timer_value parameter
- added wildcard as a prefix support for attach_reject_filter
- added priority_level, pre_emption_capability and pre_emption_vulnerability parameters to ue_modify_pdu_session remote API
- added guti_realloc remote API
- added registration remote API event

11.17 Version 2022-06-17

- OpenSSL library is upgraded to 1.1.1n
- improved GTP-U performance
- removed ue_db_filename configuration option and associated functionality
- added ipv4_local_addr, ipv6_remote_addr_prefix and ipv6_local_addr_prefix TFT components
- added new ePDG IKE-Sa and ESP-Sa algorithms and groups
- added ePDG IKE-Sa rekeying procedure
- added apn, dnn, sst and sd fields to non_ip_data remote API
- added apn_oi parameter
- added S1AP EN-DC SON Configuration Transfer support
- added start_timestamp and end_timestamp to log_get API
- added allow_apn_in_attach_req parameter
- added ike_duration parameter
- esp_duration and ike_duration parameters can be changed with config_set API
- configured TCP congestion control to bbr in lte_init.sh script
- S1AP ASN.1 is updated to v16.9.0
- added missing n13 options
- added n13, n13connect and n13disconnect remote APIs

11.18 Version 2022-03-18

- added --no-nat6 option to the lte_init.sh script
- added NAT traversal support to ePDG
- increment_serial_number optional parameter is added to pws_write remote API
- ike_generate_error configuration object is added
- eps_user_unknown_reject_cause and 5gs_user_unknown_reject_cause optional parameters are added. The default EPS reject cause for an unknown user is changed from 2 (IMSI unknown in HSS) to 8 (EPS services and non-EPS services not allowed)

11.19 Version 2021-12-17

- LCSAP and NL1 support are added
- `registration_area_alloc_ind` parameter is added to control the MICO registration area allocation
- `ike_encryption_algo_list`, `ike_integrity_algo_list`, `ike_prf_list`, `ike_dh_group_list`, `esp_encryption_algo_list`, `esp_integrity_algo_list` and `esp_dh_group_list` parameters are added to make the list of ePDG supported algorithms configurable
- `license` monitor command is added
- `config_get/config_set` remote APIs are updated to handle more logging options
- `cpu_core_list` parameter is added to control the CPUs used by LTEMME
- `ue_aggregate_max_bitrate_dl` and `ue_aggregate_max_bitrate_ul` default values are increased
- `ue_modify_bearer` and `ue_modify_pdu_session` have a new `dns` parameter
- `nr_support` parameter is renamed to `dcnr_support`. `nr_support` is still supported for backward compatibility
- `dns_addr` parameter is added to the `config_set` remote API
- `dns` parameter is added to the `ue_modify_bearer` and `ue_modify_pdu_session` remote APIs
- S1AP ASN.1 is updated to v16.7.0

11.20 Version 2021-09-17

- the minimum GLIBC version is now 2.17
- addition of control plane CIoT 5GS optimization
- logs can be displayed with microseconds precision
- `truncated_amf_set_id` and `truncated_amf_pointer` parameters are added for NB-IoT control plane CIoT 5GS reestablishment
- the former `ims_vops` parameter is now split in 3 parameters `ims_vops_eps`, `ims_vops_5gs_3gpp` and `ims_vops_5gs_n3gpp`
- `emc_n3gpp` parameter is added to control emergency support indication in non-3GPP 5GS
- `control_plane_service_request` filter is added to `5gmm_procedure_filter`
- NAI can now be configured instead of IMSI, and the remote APIs are updated accordingly
- `omit_auth_in_first_auth_rsp` ePDG option is added to workaround some buggy UEs
- the `mme-ims.cfg` configuration file now logs more network interfaces by default
- S1AP ASN.1 is updated to v16.6.0
- NGAP ASN.1 is updated to v16.6.0

12 License

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Abbreviations

5G-EIR	5G Equipment Identity Register
5GC	5G Core Network
5GS	5G System
5QI	5G QoS Identifier
AMF	Access and Mobility Management Function
APN	Access Point Name
AUSF	Authentication Server Function
DCNR	Dual Connectivity with NR
DL	Downlink
DNN	Data Network Name
E-RAB	E-UTRAN Radio Access Bearer
E-UTRA	Evolved UMTS Terrestrial Radio Access
E-UTRAN	Evolved UMTS Terrestrial Radio Access Network
EIR	Equipment Identity Register
EPC	Evolved Packet Core
ePCO	Extended Protocol Configuration Options
ePDG	evolved Packet Data Gateway
EPS	Evolved Packet System
HSS	Home Subscriber Server
IMEI	International Mobile Equipment Identity
IMSI	International Mobile Subscriber Identity
LTE	Long Term Evolution
MBS	Multicast Broadcast Service
MME	Mobility Management Entity
NAS	Non Access Stratum
NR	New Radio
PCO	Protocol Configuration Options
PCRF	Policy and Charging Enforcement Function
PDN	Packet Data Network
PDU	Protocol Data Unit
PGW	Packet Data Network Gateway
QCI	Quality of Service (QoS) Class Identifier
QoS	Quality of Service
SDU	Service Data Unit

SGW	Serving Gateway
SMF	Session Management Function
TMSI	Temporary Mobile Subscriber Identity
UDM	Unified Data Management
UE	User Equipment
UL	Uplink
UPF	User Plane Function
USIM	Universal Subscriber Identity Module
VoLTE	Voice over LTE